Communications Network

Ford Ecat Electronic Parts Catalog

Special Tool(s)

	Fluke 77-IV Digital Multimeter FLU77-4 or equivalent
	Vehicle Communication Module (VCM) and Integrated Diagnostic System (IDS) software with appropriate hardware, or equivalent scan tool
	Flex Probe Kit NUD105-R025D or equivalent

Principles of Operation

The vehicle has 3 module communication networks which are connected to the Data Link Connector (DLC) , located under the instrument panel.

Both the HS-CAN and the MS-CAN use an unshielded twisted pair cable of data (+) and data (-) circuits. The HS-CAN operates at a maximum data transfer speed of 500 Kbps and is designed for real time information transfer and control. The MS-CAN operates at a maximum data transfer speed of 125 Kbps for bus messages and is designed for general information transfer.

The I-CAN is not connected to the DLC . The I-CAN modules communicate with the scan tool over the HS-CAN . The Instrument Panel Cluster (IPC) is used as a gateway for the messages to transfer between the scan tool and the modules on the I-CAN .

Controller Area Network (CAN) Fault Tolerance

NOTE: *The oscilloscope traces below are from the Integrated Diagnostic System (IDS) oscilloscope taken using the CAN setup defaults. The traces are for both data (+) and data (-) taken simultaneously (2-channel) at a sample rate of 1 mega-sample per second (1MS/s) or greater and viewed at 20 microseconds (20µs) per division. Readings taken with a different oscilloscope will vary from those shown. Compare any suspect readings to a known good vehicle.*

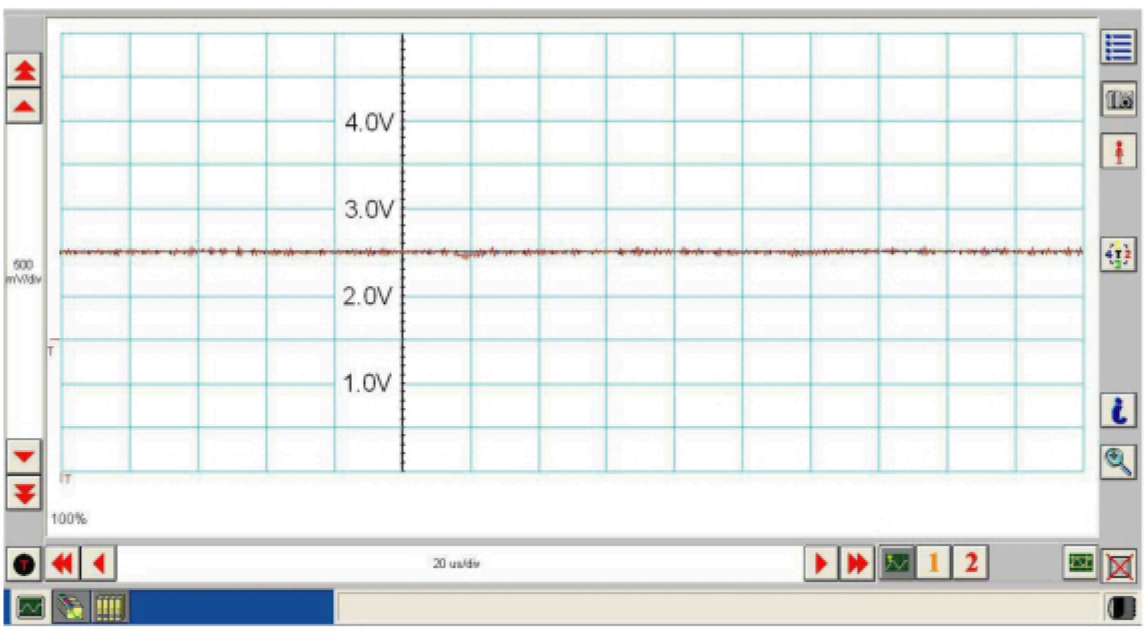
Normal CAN Operation

N0102180

The data (+) and data (-) circuits are each regulated to approximately 2.5 volts during neutral or rested network traffic. As messages are sent on the data (+) circuit, voltage is increased by approximately 1.0 volt. Inversely, the data (-) circuit is reduced by approximately 1.0 volt when a message is sent.

Successful communication of a message can usually be identified by the slight spike at the end of a message transmission. This spike is a "positive acknowledge" signal from multiple modules confirming receipt of the message sent.

CAN Circuits Shorted Together



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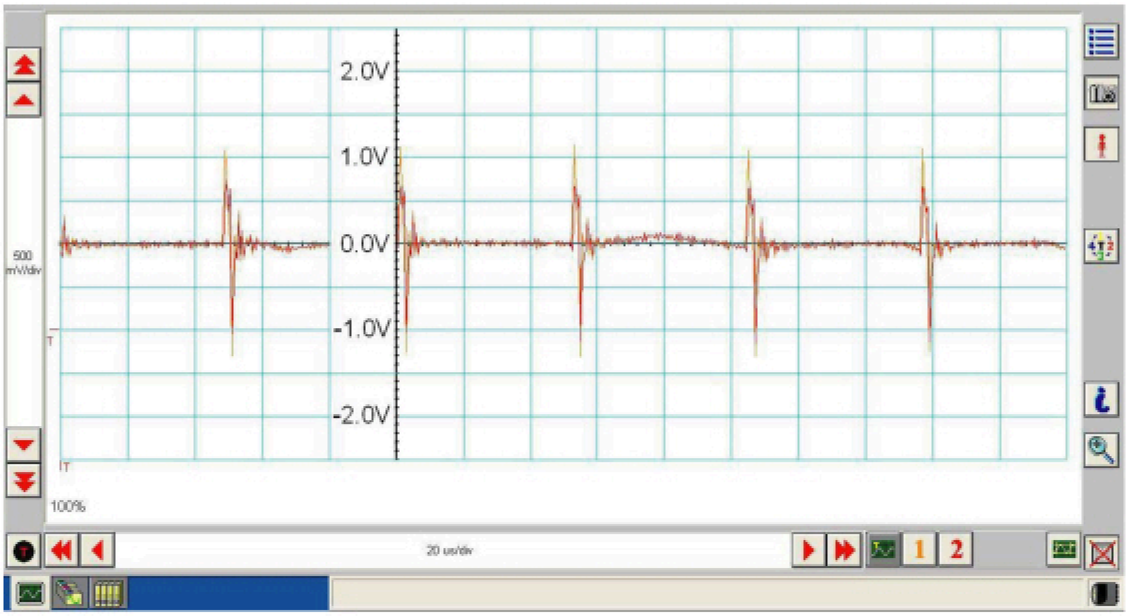
In the event the data (+) and data (-) circuits become shorted together, the signal stays at base voltage (2.5V) continuously and all communication capabilities are lost.

CAN (+) Circuit Shorted To Ground



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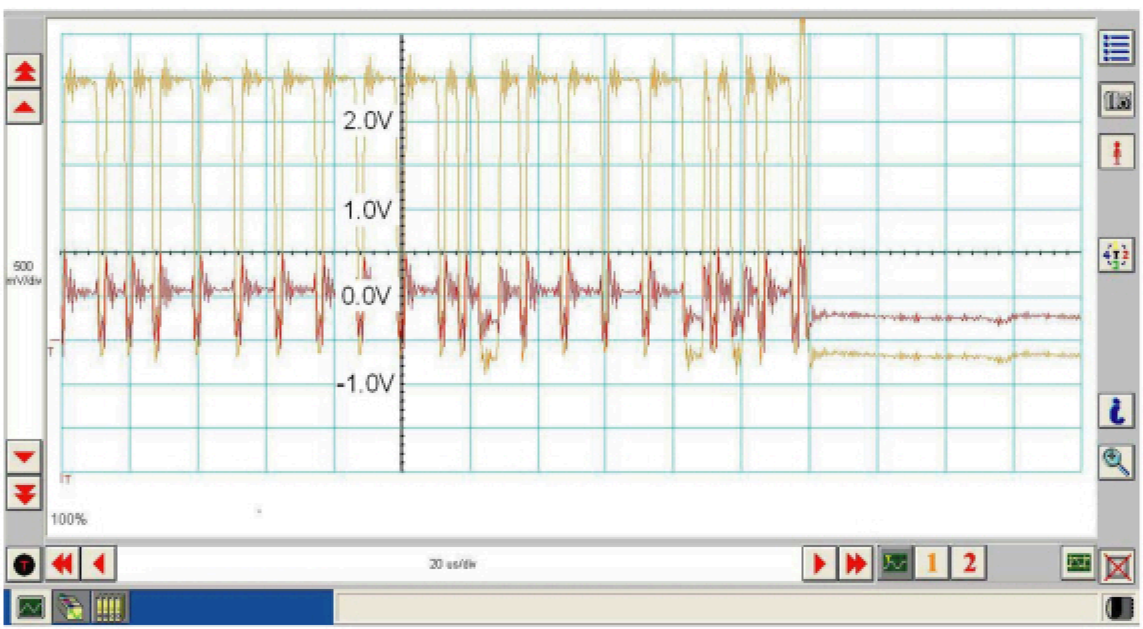
In the event the data (+) circuit becomes shorted to ground, both the data (+) and data (-) circuits are pulled low (0V) and all communication capabilities are lost.

CAN (-) Circuit Shorted To Ground



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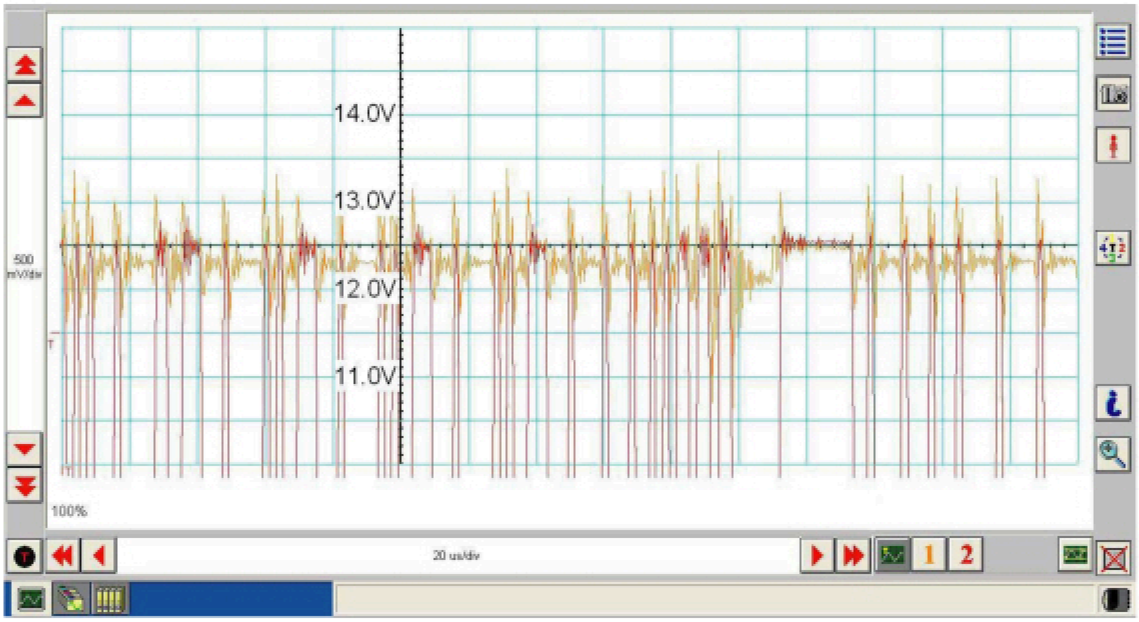
In the event the data (-) circuit becomes shorted to ground, the data (-) circuit is pulled low (0V) and the data (+) circuit reaches normal peak voltage (3.5V) during communication but falls to 0V instead of normal base voltage (2.5V). Communication can continue but at a degraded level.

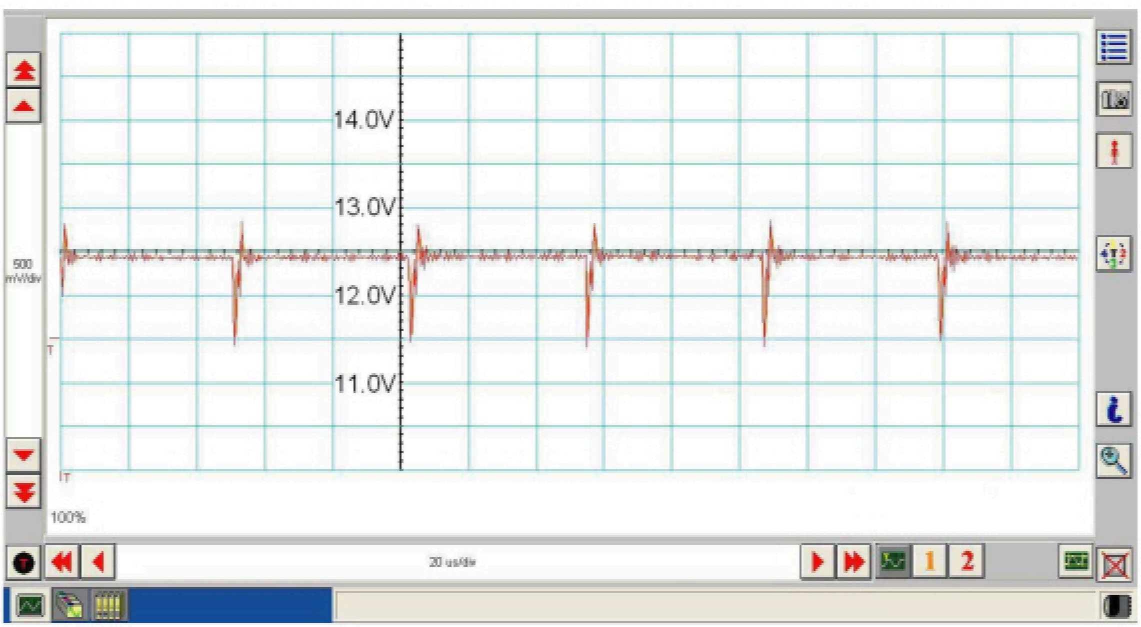
CAN (+) Circuit Shorted To Battery Voltage



N0102184

In the event the data (+) circuit becomes shorted to battery voltage, the data (+) circuit is pulled high (12V) and the data (-) circuit falls to abnormally high voltage (above 5V) during communication and reaches battery voltage (12V) for peak voltage. Communication can continue but at a degraded level.

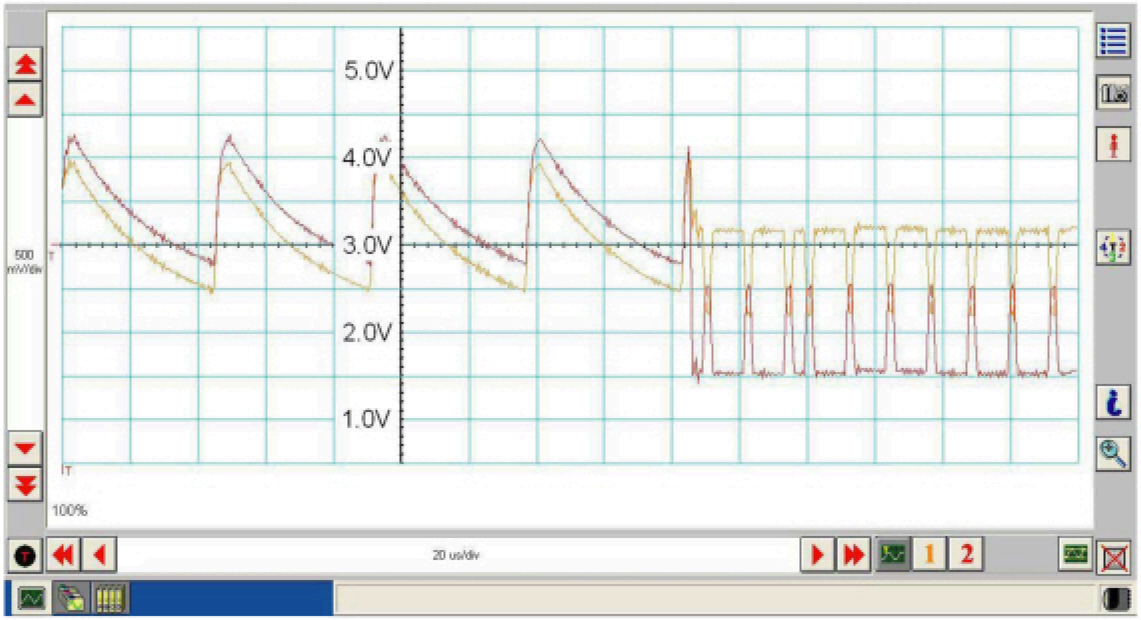
CAN (-) Circuit Shorted To Battery Voltage



N0102185

In the event the data (-) circuit becomes shorted to battery voltage, both the data (+) and data (-) circuits are pulled high (12V) and all communication capabilities are lost.

CAN Circuit Signal Corruption



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Rhythmic oscillations, inductive spikes or random interference can corrupt the network communications. The corruption signal source may be outside electrical interference such as motors or solenoids or internal interference generated from a module on the network. In some cases, an open in either the data (+) or data (-) circuit to a network module may cause the module to emit interference on the one circuit which is still connected. The trace shown is an example of a "sawtooth" pattern transmitted from a module with one open network circuit.

Other corruptions may be present when a module is intermittently powered up and down. The module on power up can initiate communication out of sync with other modules on the network causing momentary communication outages. These types of message corruptions are referred to as a Negative-Acknowledge (NACK) message and lack an acknowledgement spike at the end of the transmitted message.

Controller Area Network (CAN) Multiplex Messages

Modules on the CAN utilize simultaneous communication of 2 or more messages on the same network circuits. The following chart summarizes the messages sent and received on the network.

CAN Module Communication Message Chart

NOTE: *This chart describes the specific HS-CAN , MS-CAN and I-CAN messages broadcast by each module, and the module(s) that receive the message.*

NOTE: *The TPM module is also known as the Radio Transceiver Module (RTM).*

Broadcast Message	Originating Module	Network Type	Receiving Module(s)
4X4 indicator status	TCCM	HS-CAN	• IPC
4X4 message display	TCCM	HS-CAN	• IPC
4X4 mode	TCCM	HS-CAN	• ABS module • IPC • PCM

Broadcast Message	Originating Module	Network Type	Receiving Module(s)
			• BCM
4X4 range selection	TCCM	HS-CAN	• ABS module • PCM • BCM • PSCM • IPC
4X4 service required	TCCM	HS-CAN	• IPC • RCM
4X4 warning indicator request	TCCM	HS-CAN	• IPC
A/C clutch status	PCM	HS-CAN	• BCM
A/C clutch status (gateway)	BCM	MS-CAN	• HVAC module
A/C pressure	PCM	HS-CAN	• BCM
A/C recirc request	BCM	MS-CAN	• HVAC module
A/C request	FCIM	I-CAN	• IPC • APIM • FCDIM
A/C request (gateway)	IPC	HS-CAN	• BCM
A/C request (gateway)	BCM	MS-CAN	• HVAC module
A/C request	HVAC module	MS-CAN	• BCM
A/C request (gateway)	BCM	HS-CAN	• PCM
ABS event in progress	ABS module	HS-CAN	• PCM • RCM • TBC module
ABS warning indicator request	ABS module	HS-CAN	• IPC • RCM
ABS/stability control chime request	ABS module	HS-CAN	• IPC
Accelerator pedal position	PCM	HS-CAN	• TCCM • ABS module • BCM • RCM

Broadcast Message	Originating Module	Network Type	Receiving Module(s)
Accelerometer and yaw sensor data	ABS module	HS-CAN	• TCCM • IPC • PCM
Accessory delay status	BCM	HS-CAN	• IPC
Accessory delay status	BCM	MS-CAN	• APIM • ACM
ACM radio functions	ACM	I-CAN	• APIM • FCDIM
Airbag deployment eCall confirmation	APIM	HS-CAN	• BCM • RCM
Airbag deployment eCall notification	RCM	HS-CAN	• APIM
Airbag deployment status	RCM	HS-CAN	• BCM
Airbag deployment status enabled	RCM	HS-CAN	• OCSM
Airbag status	RCM	HS-CAN	• APIM
Airbag warning indicator request	RCM	HS-CAN	• IPC
Airbag warning indicator status	IPC	HS-CAN	• RCM
Alternator charge mode	BCM	HS-CAN	• IPC • PCM
Alternator charge voltage request	BCM	HS-CAN	• PCM
Alternator current output	PCM	HS-CAN	• BCM
Alternator duty cycle	PCM	HS-CAN	• BCM
Alternator fault	PCM	HS-CAN	• BCM
Ambient air temperature	PCM	HS-CAN	• TCCM • BCM • IPC • PAM
Ambient air temperature (gateway)	BCM	MS-CAN	• ACM • HVAC module
Ambient lighting color	APIM	HS-CAN	• BCM

Broadcast Message	Originating Module	Network Type	Receiving Module(s)
Ambient lighting color	BCM	MS-CAN	• HVAC module
Ambient lighting intensity	APIM	HS-CAN	• BCM
Ambient lighting intensity	BCM	MS-CAN	• HVAC module
Audio settings	IPC	I-CAN	• ACM
Audio source	ACM	I-CAN	• APIM • IPC • Audio DSP module • FCDIM
Audio volume status	Audio DSP module	I-CAN	• APIM
Autolamp delay status	BCM	HS-CAN	• IPC
Autolamp delay time	IPC	HS-CAN	• BCM
Back up camera command	IPC	HS-CAN	• BCM
Back up camera status	BCM	HS-CAN	• IPC
Battery charge low	BCM	HS-CAN	• IPC
Battery charge low	IPC	I-CAN	• APIM • FCDIM
Belt-minder® audio mute	IPC	I-CAN	• ACM • Audio DSP module
Beltminder programming request	RCM	HS-CAN	• IPC
Beltminder warning status	RCM	HS-CAN	• BCM • IPC
Beltminder warning status (gateway)	BCM	MS-CAN	• ACM • Audio DSP module
Brake on/off switch status	PCM	HS-CAN	• ABS module • RCM • TBC module
Brake (red) warning indicator request	ABS module	HS-CAN	• IPC • RCM
Brake fluid level low	BCM	HS-CAN	• IPC
Brake lamp switch status	PCM	HS-CAN	• ABS module

Broadcast Message	Originating Module	Network Type	Receiving Module(s)
			• RCM • TBC module
Brake required	ABS module	HS-CAN	• PCM • TBC module
Camera settings	APIM	I-CAN	• IPC
Camera settings	FCDIM	I-CAN	• IPC
Camera status	BCM	HS-CAN	• IPC
Camera status	BCM	MS-CAN	• ACM
Camera status (gateway)	IPC	I-CAN	• APIM • FCDIM
CD data	APIM	I-CAN	• ACM
CD data	FCDIM	I-CAN	• ACM
CD load/eject	ACM	I-CAN	• APIM • FCDIM
Chime selection	IPC	I-CAN	• ACM • Audio DSP module
Climate controlled seat command, front	HVAC module	MS-CAN	• DCSM
Climate controlled seat command, front	FCIM	I-CAN	• IPC
Climate controlled seat command, front (gateway)	IPC	HS-CAN	• BCM
Climate controlled seat command, front (gateway)	BCM	MS-CAN	• DCSM
Climate controlled seat status, front	DCSM	MS-CAN	• HVAC module
Climate controlled seat request, front	IPC	HS-CAN	• BCM
Climate controlled seat request, front	BCM	MS-CAN	• HVAC module
Climate controlled seat request, rear	IPC	HS-CAN	• BCM
Climate controlled seat request, rear	BCM	MS-CAN	• HVAC module
Climate controlled seat status, front	DCSM	MS-CAN	• BCM

Broadcast Message	Originating Module	Network Type	Receiving Module(s)
Climate controlled seat status, front (gateway)	BCM	HS-CAN	• IPC
Climate controlled seat status, front (gateway)	IPC	I-CAN	• FCIM
Compass display data	BCM	MS-CAN	• ACM • FDIM
Compass zone and calibration status	BCM	HS-CAN	• IPC
Compass zone and calibration status	BCM	MS-CAN	• ACM • FDIM
Courtesy delay status	BCM	MS-CAN	• HVAC module
Cruise control mode	PCM	HS-CAN	• IPC
Cruise control override	PCM	HS-CAN	• TCCM • BCM • IPC • RCM
Cruise control set speed	PCM	HS-CAN	• IPC
Cruise control status	PCM	HS-CAN	• TCCM • BCM • IPC • RCM
Date and time	IPC	I-CAN	• APIM • FCDIM
Date and time request	APIM	I-CAN	• IPC
Day/night status	BCM	HS-CAN	• TBC module • SCCM • IPC
Day/night status	BCM	MS-CAN	• HVAC module
Day/night status (gateway)	IPC	I-CAN	• APIM • FCIM • FCDIM
Differential lock disable request	ABS module	HS-CAN	• TCCM
Differential lock indicator request	TCCM	HS-CAN	• BCM • IPC
Door ajar status	BCM	HS-CAN	• ABS module • IPC

Broadcast Message	Originating Module	Network Type	Receiving Module(s)
			• OCSM • SCCM • TCCM
Door ajar status	BCM	MS-CAN	• PRB module
Door ajar status	IPC	I-CAN	• APIM • FCIM • FCDIM
Drivers seat memory command	DSM	MS-CAN	• BCM
Drivers seat status	DSM	MS-CAN	• BCM
Drivers seat status (gateway)	BCM	HS-CAN	• SCCM
DRL status	BCM	HS-CAN	• IPC
Easy entry exit command	IPC	HS-CAN	• BCM
Easy entry request	DSM	MS-CAN	• BCM
Easy entry/exit request	BCM	HS-CAN	• SCCM
Easy entry/exit status	BCM	HS-CAN	• IPC
Easy entry/exit status	DSM	MS-CAN	• BCM
Engine air intake temperature	PCM	HS-CAN	• BCM
Engine air intake temperature	BCM	MS-CAN	• HVAC module
Engine coolant temperature	PCM	HS-CAN	• BCM • IPC
Engine MIL request	PCM	HS-CAN	• BCM • IPC • RCM
Engine off	BCM	HS-CAN	• ABS module • IPC • PCM
Engine oil life percent	PCM	HS-CAN	• IPC
Engine oil life reset	IPC	HS-CAN	• PCM
Engine overheat mitigation	PCM	HS-CAN	• IPC

Broadcast Message	Originating Module	Network Type	Receiving Module(s)
Engine RPM	PCM	HS-CAN	• TCCM • ABS module • BCM • IPC • RCM
Engine RPM (gateway)	BCM	MS-CAN	• APIM
Engine RPM (gateway)	IPC	I-CAN	• APIM • ACM • Audio DSP module • FCDIM
Engine service required	PCM	HS-CAN	• IPC • RCM
Engine torque reduction request	ABS module	HS-CAN	• TCCM • PCM
English/metric display - navigation	ACM	MS-CAN	• BCM
English/metric mode	APIM	I-CAN	• IPC
English/metric mode	FCDIM	I-CAN	• IPC
English/metric mode (gateway)	IPC	HS-CAN	• BCM • PCM
EPAS calculated current flow	PSCM	HS-CAN	• BCM
EPAS fault indicator	PSCM	HS-CAN	• IPC
EPAS measured voltage	PSCM	HS-CAN	• BCM
ePRNDL mode	IPC	HS-CAN	• PCM
Gear lever position	PCM	HS-CAN	• BCM • RCM • IPC • PAM • SCCM • TCCM • ABS module
Gear lever position	BCM	MS-CAN	• APIM • GPSM • HVAC module
Gear lever position	IPC	I-CAN	• APIM • ACM • Audio DSP module

Broadcast Message	Originating Module	Network Type	Receiving Module(s)
			• FCDIM
FCIM button press data	FCIM	MS-CAN	• BCM
FCIM button press data (gateway)	BCM	HS-CAN	• IPC
FCIM button press data (4.2-inch screen, 8-inch touchscreen)	FCIM	I-CAN	• APIM • FCDIM
FCIM set volume	FCIM	I-CAN	• ACM • Audio DSP module
FDIM display status	HVAC module	MS-CAN	• FDIM
Fuel economy reset request	BCM	HS-CAN	• IPC
Fuel flow display	PCM	HS-CAN	• IPC
Fuel level status	IPC	HS-CAN	• APIM • BCM
Fuel level status (gateway)	BCM	MS-CAN	• ACM • HVAC module
Fuel level status	IPC	I-CAN	• APIM
Fuel cutoff request	RCM	HS-CAN	• PCM
GPS compass direction	GPSM	MS-CAN	• APIM • BCM
GPS compass direction (gateway)	BCM	HS-CAN	• IPC
GPS location	GPSM	MS-CAN	• APIM
Headlamp high beam indicator request	BCM	HS-CAN	• IPC
Headlamp high beam flash to pass	SCCM	HS-CAN	• BCM
Headlamp low beam status	BCM	HS-CAN	• SCCM
Headlamp switch status	SCCM	HS-CAN	• BCM
Headlamp switch status (gateway)	BCM	MS-CAN	• ACM • HVAC module
Headlamps on warning chime command	BCM	HS-CAN	• IPC

Broadcast Message	Originating Module	Network Type	Receiving Module(s)
Hill descent active	ABS module	HS-CAN	• PCM
Hill descent mode status	ABS module	HS-CAN	• IPC
Hood ajar status	BCM	HS-CAN	• IPC • OCSM • ABS module • SCCM
HVAC auto A/C indicator temperature	BCM	HS-CAN	• IPC
HVAC auto A/C indicator temperature (gateway)	IPC	I-CAN	• APIM • FCDIM
HVAC blower motor command	HVAC module	MS-CAN	• BCM
HVAC button status (with navigation)	HVAC module	MS-CAN	• ACM
HVAC button status (without navigation)	FCIM	MS-CAN	• HVAC module
HVAC button status, navigation ACM	ACM	MS-CAN	• HVAC module
HVAC floor mode request	IPC	HS-CAN	• BCM
HVAC floor mode request	BCM	MS-CAN	• HVAC module
HVAC floor mode status	HVAC module	MS-CAN	• BCM
HVAC floor mode status	BCM	HS-CAN	• IPC
HVAC floor mode status	IPC	I-CAN	• APIM • FCDIM
HVAC front temperature set point request	FCIM	I-CAN	• IPC • APIM • FCDIM
HVAC front temperature set point request (gateway)	IPC	HS-CAN	• BCM
HVAC front temperature set point request (gateway)	BCM	HS-CAN	• HVAC module
HVAC front temperature set point status	HVAC module	MS-CAN	• BCM
HVAC front temperature set point status (gateway)	BCM	HS-CAN	• IPC
HVAC indicator status	HVAC module	MS-CAN	• FCIM

Broadcast Message	Originating Module	Network Type	Receiving Module(s)
HVAC indicator status, navigation ACM	HVAC module	MS-CAN	• ACM
HVAC panel mode request	IPC	HS-CAN	• BCM
HVAC panel mode request	BCM	MS-CAN	• HVAC module
HVAC panel mode status	HVAC module	MS-CAN	• BCM
HVAC panel mode status (gateway)	BCM	HS-CAN	• IPC
HVAC panel mode status (gateway)	IPC	I-CAN	• APIM • FCDIM
HVAC rear control status	IPC	HS-CAN	• BCM
HVAC rear control status	BCM	MS-CAN	• HVAC module
HVAC rear defrost, heated mirror request	IPC	HS-CAN	• BCM
HVAC rear defrost, heated mirror request	BCM	MS-CAN	• HVAC module
HVAC rear system mode request	IPC	HS-CAN	• BCM
HVAC rear system mode request	BCM	MS-CAN	• HVAC module
HVAC rear temperature set point request	FCIM	I-CAN	• IPC • APIM • FCDIM
HVAC rear temperature set point request (gateway)	IPC	HS-CAN	• BCM
HVAC rear temperature set point request (gateway)	BCM	MS-CAN	• HVAC module
HVAC rear temperature set point status	HVAC module	MS-CAN	• BCM
HVAC rear temperature set point status (gateway)	BCM	HS-CAN	• IPC
HVAC recirc door status	HVAC module	MS-CAN	• BCM
HVAC recirc door status (gateway)	BCM	HS-CAN	• IPC
HVAC recirc door status (gateway)	IPC	I-CAN	• APIM • FCDIM
HVAC recirc request	IPC	HS-CAN	• BCM
HVAC recirc request	BCM	MS-CAN	• HVAC module

Broadcast Message	Originating Module	Network Type	Receiving Module(s)
HVAC temperature units	HVAC module	MS-CAN	• BCM
HVAC temperature units	BCM	HS-CAN	• IPC
HVAC voice request	ACM	MS-CAN	• HVAC module
Ignition key read data	BCM	HS-CAN	• PCM
Ignition key type	BCM	HS-CAN	• PCM • RCM • IPC • APIM
Ignition status	BCM	HS-CAN	• TCCM • ABS module • IPC • PAM • PCM • PSCM • SCCM • TBC module • OCSM • RCM
Ignition status	BCM	MS-CAN	• HVAC module • DSM • GPSM • ACM • PRB module
Ignition status	IPC	I-CAN	• APIM • Audio DSP module • FCIM • FCDIM
Ignition switch position	BCM	MS-CAN	• ACM • DSM • GPSM • HVAC module • PRB module
IKT programming status	BCM	HS-CAN	• IPC
Illuminated dimming level	IPC	I-CAN	• APIM • Audio ACM module • FCIM • FCDIM
Illumination display	IPC	I-CAN	• APIM • FCIM • FCDIM

Broadcast Message	Originating Module	Network Type	Receiving Module(s)
Illuminated entry status	BCM	MS-CAN	• HVAC module
Illuminated exit status	BCM	MS-CAN	• HVAC module
Illumination dimming level	BCM	HS-CAN	• IPC • TBC module
Illumination dimming level	BCM	MS-CAN	• ACM • HVAC module
In-car temperature sensor	HVAC module	MS-CAN	• BCM
IPC chime	IPC	I-CAN	• ACM • Audio DSP module
Keycode status	BCM	HS-CAN	• IPC
Key-in-ignition status	BCM	HS-CAN	• IPC
Key-in-ignition status	BCM	MS-CAN	• ACM • DSM
Key-in-ignition warning chime command	BCM	HS-CAN	• IPC
Language selection	IPC	I-CAN	• APIM • FCDIM
Light sensor status	BCM	HS-CAN	• IPC • SCCM • TBC module
Media player display request	APIM	I-CAN	• ACM
Media player display request	FCDIM	I-CAN	• ACM
Memory feedback request	DSM	MS-CAN	• BCM
Memory feedback request (gateway)	BCM	HS-CAN	• IPC
Memory switch status	DSM	MS-CAN	• BCM
Message center display	IPC	HS-CAN	• APIM • BCM • SCCM
Message center display (gateway)	BCM	MS-CAN	• DSM • HVAC module

Broadcast Message	Originating Module	Network Type	Receiving Module(s)
Multifunction button push display	HVAC module	MS-CAN	• BCM
Multifunction button push display (gateway)	BCM	HS-CAN	• IPC
Multimedia system status	IPC	HS-CAN	• SCCM
MyKey® volume limit	IPC	I-CAN	• ACM • Audio DSP module
Navigation audio settings	APIM	I-CAN	• ACM • Audio DSP module
Navigation audio settings	FCDIM	I-CAN	• ACM • Audio DSP module
Navigation rolling wheel count	ABS module	HS-CAN	• BCM • IPC
Navigation rolling wheel count (gateway)	BCM	MS-CAN	• APIM • ACM • GPSM
Navigation rolling wheel count (gateway)	IPC	I-CAN	• APIM
Navigation unit settings	BCM	HS-CAN	• IPC
Navigation wheel direction	ABS module	HS-CAN	• BCM • IPC
Navigation wheel direction (gateway)	BCM	MS-CAN	• APIM • ACM • GPSM
OCS calibration data	OCSM	HS-CAN	• RCM
OCS fault status	OCSM	HS-CAN	• RCM
OCS object entrapped message request	OCSM	HS-CAN	• IPC
OCS sensor data	OCSM	HS-CAN	• RCM
OCS serial number	OCSM	HS-CAN	• RCM
Odometer count	PCM	HS-CAN	• IPC
Odometer master value	IPC	HS-CAN	• BCM • PCM

Broadcast Message	Originating Module	Network Type	Receiving Module(s)
Oil pressure warning	PCM	HS-CAN	• IPC
Outside air temperature	HVAC module	MS-CAN	• BCM
Outside air temperature (gateway)	BCM	HS-CAN	• IPC • PAM
Outside air temperature (gateway)	IPC	I-CAN	• APIM • Audio DSP module • FCIM • FCDIM
Park brake chime request	BCM	HS-CAN	• IPC
Parking aid audible warning indicator	PAM	HS-CAN	• IPC
Parking aid enabled status	PAM	HS-CAN	• IPC
Parking aid enabled status (gateway)	IPC	I-CAN	• APIM • FCDIM
Parking aid fault status	PAM	HS-CAN	• BCM
Parking aid range to object	PAM	HS-CAN	• BCM • IPC
Parking aid range to object (gateway)	IPC	I-CAN	• APIM • FCDIM
Parking aid sensors status	PAM	HS-CAN	• BCM • IPC
Parking aid sensors status (gateway)	IPC	I-CAN	• APIM • FCDIM
Parking aid volume cutback request	PAM	HS-CAN	• BCM • IPC
Parking aid volume cutback request (gateway)	BCM	MS-CAN	• ACM
Parking aid warning command	IPC	HS-CAN	• PAM
Parking brake status	BCM	HS-CAN	• IPC • PCM • ABS module • TCCM
Parking brake status	BCM	MS-CAN	• ACM • HVAC module

Broadcast Message	Originating Module	Network Type	Receiving Module(s)
Parklamp status	BCM	HS-CAN	• IPC • TBC module
Parklamp status	BCM	MS-CAN	• ACM • HVAC module • TBC module
Passenger restraint indictor status	FCIM	I-CAN	• IPC
Passenger seat belt buckle status	RCM	HS-CAN	• OCS module
Passive entry lock request	BCM	HS-CAN	• IPC
Passive entry lock status	BCM	HS-CAN	• IPC
PATS security data	BCM	HS-CAN	• PCM
PATS security status	PCM	HS-CAN	• BCM
Perimeter alarm chime request	BCM	HS-CAN	• IPC
Power running board enable command	IPC	HS-CAN	• BCM
Power running board enable command (gateway)	BCM	MS-CAN	• PRB module
Power running board status	PRB module	MS-CAN	• BCM
Power running board status (gateway)	BCM	HS-CAN	• IPC
PSCM fault status	PSCM	HS-CAN	• IPC
RCM serial number	RCM	HS-CAN	• ABS module
Remote start quiet mode	HVAC module	MS-CAN	• BCM • TPM module (Radio Transceiver Module [RTM])
Remote start quiet mode	BCM	HS-CAN	• IPC
Remote start status	BCM	MS-CAN	• APIM • HVAC module • TPM module (Radio Transceiver Module [RTM])
Remote start status	BCM	HS-CAN	• IPC
Remote start warning	BCM	HS-CAN	• IPC

Broadcast Message	Originating Module	Network Type	Receiving Module(s)
Reverse mirror command	IPC	HS-CAN	• BCM
Reverse mirror command (gateway)	BCM	MS-CAN	• DSM
Reverse mirror status	DSM	MS-CAN	• BCM
Reverse mirror status (gateway)	BCM	HS-CAN	• IPC
RKE configuration	TPM module (Radio Transceiver Module [RTM])	MS-CAN	• BCM
RKE data	BCM	MS-CAN	• TPM module (Radio Transceiver Module [RTM])
Seat belt buckle status	RCM	HS-CAN	• APIM
Seat belt indicator request	RCM	HS-CAN	• IPC
Seat belt indicator status	IPC	HS-CAN	• RCM
Seat belt warning chime request	RCM	HS-CAN	• IPC
Seat belt warning chime status	IPC	HS-CAN	• RCM
Selector lever (PRNDL) status	PCM	HS-CAN	• TCCM • ABS module • BCM • IPC • PAM • RCM • SCCM
Selector lever (PRNDL) status (gateway)	BCM	MS-CAN	• APIM • ACM • DSM • GPSM
Speed control deactivation request	ABS module	HS-CAN	• PCM
Stability control event in progress	ABS module	HS-CAN	• TCCM • PCM • RCM • TBC module
Starter crank request	BCM	HS-CAN	• PCM
Steering angle calibration (hydraulic power steering)	SCCM	HS-CAN	• BCM • ABS module

Broadcast Message	Originating Module	Network Type	Receiving Module(s)
Steering angle calibration (Electronic Power Assist Steering (EPAS))	PSCM	HS-CAN	• BCM • ABS module
Steering angle count (hydraulic power steering)	SCCM	HS-CAN	• BCM • ABS module
Steering angle count (EPAS)	PSCM	HS-CAN	• BCM • ABS module • SCCM
Steering angle initialization (hydraulic power steering)	SCCM	HS-CAN	• BCM
Steering angle initialization (EPAS)	PSCM	HS-CAN	• BCM • SCCM
Steering column manual override	SCCM	HS-CAN	• BCM
Steering column manual override (gateway)	BCM	MS-CAN	• DSM
Steering column status	SCCM	HS-CAN	• BCM
Steering column status (gateway)	BCM	MS-CAN	• DSM
Steering wheel angle offset	ABS module	HS-CAN	• TCCM • IPC • PSCM • PCM • RCM
Steering wheel switch control mode	IPC	I-CAN	• APIM • FCDIM
Steering wheel switch_flash to pass	SCCM	HS-CAN	• BCM
Steering wheel switch_media	SCCM	HS-CAN	• BCM • IPC
Steering wheel switch_media (gateway)	BCM	MS-CAN	• ACM
Steering wheel switch_media (gateway)	IPC	I-CAN	• APIM • FCDIM
Steering wheel switch_OK	SCCM	HS-CAN	• BCM • IPC
Steering wheel switch_OK (gateway)	BCM	MS-CAN	• APIM
Steering wheel switch_OK (gateway)	IPC	I-CAN	• APIM • FCDIM

Broadcast Message	Originating Module	Network Type	Receiving Module(s)
Steering wheel switch_phone	SCCM	HS-CAN	• BCM • IPC
Steering wheel switch_phone (gateway)	BCM	MS-CAN	• APIM • ACM
Steering wheel switch_seek left	SCCM	HS-CAN	• BCM • IPC
Steering wheel switch_seek left (gateway)	BCM	MS-CAN	• ACM • APIM
Steering wheel switch_seek left (gateway)	IPC	I-CAN	• APIM • FCDIM
Steering wheel switch_seek right	SCCM	HS-CAN	• BCM • IPC
Steering wheel switch_seek right (gateway)	BCM	MS-CAN	• ACM • APIM
Steering wheel switch_seek right (gateway)	IPC	I-CAN	• APIM • FCDIM
Steering wheel switch_speed control	SCCM	HS-CAN	• PCM
Steering wheel switch_turn signal	SCCM	HS-CAN	• BCM • IPC
Steering wheel switch_voice	SCCM	HS-CAN	• BCM • IPC
Steering wheel switch_voice (gateway)	BCM	MS-CAN	• APIM • ACM
Steering wheel switch_voice (gateway)	IPC	I-CAN	• APIM • FCDIM
Steering wheel switch_volume down	SCCM	HS-CAN	• BCM • IPC
Steering wheel switch_volume down (gateway)	BCM	MS-CAN	• ACM
Steering wheel switch_volume down (gateway)	IPC	I-CAN	• APIM • FCDIM
Steering wheel switch_volume up	SCCM	HS-CAN	• BCM • IPC

Broadcast Message	Originating Module	Network Type	Receiving Module(s)
Steering wheel switch_volume up (gateway)	BCM	MS-CAN	• ACM
Steering wheel switch_volume up (gateway)	IPC	I-CAN	• APIM • FCDIM
Steering wheel switch_wiper switch	SCCM	HS-CAN	• BCM • PCM
Steering wheel switch_wiper wash	SCCM	HS-CAN	• PCM
Steering wheel switch_wiper wash	SCCM	HS-CAN	• PCM
Stoplamp request	ABS module	HS-CAN	• BCM
Sunload sensor data	HVAC module	MS-CAN	• BCM
SYNC® alerts	APIM	I-CAN	• ACM • Audio DSP module
Tailgate ajar status	BCM	HS-CAN	• ABS module • IPC • OCSM • SCCM
Tire pressure data	BCM	HS-CAN	• APIM • IPC
Tire pressure status	BCM	HS-CAN	• APIM • IPC
Torque transfer percent command	TCCM	HS-CAN	• ABS module • IPC
TPMS data	TPM module (Radio Transceiver Module [RTM])	MS-CAN	• BCM
TPMS pressure information	BCM	MS-CAN	• TPM module (Radio Transceiver Module [RTM])
Traction control event in progress	ABS module	HS-CAN	• TCCM • PCM • RCM
Traction control mode	ABS module	HS-CAN	• RCM
Trailer brake configuration	IPC	HS-CAN	• TBC module
Trailer brake fault status	TBC module	HS-CAN	• IPC

Broadcast Message	Originating Module	Network Type	Receiving Module(s)
Trailer brake gain display request	TBC module	HS-CAN	• IPC
Trailer brake gain setting	TBC module	HS-CAN	• IPC
Trailer brake lamp request	TBC module	HS-CAN	• IPC
Trailer brake output request	ABS module	HS-CAN	• TBC module
Trailer brake stoplamp request	TBC module	HS-CAN	• BCM
Trailer brake type (electric/electric over hydraulic)	IPC	HS-CAN	• TBC module
Trailer connect/disconnect status	TBC module	HS-CAN	• ABS module • BCM • IPC • PAM • SCCM
Trailer connect/disconnect status (gateway)	IPC	I-CAN	• APIM • FCDIM
Trailer sway event in progress	ABS module	HS-CAN	• IPC
Transfer case maximum torque required	ABS module	HS-CAN	• TCCM
Transfer case minimum torque required	ABS module	HS-CAN	• TCCM
Transmission gear display	PCM	HS-CAN	• IPC
Transmission in reverse	PCM	HS-CAN	• BCM • RCM • SCCM
Transmission malfunction indicator request	PCM	HS-CAN	• IPC • RCM
Transmission mode	PCM	HS-CAN	• IPC
Transmission oil temperature	PCM	HS-CAN	• TCCM • IPC
Transmission shift in progress	PCM	HS-CAN	• ABS module
Transmission shift mode	PCM	HS-CAN	• IPC • ABS module
Turn indicator command	BCM	HS-CAN	• IPC • SCCM

Broadcast Message	Originating Module	Network Type	Receiving Module(s)
Vehicle lateral acceleration	RCM	HS-CAN	• PSCM • PCM • OCSM
Vehicle lock status	BCM	HS-CAN	• IPC
Vehicle longitudinal acceleration	RCM	HS-CAN	• PCM • OCSM
Vehicle roll sense	RCM	HS-CAN	• PCM
Vehicle speed	PCM	HS-CAN	• ABS module • BCM • IPC • OCSM • PAM • PSCM • RCM • SCCM
Vehicle speed (gateway)	BCM	MS-CAN	• APIM • ACM • GPSM • PRB module
Vehicle yaw rate (gateway)	BCM	MS-CAN	• GPSM
Vehicle yaw rate	RCM	HS-CAN	• BCM • PCM
Voice commands	APIM	I-CAN	• IPC
Voice recognition status	APIM	I-CAN	• FCDIM
Wheel speed data	ABS module	HS-CAN	• TCCM • BCM • IPC • PSCM • PCM
Wiper motor status	BCM	MS-CAN	• HVAC module

Inspection and Verification

1. Verify the customer concern.
2. Visually inspect for obvious signs of electrical damage.
   - If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step.

Visual Inspection Chart

Electrical
• Battery Junction Box (BJB) fuse(s): ▪ High current 1 (125A) (PSCM) ▪ 11 (30A) (PRB module) ▪ 17 (30A) (TBC module) ▪ 20 (20A) (TCCM ESOF) ▪ 23 (15A) (SCCM) ▪ 24 (15A) (SCCM) ▪ 26 (10A) (PCM) ▪ 36 (30A) (ABS module) ▪ 45 (10A) BCM ▪ 47 (60A) (ABS module) ▪ 52 (5A) (PSCM) ▪ 54 (5A) (TCCM , ABS module) ▪ 68 (25A) (TCCM) ▪ 69 (30A) (DCSM) ▪ 75 (25A) (3.5L PCM) ▪ 75 (15A) (3.7L, 5.0L and 6.2L PCM) • Body Control Module (BCM) fuse(s): ▪ 2 (15A) (APIM) ▪ 7 (7.5A) (DSM) ▪ 9 (10A) (FCIM , FDIM , GPSM) ▪ 11 (10A) (IPC) ▪ 19 (20A) (audio DSP module) ▪ 24 (15A) (DLC) ▪ 26 (5A) (TPM module) (Radio Transceiver Module [RTM]) ▪ 29 (20A) (ACM) ▪ 34 (10A) (PAM) ▪ 36 (10A) (RCM and OCSM) ▪ 37 (10A) (TBC module) ▪ 46 (10A) (HVAC module) • Data Link Connector (DLC) • Wiring harness • Wiring, terminals or connectors

3. Connect the scan tool to the DLC .

- **NOTE:** *Make sure to use the latest scan tool software release.*

  NOTE: *The Vehicle Communication Module (VCM) LED prove-out confirms power and ground from the DLC are provided to the VCM .*

  If the Integrated Diagnostic System (IDS) does not communicate with the VCM :
  - check the VCM connection to the vehicle.
  - check the scan tool connection to the VCM .
  - GO to Pinpoint Test AA to diagnose No Power To The Scan Tool.

4. Establish a scan tool session.

- **NOTE:** *The scan tool first attempts to communicate with the PCM. After establishing communication with the PCM, the scan tool then attempts to communicate with all other modules on the vehicle.*

  If an IDS session cannot be established with the vehicle, (IDS may state "No communication can be established with the PCM"):
  - Choose "NO" when the scan tool prompts whether or not to retry communication.
  - Enter either a PCM part number, tear tag or calibration number to identify the vehicle and start a session (the PCM part number and 4-character tear tag are printed on the PCM label).
  - GO to Pinpoint Test A to diagnose The PCM Does Not Respond To The Scan Tool.

5. Carry out the network test.
   - If the network test passes, retrieve and record the Continuous Memory Diagnostic Trouble Codes (CMDTCs) and proceed to Step 6.
   - If the network test fails, GO to Symptom Chart to diagnose the failed communication network.
   - If a module fails to communicate during the network test, GO to Symptom Chart.

6. Retrieve and review the DTCs.
   - If the DTCs retrieved are related to the concern, go to DTC Charts. Follow the non-network DTC diagnostics (B-codes, C-codes, P-codes) prior to the network DTC diagnostics (U-codes). For all other DTCs, refer to the Diagnostic Trouble Code (DTC) Chart in Section 419-10.
   - If no DTCs related to the concern are retrieved, GO to Symptom Chart.

DTC Charts

NOTE: *Network DTCs (U-codes) are often a result of intermittent concerns such as damaged wiring or low battery voltage occurrences. Additionally, vehicle repair procedures such as module reprogramming often set network DTCs. Replacing a module to resolve a network DTC is unlikely to resolve the concern. To prevent repeat network DTC concerns, inspect all network wiring, especially connectors. Test the vehicle battery, refer to Section 414-01.*

NOTE: *Some modules on this vehicle utilize a 5-character DTC followed by a 2-character failure-type code. The failure-type code provides information about specific fault conditions such as opens, or shorts to ground. Continuous memory DTCs have an additional 2-character DTC status code suffix to assist in determining DTC history.*

Communication Network DTC Chart

DTC	Description	Source	Action
U0001:88	High Speed CAN Communication Bus: Bus off	Power Steering Control Module (PSCM)	The module could not communicate on the network at a point in time. The fault is currently not present since the module is currently communicating with the scan tool reporting this DTC. CLEAR the DTC. REPEAT the network test with the scan tool. VERIFY the integrity of the connectors and wiring. Refer to Wiring Diagrams Cell 14 , Module Communications Network for schematic and connector information.
U0028:08	Vehicle Communication Bus A: Bus Signal / Message Failures	Restraints Control Module (RCM)	The module could not communicate on the network at a point in time. The fault is currently not present since the module is currently communicating with the scan tool reporting this DTC. CLEAR the DTC. REPEAT the network test with the scan tool. VERIFY the integrity of the connectors and wiring. Refer to Wiring Diagrams Cell 14 , Module Communications Network for schematic and connector information.
U0028:88	Vehicle Communication Bus A: Bus Off	Restraints Control Module (RCM)	The module could not communicate on the network at a point in time. The fault is currently not present since the module is currently communicating with the scan tool reporting this DTC. CLEAR the DTC. REPEAT the network test with the scan tool.

DTC	Description	Source	Action
			VERIFY the integrity of the connectors and wiring. Refer to Wiring Diagrams Cell 14 , Module Communications Network for schematic and connector information.
U0073	Vehicle Communication Bus A "Off"	ABS module	The module could not communicate on the network at a point in time. The fault is currently not present since the module is currently communicating with the scan tool reporting this DTC. CLEAR the DTC. REPEAT the network test with the scan tool. VERIFY the integrity of the connectors and wiring. Refer to Wiring Diagrams Cell 14 , Module Communications Network for schematic and connector information.
U1900	CAN Communication Bus Fault - Receive Error	Power Running Board (PRB) module	CARRY out the network test. GO to Symptom Chart for the module that fails the network test.
U3000:88	Control Module: Bus Off	Accessory Protocol Interface Module (APIM)	The module could not communicate on the network at a point in time. The fault is currently not present since the module is currently communicating with the scan tool reporting this DTC. CLEAR the DTC. REPEAT the network test with the scan tool. VERIFY the integrity of the connectors and wiring. Refer to Wiring Diagrams Cell 14 , Module Communications Network for schematic and connector information.

Symptom Chart

NOTE: *The TPM module is also known as the Radio Transceiver Module (RTM).*

Symptom Chart

Condition	Possible Causes	Action
• The vehicle does not start with the scan tool connected to the Data Link Connector (DLC) and/or multiple malfunction indicators are only on when the scan tool is connected to the DLC	• Wiring, terminals or connectors • Scan tool	• GO to Pinpoint Test AA.
• The PCM does not respond to the scan tool	• Wiring, terminals or connectors • PCM	• REFER to the Powertrain Control/Emissions Diagnosis (PC/ED) manual, Section 5, pinpoint test QA before proceeding to Pinpoint Test A. If pinpoint test QA has been completed, GO to Pinpoint Test A.
• The ABS module does not respond to the scan tool	• Fuse • Wiring, terminals or connectors • ABS module	• GO to Pinpoint Test B.

• The Instrument Panel Cluster (IPC) does not respond to the scan tool	• Fuse • Wiring, terminals or connectors • IPC	• GO to Pinpoint Test C.
• The Restraints Control Module (RCM) does not respond to the scan tool	• Fuse • Wiring, terminals or connectors • RCM	• GO to Pinpoint Test D.
• The OCSM does not respond to the scan tool	• Fuse • Wiring, terminals or connectors • OCSM	• GO to Pinpoint Test E.
• The TCCM does not respond to the scan tool	• Fuse(s) • Wiring, terminals or connectors • TCCM	• GO to Pinpoint Test F.
• The Trailer Brake Control (TBC) module does not respond to the scan tool	• Fuse • Wiring, terminals or connectors • TBC module	• GO to Pinpoint Test G.
• The Power Steering Control Module (PSCM) does not respond to the scan tool	• Fuse • Wiring, terminals or connectors • PSCM	• GO to Pinpoint Test H.
• The Body Control Module (BCM) does not respond to the scan tool	• Fuse • Wiring, terminals or connectors • BCM	• GO to Pinpoint Test I.
• The HVAC module does not respond to the scan tool	• Fuse • Wiring, terminals or connectors • HVAC module	• GO to Pinpoint Test J.

• The Audio Front Control Module (ACM) does not respond to the scan tool	• Fuse • Wiring, terminals or connectors • ACM	• GO to Pinpoint Test K.
• The audio Digital Signal Processing (DSP) module does not respond to the scan tool	• Fuse • Wiring, terminals or connectors • Audio DSP module	• GO to Pinpoint Test L.
• The Front Controls Interface Module (FCIM) does not respond to the scan tool	• Fuse • Wiring, terminals or connectors • FCIM	• GO to Pinpoint Test M.
• The Front Display Interface Module (FDIM) does not respond to the scan tool	• Fuse • Wiring, terminals or connectors • Front Display Interface Module (FDIM)	• GO to Pinpoint Test N.
• The FCDIM does not respond to the scan tool	• Fuse • Wiring, terminals or connectors • FCDIM	• GO to Pinpoint Test O
• The Driver Seat Module (DSM) does not respond to the scan tool	• Fuse • Wiring, terminals or connectors • DSM	• GO to Pinpoint Test P.
• The Dual Climate Controlled Seat Module (DCSM) does not respond to the scan tool	• Fuse • Wiring, terminals or connectors • DCSM	• GO to Pinpoint Test Q.
• The Power Running Board (PRB) module does not respond to the scan tool	• Fuse • Wiring, terminals or connectors • PRB module	• GO to Pinpoint Test R.

• The Parking Aid Module (PAM) does not respond to the scan tool	• Fuse • Wiring, terminals or connectors • PAM	• GO to Pinpoint Test S.
• The Global Positioning System Module (GPSM) does not respond to the scan tool	• Fuse • Wiring, terminals or connectors • GPSM	• GO to Pinpoint Test T.
• The Accessory Protocol Interface Module (APIM) does not respond to the scan tool	• Fuse • Wiring, terminals or connectors • APIM	• GO to Pinpoint Test U.
• The Steering Column Control Module (SCCM) does not respond to the scan tool	• Fuse • Wiring, terminals or connectors • SCCM	• GO to Pinpoint Test V.
• The Tire Pressure Monitor (TPM) module (Radio Transceiver Module [RTM]) does not respond to the scan tool	• Fuse • Wiring, terminals or connectors • TPM module (Radio Transceiver Module [RTM])	• GO to Pinpoint Test W.
• No Medium Speed Controller Area Network (MS-CAN) communication, all modules are not responding	• Wiring, terminals or connectors • APIM (if equipped) • ACM (if equipped) • BCM • DSM (if equipped) • DCSM (if equipped) • FCIM (if equipped) • FDIM (if equipped)	• GO to Pinpoint Test X.

	• GPSM (if equipped) • HVAC module • PRB module (if equipped) • TPM module (Radio Transceiver Module [RTM])	
• No High Speed Controller Area Network (HS-CAN) communication, all modules are not responding	• Wiring, terminals or connectors • TCCM (if equipped) • ABS module • APIM (if equipped) • BCM • IPC • OCSM • PAM (if equipped) • PCM • PSCM (if equipped) • RCM • SCCM • TBC module (if equipped)	• GO to Pinpoint Test Y.
• No Infotainment Controller Area Network (I-CAN) communication, all modules are not responding	• ACM (if equipped) • APIM (if equipped) • Audio DSP module (if equipped) • FCIM (if equipped) • FCDIM (if equipped) • IPC	• GO to Pinpoint Test Z

• No power to the scan tool	• Fuse • Wiring, terminals or connectors • DLC • Scan tool	• GO to Pinpoint Test AA.

Pinpoint Tests

Pinpoint Test A: The PCM Does Not Respond To The Scan Tool

Refer to Wiring Diagrams Cell 14 , Module Communications Network for schematic and connector information.

Refer to Wiring Diagrams Cell 23 , Electronic Engine Controls - 3.7L for schematic and connector information.

Refer to Wiring Diagrams Cell 24 , Electronic Engine Controls - 5.0L for schematic and connector information.

Refer to Wiring Diagrams Cell 25 , Electronic Engine Controls - 3.5L for schematic and connector information.

Refer to Wiring Diagrams Cell 26 , Electronic Engine Controls - 6.2L for schematic and connector information.

Normal Operation

The PCM communicates with the scan tool through the High Speed Controller Area Network (HS-CAN) .

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- PCM

PINPOINT TEST A : THE PCM DOES NOT RESPOND TO THE SCAN TOOL

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

NOTE: *Failure to disconnect the battery when instructed results in false resistance readings. Refer to Section 414-01.*

A1 VERIFY IF OTHER HS-CAN MODULES PASS THE NETWORK TEST

- Enter the following diagnostic mode on the scan tool: Network Test .
- In the LH pane of the IDS network test display screen, verify whether any HS-CAN modules passed the network test.

Is the text "pass" or a DTC listed next to any of the following modules (if equipped): TCCM , ABS module, Accessory Protocol Interface Module (APIM) , Body Control Module (BCM) , Instrument Panel Cluster (IPC) , Occupant Classification System (OCS) module, Parking Aid Module (PAM) , Power Steering Control Module (PSCM) , Restraints Control Module (RCM) , Steering Column Control Module (SCCM) or Trailer Brake Control (TBC) module or PCM?

Yes	If "pass" or a DTC was listed next to the PCM, a network fault is not currently present. If "pass" or a DTC was listed next to one or more modules other than the PCM, GO to A2.

No	No modules are currently communicating on the HS-CAN . GO to Pinpoint Test Y to diagnose no HS-CAN communication.

A2 VERIFY THE PC/ED MANUAL PINPOINT TEST QA HAS BEEN CARRIED OUT

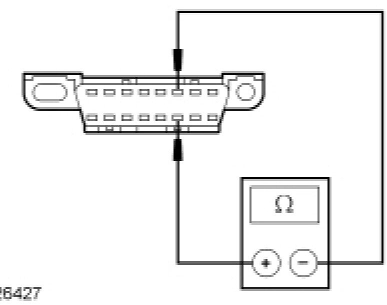
- Verify that pinpoint test QA has been carried out.

Has pinpoint test QA been carried out?

Yes	GO to A3.
No	REFER to the Powertrain Control/Emissions Diagnosis (PC/ED) manual, Section 5, pinpoint test QA to diagnose no communication with the PCM.

A3 CHECK THE HS-CAN TERMINATION RESISTANCE

- Ignition OFF.
- Disconnect: Negative Battery Cable .
- Measure the resistance between the Data Link Connector (DLC) C251 Pin 6, circuit VDB04 (WH/BU), harness side and the DLC C251 Pin 14, circuit VDB05 (WH), harness side.



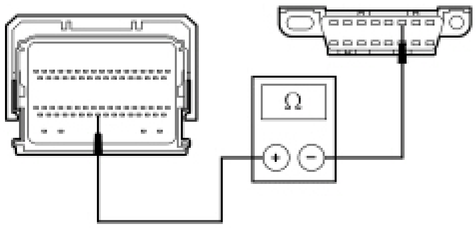
- N0026427

Is the resistance between 54 and 66 ohms?

Yes	CONNECT the negative battery cable. GO to A6.
No	For the 3.7L, 5.0L or 6.2L engine GO to A4. For the 3.5L engine GO to A5.

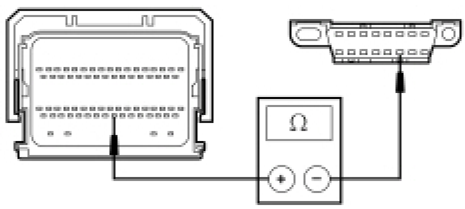
A4 CHECK THE HS-CAN CIRCUITS BETWEEN THE PCM AND THE DLC FOR AN OPEN

- Disconnect: PCM C175B or C1381B.
- Measure the resistance between the PCM C175B Pin 59 or C1381B Pin 59, circuit VDB04 (WH/BU), harness side and the DLC C251 Pin 6, circuit VDB04 (WH/BU), harness side.

- 

  N0077607

- Measure the resistance between the PCM C175B Pin 58 or C1381B Pin 58, circuit VDB05 (WH), harness side and the DLC C251 Pin 14, circuit VDB05 (WH), harness side.

- 

  N0113941

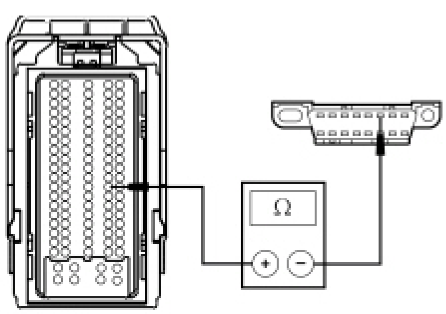
Are the resistances less than 5 ohms?

Yes	CONNECT the negative battery cable. GO to A6.
No	REPAIR the circuit in question. CONNECT the negative battery cable. CLEAR the DTCs. REPEAT the network test with the scan tool.

A5 CHECK THE HS-CAN CIRCUITS BETWEEN THE PCM AND THE DLC FOR AN OPEN

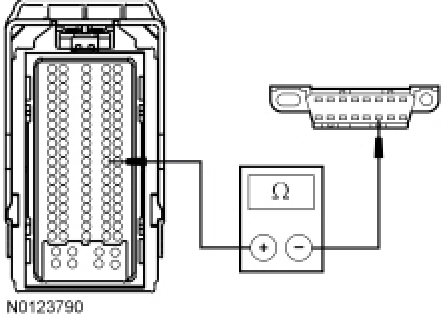
- Disconnect: PCM C1551B.

- Measure the resistance between the PCM C1551B Pin 69, circuit VDB04 (WH/BU), harness side and the DLC C251 Pin 6, circuit VDB04 (WH/BU), harness side.

- 

  N0123791

- Measure the resistance between the PCM C1551B Pin 68, circuit VDB05 (WH), harness side and the DLC C251 Pin 14, circuit VDB05 (WH), harness side.



Are the resistances less than 5 ohms?

Yes	CONNECT the negative battery cable. GO to A6.
No	REPAIR the circuit in question. CONNECT the negative battery cable. CLEAR the DTCs. REPEAT the network test with the scan tool.

A6 CHECK FOR CORRECT PCM OPERATION

- Disconnect all the PCM connectors.
- Check for:
  - corrosion
  - damaged pins
  - pushed-out pins
- Connect all the PCM connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new PCM. REFER to Section 303-14. CLEAR the DTCs. REPEAT the network test with the scan tool.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test B: The ABS Module Does Not Respond To The Scan Tool

Refer to Wiring Diagrams Cell 14 , Module Communications Network for schematic and connector information.

Refer to Wiring Diagrams Cell 42 , Vehicle Dynamic Systems for schematic and connector information.

Normal Operation

The ABS module communicates with the scan tool through the High Speed Controller Area Network (HS-CAN) .

This pinpoint test is intended to diagnose the following:

- Fuse

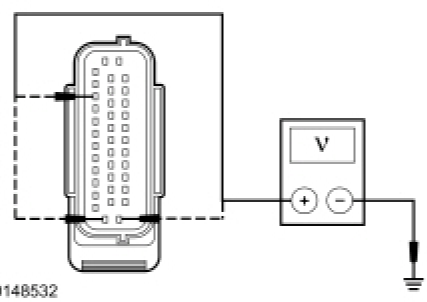
- Wiring, terminals or connectors
- ABS module

PINPOINT TEST B : THE ABS MODULE DOES NOT RESPOND TO THE SCAN TOOL

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

NOTE: *Failure to disconnect the battery when instructed results in false resistance readings. Refer to Section 414-01.*

B1 CHECK THE ABS MODULE VOLTAGE SUPPLY CIRCUIT FOR AN OPEN

- Ignition OFF.
- Disconnect: ABS Module C135.
- Ignition ON.
- Measure the voltage between the ABS module C135 Pin 1, circuit SBB47 (WH/RD), harness side and ground; between the ABS module C135 Pin 25, circuit SBB36 (GN/RD), harness side and ground; and between the ABS module C135 Pin 35, circuit CBB54 (VT/OG), harness side and ground.
- 

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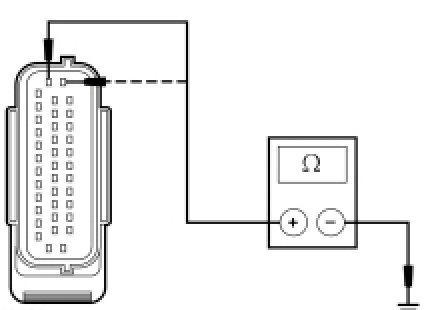
Is the voltage greater than 10 volts?

Yes	GO to B2.
No	VERIFY the Battery Junction Box (BJB) fuse 36 (30A), 47 (60A) or 54 (5A) is OK. If OK, REPAIR the circuit in question. If not OK, REFER to the Wiring Diagrams manual to identify the possible causes of the circuit short. CLEAR the DTCs. REPEAT the network test with the scan tool.

B2 CHECK THE ABS MODULE GROUND CIRCUIT FOR AN OPEN

- Ignition OFF.
- Disconnect: Negative Battery Cable .
- Measure the resistance between the ABS module C135 Pin 38, circuit GD121 (BK/YE), harness side and ground; and between the ABS module C135 Pin 13, circuit GD121 (BK/YE), harness side and ground.
- 

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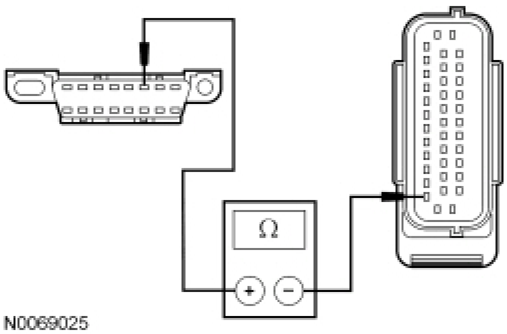
Is the resistance less than 5 ohms?

Yes	GO to B3.
No	REPAIR the circuit. CONNECT the negative battery cable. CLEAR the DTCs. REPEAT the network test with the scan tool.

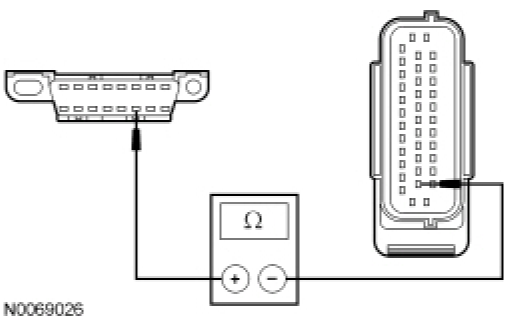
B3 CHECK THE HS-CAN CIRCUITS BETWEEN THE DLC AND THE ABS MODULE FOR AN OPEN

- Measure the resistance between the Data Link Connector (DLC) C251 Pin 6, circuit VDB04 (WH/BU), harness side and the ABS module C135 Pin 26, circuit VDB04 (WH/BU), harness side.



- N0069025

- Measure the resistance between the DLC C251 Pin 14, circuit VDB05 (WH), harness side and the ABS module C135 Pin 14, circuit VDB05 (WH), harness side.



- N0069026

Are the resistances less than 5 ohms?

Yes	CONNECT the negative battery cable. GO to B4.
No	REPAIR the circuit in question. CONNECT the negative battery cable. CLEAR the DTCs. REPEAT the network test with the scan tool.

B4 CHECK FOR CORRECT ABS MODULE OPERATION

- Disconnect the ABS module connector.
- Check for:
  - corrosion
  - damaged pins
  - pushed-out pins
- Connect the ABS module connector and make sure it seats correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new ABS module. REFER to Section 206-09. CLEAR the DTCs. REPEAT the network test with the scan tool.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test C: The Instrument Panel Cluster (IPC) Does Not Respond To The Scan Tool

Refer to Wiring Diagrams Cell 14 , Module Communications Network for schematic and connector information.

Refer to Wiring Diagrams Cell 60 , Instrument Cluster for schematic and connector information.

Normal Operation

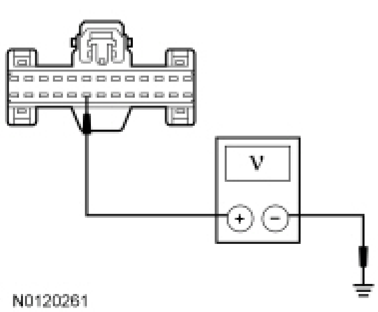
The Instrument Panel Cluster (IPC) communicates with the scan tool through the High Speed Controller Area Network (HS-CAN) . For vehicles equipped with a 4.2-inch screen or 8-inch touchscreen audio system, the IPC is a gateway for messages sent between the scan tool and the modules on the Infotainment Controller Area Network (I-CAN) .

This pinpoint test is intended to diagnose the following:

- Fuse
- Wiring, terminals or connectors
- IPC

PINPOINT TEST C : THE IPC DOES NOT RESPOND TO THE SCAN TOOL

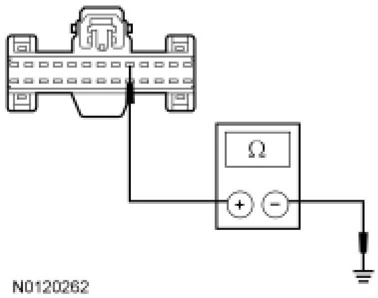
NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.
NOTE: *Failure to disconnect the battery when instructed results in false resistance readings. Refer to Section 414-01.*
C1 CHECK THE IPC VOLTAGE SUPPLY CIRCUITS FOR AN OPEN

- Disconnect: IPC C220 .
- Ignition ON.
- Measure the voltage between the IPC C220 Pin 21, circuit SBP11 (BU/RD), harness side and ground.
- 

Is the voltage greater than 10 volts?

Yes	GO to C2
No	VERIFY the Body Control Module (BCM) fuse 11 (10A) is OK. If OK, REPAIR the circuit. If not OK, REFER to the Wiring Diagrams manual to identify the possible causes of the circuit short. CLEAR the DTCs. REPEAT the network test with the scan tool.

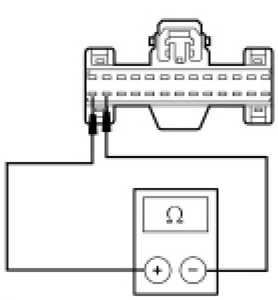
C2 CHECK THE IPC GROUND CIRCUIT FOR AN OPEN

- Ignition OFF.
- Disconnect: Negative Battery Cable .
- Measure the resistance between the IPC C220 Pin 5, circuit GD133 (BK), harness side and ground.
- 
  N0120262

Is the resistance less than 5 ohms?

Yes	CONNECT the negative battery cable. If equipped with 4.2-inch screen or 8-inch touchscreen, GO to C3. If not equipped with 4.2-inch screen or 8-inch touchscreen, GO to C5.
No	REPAIR the circuit. CONNECT the negative battery cable. CLEAR the DTCs. REPEAT the network test with the scan tool.

C3 CHECK THE I-CAN TERMINATION RESISTANCE

- Ignition OFF.
- Disconnect: Negative Battery Cable .
- Measure the resistance between the IPC C220 Pin 25, circuit VDB13 (BU/GY) and IPC C220 Pin 26, circuit VDB14 (VT/GY), harness side.
- 
  N0150481

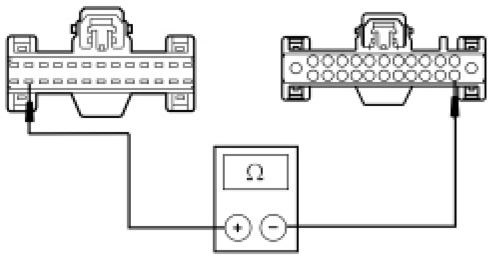
Is the resistance between 108 and 132 ohms?

Yes	CONNECT the battery cable. GO to C5.

No	GO to C4.

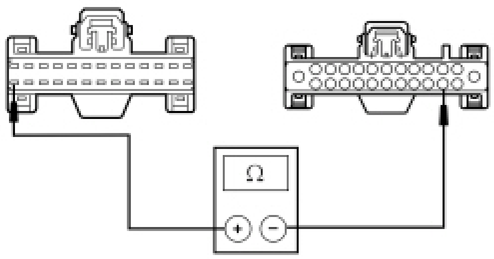
C4 CHECK THE I-CAN CIRCUITS BETWEEN IPC AND THE ACM FOR AN OPEN

- Disconnect: ACM C240A .
- Measure the resistance between the IPC C220 Pin 25, circuit VDB13 (BU/GY), harness side and the ACM C240A Pin 14, circuit VDB13 (BU/GY), harness side.



- N0150482

- Measure the resistance between the IPC C220 Pin 26, circuit VDB14 (VT/GY), harness side and the ACM C240A Pin 15, circuit VDB14 (VT/GY), harness side.



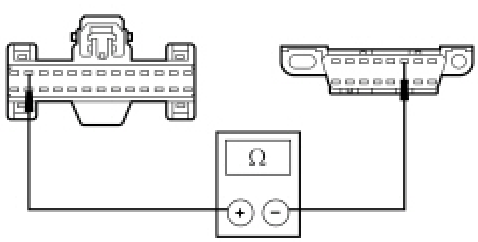
- N0150483

Are the resistances less than 5 ohms?

Yes	GO to C5.
No	REPAIR the circuit in question. CONNECT the negative battery cable. CLEAR the DTCs. REPEAT the network test with the scan tool.

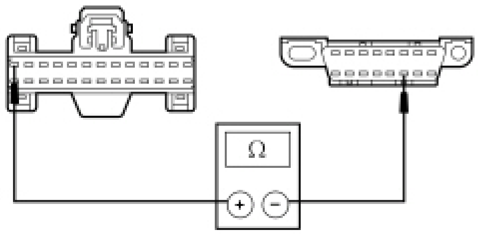
C5 CHECK THE HS-CAN CIRCUITS BETWEEN IPC AND THE DLC FOR AN OPEN

- Measure the resistance between the IPC C220 Pin 12, circuit VDB04 (WH/BU), harness side and the DLC C251 Pin 6, circuit VDB04 (WH/BU), harness side.



- N0120263

- Measure the resistance between the IPC C220 Pin 13, circuit VDB05 (WH), harness side and the DLC C251 Pin 14, circuit VDB05 (WH), harness side.



- N0120264

Are the resistances less than 5 ohms?

Yes	CONNECT the negative battery cable. GO to C6.
No	REPAIR the circuit in question. CONNECT the negative battery cable. CLEAR the DTCs. REPEAT the network test with the scan tool.

C6 CHECK FOR CORRECT IPC OPERATION

- Disconnect the IPC connector.
- Check for:
  - corrosion
  - damaged pins
  - pushed-out pins
- Connect the IPC connector and make sure it seats correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. CLEAR the DTCs. REPEAT the network test with the scan tool.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test D: The Restraints Control Module (RCM) Does Not Respond To The Scan Tool

Refer to Wiring Diagrams Cell 14 , Module Communications Network for schematic and connector information.

Refer to Wiring Diagrams Cell 46 , Supplemental Restraint System for schematic and connector information.

Normal Operation

The Restraints Control Module (RCM) communicates with the scan tool through the High Speed Controller Area Network (HS-CAN) .

This pinpoint test is intended to diagnose the following:

- Fuse
- Wiring, terminals or connectors
- RCM

PINPOINT TEST D : THE RCM DOES NOT RESPOND TO THE SCAN TOOL

WARNING: Never disassemble or tamper with seatbelt deployable components, including pretensioners, load limiters and inflators. Never back probe deployable device electrical connectors. Tampering or back probing may cause an accidental deployment and result in personal injury or death.

WARNING: Never probe the electrical connectors on airbag, Safety Canopy or side air curtain assemblies. Failure to follow this instruction may result in the accidental deployment of these assemblies, which increases the risk of serious personal injury or death.

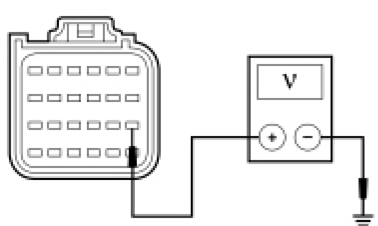
NOTE: *The Supplemental Restraint System (SRS) must be fully operational and free of faults before releasing the vehicle to the customer.*

NOTE: *Most faults are due to connector and/or wiring concerns. Carry out a thorough inspection and verification before proceeding with the pinpoint test.*

NOTE: *Failure to disconnect the battery when instructed results in false resistance readings. Refer to Section 414-01.*

D1 CHECK THE RCM VOLTAGE SUPPLY CIRCUIT FOR AN OPEN

- Depower the SRS . Refer to Section 501-20B.
- Disconnect: RCM C310A .
- Repower the SRS . Refer to Section 501-20B.
- Ignition ON.
- Measure the voltage between the RCM C310A Pin 13, circuit CBP36 (BU/BN), harness side and ground.



- N0117680

Is the voltage greater than 10 volts?

Yes	GO to D2.
No	VERIFY the Body Control Module (BCM) fuse 36 (10A) is OK. If OK, REPAIR the circuit. If not OK, REFER to the Wiring Diagrams manual to identify the possible causes of the circuit short. CLEAR the DTCs. REPEAT the network test with the scan tool.

D2 CHECK THE RCM CASE GROUND

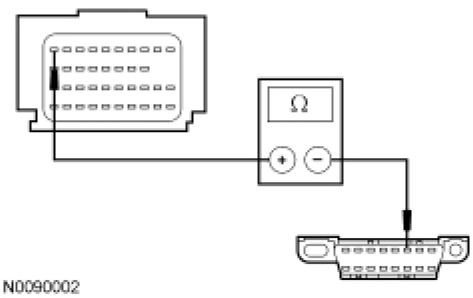
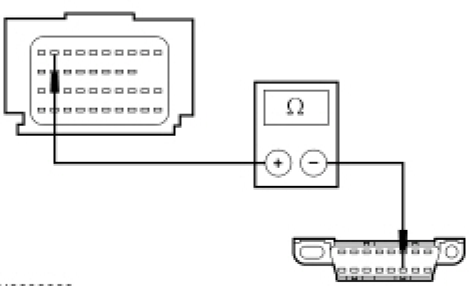
- Ignition OFF.
- Disconnect: Negative Battery Cable .

- Measure the resistance between the RCM case and a good chassis ground.

Is the resistance less than 5 ohms?

Yes	GO to D3.
No	REPAIR the RCM case ground as necessary. CONNECT the negative battery cable. CLEAR the DTCs. REPEAT the network test with the scan tool.

D3 CHECK THE HS-CAN CIRCUITS BETWEEN THE RCM AND THE DLC FOR AN OPEN

- Disconnect: RCM C310B .
- Measure the resistance between the RCM C310B Pin 10, circuit VDB04 (WH/BU), harness side and the Data Link Connector (DLC) C251 Pin 6, circuit VDB04 (WH/BU), harness side.
- 
N0090002
- Measure the resistance between the RCM C310B Pin 9, circuit VDB05 (WH), harness side and the DLC C251 Pin 14, circuit VDB05 (WH), harness side.
- 
N0090003

Are the resistances less than 5 ohms?

Yes	CONNECT the negative battery cable. GO to D4.
No	REPAIR the circuit in question. CONNECT the negative battery cable. CLEAR the DTCs. REPEAT the network test with the scan tool.

D4 CHECK FOR CORRECT RCM OPERATION

- Disconnect all the RCM connectors.
- Check for:
  - corrosion
  - damaged pins
  - pushed-out pins

- Connect all the RCM connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new RCM . REFER to Section 501-20B. CLEAR the DTCs. REPEAT the network test with the scan tool.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test E: The Occupant Classification System Module (OCSM) Does Not Respond To The Scan Tool

Refer to Wiring Diagrams Cell 14 , Module Communications Network for schematic and connector information.

Refer to Wiring Diagrams Cell 46 , Supplemental Restraint System for schematic and connector information.

Normal Operation

The Occupant Classification System Module (OCSM) communicates with the scan tool through the High Speed Controller Area Network (HS-CAN) .

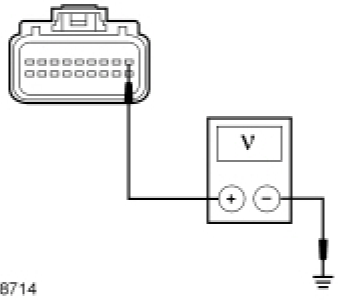
This pinpoint test is intended to diagnose the following:

- Fuse
- Wiring, terminals or connectors
- OCSM

PINPOINT TEST E : THE OCSM DOES NOT RESPOND TO THE SCAN TOOL

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.
NOTE: *The Supplemental Restraint System (SRS) must be fully operational and free of faults before releasing the vehicle to the customer.*
NOTE: *Most faults are due to connector and/or wiring concerns. Carry out a thorough inspection and verification before proceeding with the pinpoint test.*
NOTE: *Failure to disconnect the battery when instructed results in false resistance readings. Refer to Section 414-01.*
E1 CHECK THE OCSM VOLTAGE SUPPLY CIRCUIT FOR AN OPEN

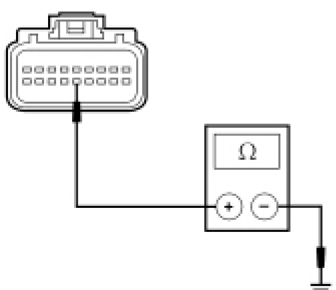
- Ignition OFF.
- Disconnect: OCSM C3159 .
- Ignition ON.
- Measure the voltage between the OCSM C3159 Pin 1, circuit CBX02 (BN/RD), harness side and ground.

- 

Is the voltage greater than 10 volts?

Yes	GO to E2.
No	VERIFY the Body Control Module (BCM) fuse 36 (10A) is OK. If OK, REPAIR the circuit. If not OK, REFER to the Wiring Diagrams manual to identify the possible causes of the circuit short. CONNECT the OCSM . CLEAR the DTCs. REPEAT the network test with the scan tool.

E2 CHECK THE OCSM GROUND CIRCUIT FOR AN OPEN

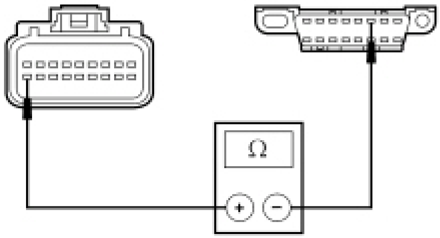
- Ignition OFF.
- Disconnect: Negative Battery Cable .
- Measure the resistance between the OCSM C3159 Pin 14, circuit GD901 (BK), harness side and ground.
- 

Is the resistance less than 5 ohms?

Yes	GO to E3.
No	REPAIR the circuit. CONNECT the negative battery cable. CLEAR the DTCs. REPEAT the network test with the scan tool.

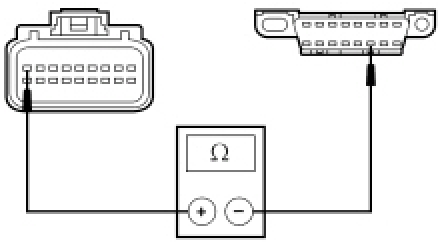
E3 CHECK THE HS-CAN CIRCUITS BETWEEN THE OCSM AND THE DLC FOR AN OPEN

- Measure the resistance between the OCSM C3159 Pin 18, circuit VDB04 (WH/BU), harness side and the Data Link Connector (DLC) C251 Pin 6, circuit VDB04 (WH/BU), harness side.



- N0026670

- Measure the resistance between the OCSM C3159 Pin 9, circuit VDB05 (WH), harness side and the DLC C251 Pin 14, circuit VDB05 (WH), harness side.



- N0026671

Are the resistances less than 5 ohms?

Yes	CONNECT the negative battery cable. GO to E4.
No	REPAIR the circuit. CONNECT the negative battery cable. CLEAR the DTCs. REPEAT the network test with the scan tool.

E4 CHECK FOR CORRECT OCSM OPERATION

- Disconnect the OCSM connector.
- Check for:
  - corrosion
  - damaged pins
  - pushed-out pins
- Connect the OCSM connector and make sure it seats correctly.
- Verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new OCSM . REFER to Section 501-20B. CLEAR the DTCs. REPEAT the network test with the scan tool.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test F: The Transfer Case Control Module (TCCM) Does Not Respond To The Scan Tool

Refer to Wiring Diagrams Cell 14 , Module Communications Network for schematic and connector information.

Refer to Wiring Diagrams Cell 34 , All Wheel Drive (AWD) for schematic and connector information.

Normal Operation

The Transfer Case Control Module (TCCM) communicates with the scan tool through the High Speed Controller Area Network (HS-CAN) .

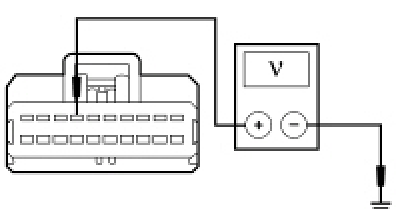
This pinpoint test is intended to diagnose the following:

- Fuse(s)
- Wiring, terminals or connectors
- TCCM

PINPOINT TEST F : THE TCCM DOES NOT RESPOND TO THE SCAN TOOL

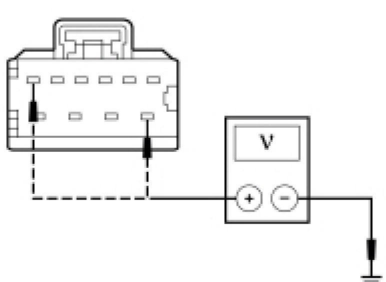
NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.
NOTE: *Failure to disconnect the battery when instructed results in false resistance readings. Refer to Section 414-01.*
F1 CHECK THE TCCM VOLTAGE SUPPLY CIRCUIT FOR AN OPEN

- Ignition OFF.
- Disconnect: TCCM C2371A .
- Disconnect: TCCM C2371B .
- Ignition ON.
- Measure the voltage between the TCCM C2371A Pin 7, circuit CBB54 (VT/OG), harness side and ground.



- N0090212

- Measure the voltage between the TCCM C2371B Pin 6 (Electronic Shift-On-The-Fly (ESOF)), circuit SBB20 (GN/RD), harness side and ground; and between the TCCM C2371B Pin 7, circuit SBB68 (GN/RD), harness side and ground.



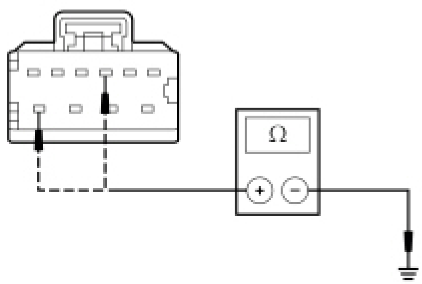
- N0122241

Are the voltages greater than 10 volts?

Yes	GO to F2.
No	VERIFY the Battery Junction Box (BJB) fuses 20 (20A) (ESOF only), 54 (5A) and 68 (25A) are OK. If OK, REPAIR the circuit in question. If not OK, REFER to the Wiring Diagrams manual to identify the possible causes of the circuit short. CLEAR the DTCs. REPEAT the network test with the scan tool.

F2 CHECK THE TCCM GROUND CIRCUIT FOR AN OPEN

- Ignition OFF.
- Disconnect: Negative Battery Cable .
- Measure the resistance between the TCCM C2371B Pin 3, circuit GD123 (BK/GY), harness side and ground; and between the TCCM C2371B Pin 10, circuit GD123 (BK/GY), harness side and ground.



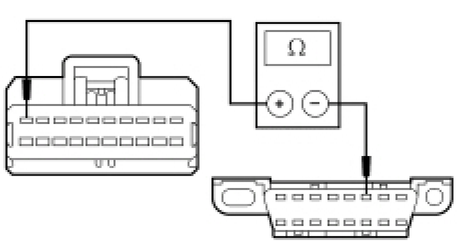
- N0122242

Is the resistance less than 5 ohms?

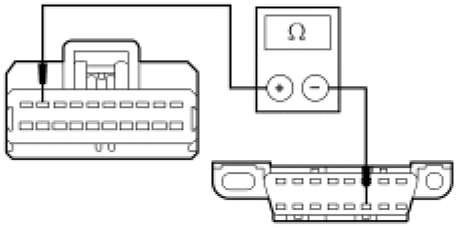
Yes	GO to F3.
No	REPAIR the circuit. CONNECT the negative battery cable. CLEAR the DTCs. REPEAT the network test with the scan tool.

F3 CHECK THE HS-CAN CIRCUITS BETWEEN THE TCCM AND THE DLC FOR AN OPEN

- Measure the resistance between the TCCM C2371A Pin 10, circuit VDB04 (WH/BU), harness side and the DLC C251 Pin 6, circuit VDB04 (WH/BU), harness side.



- N0053241
- Measure the resistance between the TCCM C2371A Pin 9, circuit VDB05 (WH), harness side and the DLC C251 Pin 14, circuit VDB05 (WH), harness side.



- N0053242

Are the resistances less than 5 ohms?

Yes	CONNECT the negative battery cable. GO to F4.
No	REPAIR the circuit in question. CONNECT the negative battery cable. CLEAR the DTCs. REPEAT the network test with the scan tool.

F4 CHECK FOR CORRECT TCCM OPERATION

- Disconnect all the TCCM connectors.
- Check for:
  - corrosion
  - damaged pins
  - pushed-out pins
- Connect all the TCCM connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new TCCM . REFER to Section 308-07A. CLEAR the DTCs. REPEAT the network test with the scan tool.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test G: The Trailer Brake Control (TBC) Module Does Not Respond To The Scan Tool

Refer to Wiring Diagrams Cell 14 , Module Communications Network for schematic and connector information.

Refer to Wiring Diagrams Cell 95 , Trailer/Camper Adapter for schematic and connector information.

Normal Operation

The Trailer Brake Control (TBC) module communicates with the scan tool through the High Speed Controller Area Network (HS-CAN) .

This pinpoint test is intended to diagnose the following:

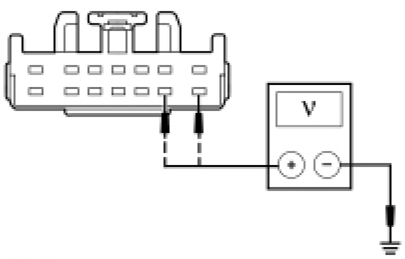
- Fuse
- Wiring, terminals or connectors
- TBC module

PINPOINT TEST G : THE TBC MODULE DOES NOT RESPOND TO THE SCAN TOOL

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

NOTE: *Failure to disconnect the battery when instructed results in false resistance readings. Refer to Section 414-01.*

G1 CHECK THE TBC MODULE VOLTAGE SUPPLY CIRCUITS FOR AN OPEN

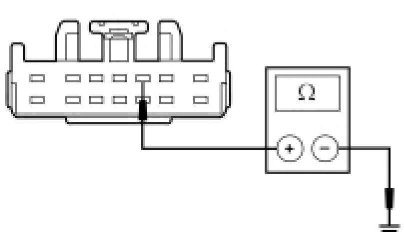
- Ignition OFF.
- Disconnect: TBC Module C2142.
- Ignition ON.
- Measure the voltage between the TBC module C2142 Pin 9, circuit CBP37 (WH), harness side and ground, and measure the voltage between the TBC module C2142 Pin 8, circuit SBB17 (RD), harness side and ground.
- 

N0131577

Are the voltages greater than 10 volts?

Yes	GO to G2.
No	VERIFY Body Control Module (BCM) fuse 37 (10A) and BJB fuse 17 (30A) are OK. If OK, REPAIR the circuit. If not OK, REFER to the Wiring Diagrams manual to identify the possible causes of the circuit short. CLEAR the DTCs. REPEAT the network test with the scan tool.

G2 CHECK THE TBC MODULE GROUND CIRCUIT FOR AN OPEN

- Ignition OFF.
- Disconnect: Negative Battery Cable .
- Measure the resistance between the TBC module C2142 Pin 3, circuit GD138 (BK/WH), harness side and ground.
- 

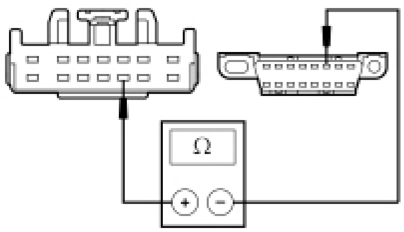
N0122244

Is the resistance less than 5 ohms?

Yes	GO to G3.
No	REPAIR the circuit. CONNECT the negative battery cable. CLEAR the DTCs. REPEAT the network test with the scan tool.

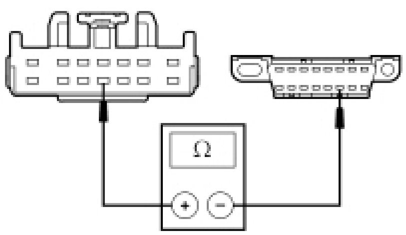
G3 CHECK THE HS-CAN CIRCUITS BETWEEN THE TBC MODULE AND THE DLC FOR AN OPEN

- Measure the resistance between the TBC module C2142 Pin 10, circuit VDB04 (WH/BU), harness side and the Data Link Connector (DLC) C251 Pin 6, circuit VDB04 (WH/BU), harness side.



- N0122245

- Measure the resistance between the TBC module C2142 Pin 11, circuit VDB05 (WH), harness side and the DLC C251 Pin 14, circuit VDB05 (WH), harness side.



- N0090216

Are the resistances less than 5 ohms?

Yes	CONNECT the negative battery cable. GO to G4.
No	REPAIR the circuit. CONNECT the negative battery cable. CLEAR the DTCs. REPEAT the network test with the scan tool.

G4 CHECK FOR CORRECT TBC MODULE OPERATION

- Disconnect the TBC module connector.
- Check for:
  - corrosion
  - damaged pins
  - pushed-out pins
- Connect the TBC module connector and make sure it seats correctly.
- Verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new TBC module. REFER to Section 206-10. CLEAR the DTCs. REPEAT the network test with the scan tool.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test H: The Power Steering Control Module (PSCM) Does Not Respond To The Scan Tool

Refer to Wiring Diagrams Cell 14 , Module Communications Network for schematic and connector information.

Refer to Wiring Diagrams Cell 43 , Power Steering Controls for schematic and connector information.

Normal Operation

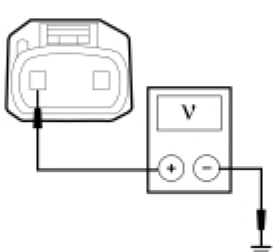
The Power Steering Control Module (PSCM) communicates with the scan tool through the High Speed Controller Area Network (HS-CAN) .

This pinpoint test is intended to diagnose the following:

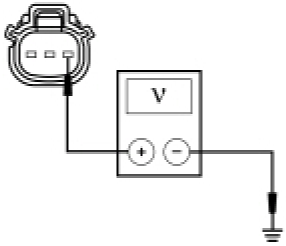
- Fuse
- Wiring, terminals or connectors
- PSCM

PINPOINT TEST H : THE PSCM DOES NOT RESPOND TO THE SCAN TOOL

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.
NOTE: *Failure to disconnect the battery when instructed results in false resistance readings. Refer to Section 414-01.*
H1 CHECK THE PSCM VOLTAGE SUPPLY CIRCUIT FOR AN OPEN

- Disconnect: PSCM C1463A and C1463B.
- Ignition ON.
- Measure the voltage between the PSCM C1463A Pin 2, circuit SDC02 (RD), harness side and ground.
- 

N0148536
- Measure the voltage between the PSCM C1463B Pin 1, circuit CBB52 (GN/WH), harness side and ground.



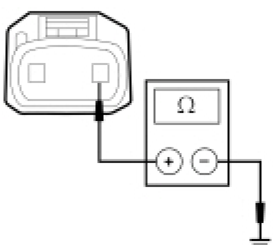
- N0122248

Is the voltage greater than 10 volts?

Yes	GO to H2.
No	VERIFY the High Current Battery Junction Box (BJB) fuse 1 (125A) or BJB fuse 52 (5A) is OK. If OK, REPAIR the circuit. CLEAR the DTCs. REPEAT the network test with the scan tool. If not OK, REFER to the Wiring Diagrams manual to identify the possible causes of the circuit short.

H2 CHECK THE PSCM GROUND CIRCUIT FOR AN OPEN

- Ignition OFF.
- Disconnect: Negative Battery Cable .
- Measure the resistance between the PSCM C1463A Pin 1, circuit GD108 (BK/VT), harness side and ground



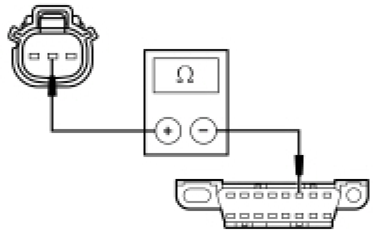
- N0113935

Are the resistances less than 1 ohms?

Yes	GO to H3.
No	REPAIR the circuit. CONNECT the negative battery cable. CLEAR the DTCs. REPEAT the network test with the scan tool.

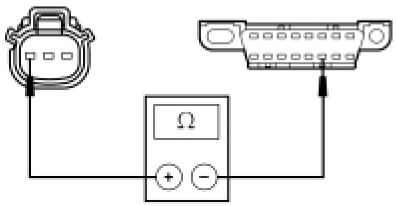
H3 CHECK THE HS-CAN CIRCUITS BETWEEN THE PSCM AND THE DLC FOR AN OPEN

- Measure the resistance between the PSCM C1463B Pin 2, circuit VDB04 (WH/BU), harness side and the Data Link Connector (DLC) C251 Pin 6, circuit VDB04 (WH/BU), harness side.



- N0122247

- Measure the resistance between the PSCM C1463B Pin 3, circuit VDB05 (WH), harness side and the DLC C251 Pin 14, circuit VDB05 (WH), harness side.



- N0122246

Are the resistances less than 5 ohms?

Yes	GO to H4.
No	REPAIR the circuit in question. CONNECT the negative battery cable. CLEAR the DTCs. REPEAT the network test with the scan tool.

H4 CHECK FOR CORRECT PSCM OPERATION

- Disconnect all the PSCM connectors.
- Check for:
  - corrosion
  - damaged pins
  - pushed-out pins
- Connect all the PSCM connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new EPAS steering gear. REFER to Section 211-02. CLEAR the DTCs. REPEAT the network test with the scan tool.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. CLEAR the DTCs. REPEAT the network test with the scan tool.

Pinpoint Test I: The Body Control Module (BCM) Does Not Respond To The Scan Tool

Refer to Wiring Diagrams Cell 13 , Power Distribution/BCM for schematic and connector information.

Refer to Wiring Diagrams Cell 14 , Module Communications Network for schematic and connector information.

Normal Operation

The Body Control Module (BCM) communicates with the scan tool through the Medium Speed Controller Area Network (MS-CAN) .

This pinpoint test is intended to diagnose the following:

- Fuse
- Wiring, terminals or connectors
- BCM

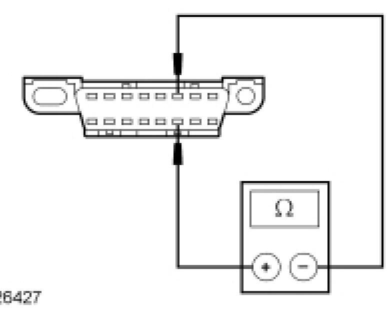
PINPOINT TEST I : THE BCM DOES NOT RESPOND TO THE SCAN TOOL

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

NOTE: *Failure to disconnect the battery when instructed results in false resistance readings. Refer to Section 414-01.*

I1 CHECK THE HS-CAN TERMINATION RESISTANCE

- Ignition OFF.
- Disconnect: Negative Battery Cable .
- Disconnect the scan tool cable from the Data Link Connector (DLC) .
- Measure the resistance between the DLC C251 Pin 6, circuit VDB04 (WH/BU), harness side and the DLC C251 Pin 14, circuit VDB05 (WH), harness side.



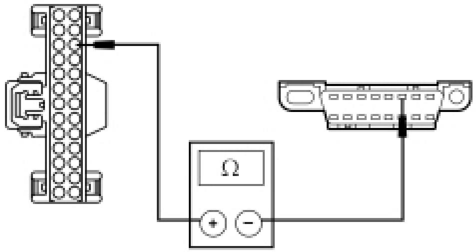
- N0026427

Is the resistance between 54 and 66 ohms?

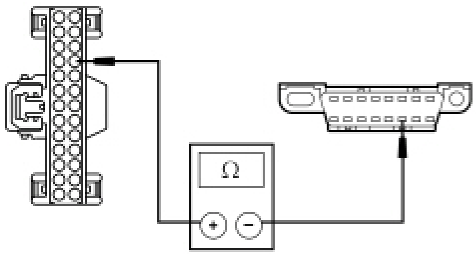
Yes	CONNECT the negative battery cable. GO to I3.
No	GO to I2.

I2 CHECK THE HS-CAN CIRCUITS BETWEEN THE BCM AND THE DLC FOR AN OPEN

- Disconnect: BCM C2280F .
- Measure the resistance between the BCM C2280F Pin 16, circuit VDB04 (WH/BU), harness side and the Data Link Connector (DLC) C251 Pin 6, circuit VDB04 (WH/BU), harness side.



- N0115338
- Measure the resistance between the BCM C2280F Pin 17, circuit VDB05 (WH), harness side and the DLC C251 Pin 14, circuit VDB05 (WH), harness side.



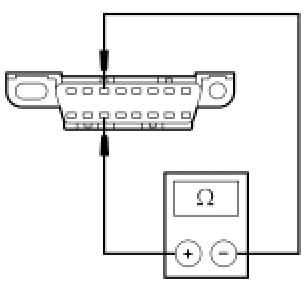
- N0115339

Are the resistances less than 5 ohms?

Yes	GO to I7.
No	REPAIR the circuit in question. CONNECT the negative battery cable. CLEAR the DTCs. REPEAT the network test with the scan tool.

I3 CHECK THE MS-CAN TERMINATION RESISTANCE

- Measure the resistance between the DLC C251 Pin 3, circuit VDB06 (GY/OG), harness side and the DLC C251 Pin 11, circuit VDB07 (VT/OG), harness side.



- N0050701

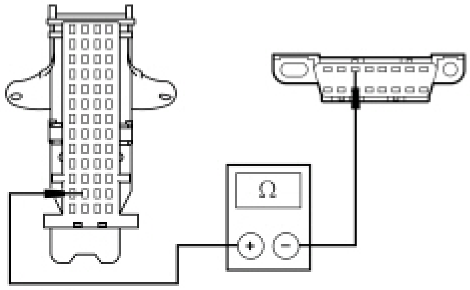
Is the resistance between 54 and 66 ohms?

Yes	CONNECT the negative battery cable. GO to I5.
No	GO to I4.

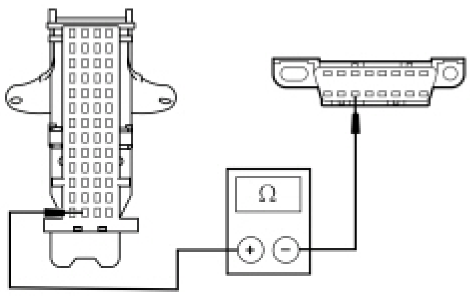
I4 CHECK THE MS-CAN CIRCUITS BETWEEN THE BCM AND THE DLC FOR AN OPEN

- Disconnect: BCM C2280B .

- Measure the resistance between the BCM C2280B Pin 38, circuit VDB06 (GY/OG), harness side and the Data Link Connector (DLC) C251 Pin 3, circuit VDB06 (GY/OG), harness side.

- 
N0114057

- Measure the resistance between the BCM C2280B Pin 39, circuit VDB07 (VT/OG), harness side and the DLC C251 Pin 11, circuit VDB07 (VT/OG), harness side.

- 
N0114058

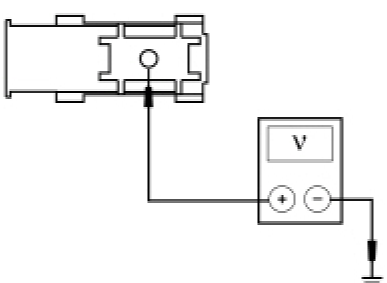
Are the resistances less than 5 ohms?

Yes	CONNECT the negative battery cable. GO to I7.
No	REPAIR the circuit in question. CONNECT the negative battery cable. CLEAR the DTCs. REPEAT the network test with the scan tool.

I5 CHECK THE BCM VOLTAGE SUPPLY CIRCUIT FOR AN OPEN

NOTE: *Measurements are taken with the fuse installed and probes placed at the terminals on the back of the fuse blade.*

- Disconnect: BCM C2280G .
- Connect: Negative Battery Cable .
- Ignition ON.
- Measure the voltage between the BCM C2280G Pin 1, SDC04 (RD), and ground.

- 
N0148537

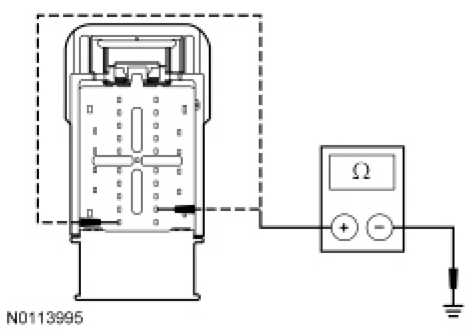
Is the voltage greater than 10 volts?

Yes	GO to I6.
No	VERIFY the BJB fuse 45 (10A) is OK. If OK, GO to I6.

I6 CHECK THE BCM GROUND CIRCUIT FOR AN OPEN

- Disconnect: Negative Battery Cable .
- Disconnect: BCM C2280D .
- Measure the resistance between the BCM , harness side and ground as follows:

Connector-Pin	Circuit
C2280D Pin 16	GD139 (BK/YE)
C2280D Pin 25	GD139 (BK/YE)

- 

Are the resistances less than 5 ohms?

Yes	GO to I7.
No	REPAIR the circuit. CONNECT the negative battery cable. CLEAR the DTCs. REPEAT the network test with the scan tool.

I7 CHECK FOR CORRECT BCM OPERATION

- Disconnect all the BCM connectors.
- Check for:
  - corrosion
  - damaged pins
  - pushed-out pins
- Connect all the BCM connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . CLEAR the DTCs. REPEAT the network test with the scan tool.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test J: The HVAC Module Does Not Respond To The Scan Tool

Refer to Wiring Diagrams Cell 14 , Module Communications Network for schematic and connector information.

Refer to Wiring Diagrams Cell 54 , Manual Climate Control System for schematic and connector information.

Refer to Wiring Diagrams Cell 55 , Automatic Climate Control System for schematic and connector information.

Normal Operation

The HVAC module communicates with the scan tool through the Medium Speed Controller Area Network (MS-CAN) .

This pinpoint test is intended to diagnose the following:

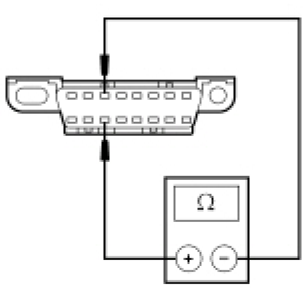
- Fuse
- Wiring, terminals or connectors
- HVAC module

PINPOINT TEST J : THE HVAC MODULE DOES NOT RESPOND TO THE SCAN TOOL

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

NOTE: *Failure to disconnect the battery when instructed results in false resistance readings. Refer to Section 414-01.*

J1 CHECK THE MS-CAN TERMINATION RESISTANCE

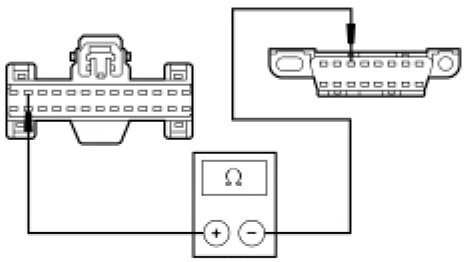
- Ignition OFF.
- Disconnect: Negative battery cable .
- Disconnect the scan tool cable from the DLC .
- Measure the resistance between the DLC C251 Pin 3, circuit VDB06 (GY/OG), harness side and the DLC C251 Pin 11, circuit VDB07 (VT/OG), harness side.
- 

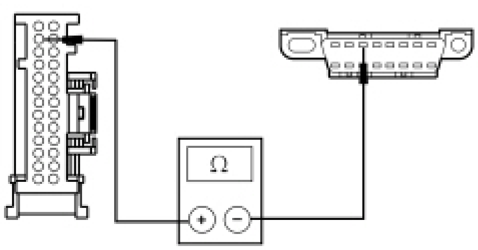
N0050701

Is the resistance between 54 and 66 ohms?

Yes	CONNECT the negative battery cable. GO to J4.
No	GO to J2.

J2 CHECK THE MS-CAN (+) CIRCUIT BETWEEN THE HVAC MODULE AND THE DLC FOR AN OPEN

- Ignition OFF.
- Disconnect: HVAC Module C294A (Electronic Manual Temperature Control (EMTC)) or C228A (EATC) .
- For EMTC , measure the resistance between the HVAC module C294A Pin 12, circuit VDB06 (GY/OG), harness side and the DLC C251 Pin 3, circuit VDB06 (GY/OG), harness side.
- 

N0104911
- For EATC , measure the resistance between the HVAC module C228A Pin 12, circuit VDB06 (GY/OG), harness side and the DLC C251 Pin 3, circuit VDB06 (GY/OG), harness side.
- 

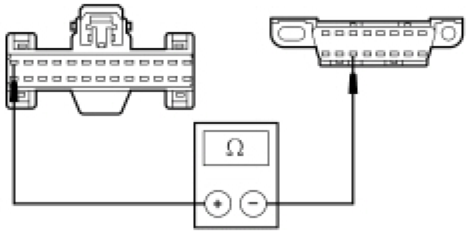
N0177348

Is the resistance less than 5 ohms?

Yes	GO to J3.
No	REPAIR the circuit. CONNECT the negative battery cable. CLEAR the DTCs. REPEAT the network test with the scan tool.

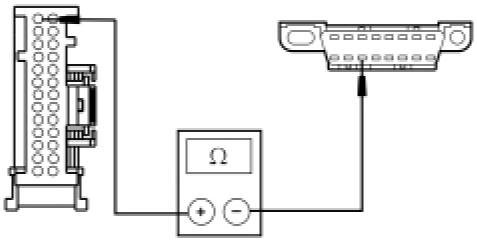
J3 CHECK THE MS-CAN (-) CIRCUIT BETWEEN THE HVAC MODULE AND THE DLC FOR AN OPEN

- For EMTC , measure the resistance between the HVAC module C294A Pin 13, circuit VDB07 (VT/OG), harness side and the DLC C251 Pin 11, circuit VDB07 (VT/OG), harness side.



- N0104912

- For EATC , measure the resistance between the HVAC module C228A Pin 13, circuit VDB07 (VT/OG), harness side and the DLC C251 Pin 11, circuit VDB07 (VT/OG), harness side.



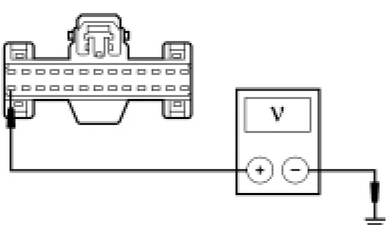
- N0177349

Is the resistance less than 5 ohms?

Yes	CONNECT the negative battery cable. GO to J6.
No	REPAIR the circuit. CONNECT the negative battery cable. CLEAR the DTCs. REPEAT the network test with the scan tool.

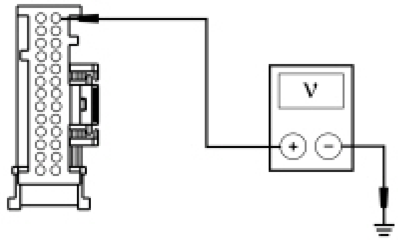
J4 CHECK THE HVAC MODULE VOLTAGE SUPPLY CIRCUITS FOR AN OPEN

- Disconnect: HVAC Module C294A (EMTC) or C228A (EATC .
- Ignition ON.
- For EMTC , measure the voltage between the HVAC module C294A Pin 26, circuit SBP46 (BN/RD), harness side and ground.



- N0104909

- For EATC , measure the voltage between the HVAC module C228A Pin 26, circuit SBP46 (BN/RD), harness side and ground.



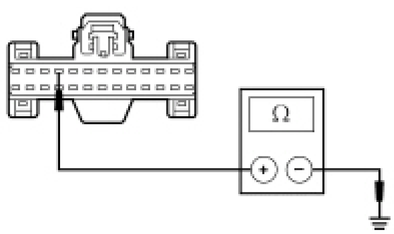
- N0177350

Is the voltage greater than 10 volts?

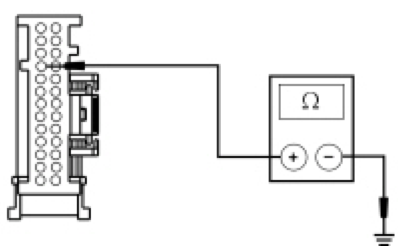
Yes	GO to J5.
No	VERIFY the Body Control Module (BCM) fuse 46 (10A) is OK. If OK, REPAIR the circuit. If not OK, REFER to the Wiring Diagrams manual to identify the possible causes of the circuit short. CLEAR the DTCs. REPEAT the network test with the scan tool.

J5 CHECK THE HVAC MODULE GROUND CIRCUIT FOR AN OPEN

- Ignition OFF.
- Disconnect: Negative Battery Cable .
- For EMTC , measure the resistance between the HVAC module C294A Pin 10, circuit GD133 (BK), harness side and ground.



- N0104910
- For EATC , measure the resistance between the HVAC module C228A Pin 10, circuit GD133 (BK), harness side and ground.



- N0177351

Is the resistance less than 5 ohms?

Yes	GO to J6.
No	REPAIR the circuit. CONNECT the negative battery cable. CLEAR the DTCs. REPEAT the

	network test with the scan tool.

J6 CHECK FOR CORRECT HVAC MODULE OPERATION

• Disconnect all the HVAC module connectors.
• Check for:
  ▪ corrosion
  ▪ damaged pins
  ▪ pushed-out pins
• Connect all the HVAC module connectors and make sure they seat correctly.
• Operate the system and verify the concern is still present.
 	
Is the concern still present?

Yes	INSTALL a new HVAC module. REFER to Section 412-01. CLEAR the DTCs. REPEAT the network test with the scan tool.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test K: The Audio Front Control Module (ACM) Does Not Respond To The Scan Tool

Refer to Wiring Diagrams Cell 14 , Module Communications Network for schematic and connector information.

Refer to Wiring Diagrams Cell 130 , Audio System/Navigation for schematic and connector information.

Normal Operation

The base and premium audio system Audio Front Control Module (ACM) communicates with the scan tool through the Medium Speed Controller Area Network (MS-CAN) . For vehicles equipped with 4.2-inch screen or 8-inch touchscreen, the Audio Front Control Module (ACM) communicates with the scan tool through the IPC (gateway) on the HS-CAN .

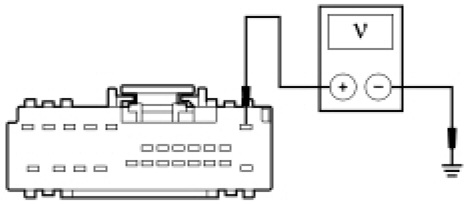
This pinpoint test is intended to diagnose the following:

- Fuse
- Wiring, terminals or connectors
- ACM

PINPOINT TEST K : THE ACM DOES NOT RESPOND TO THE SCAN TOOL

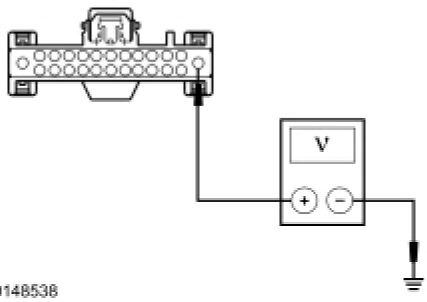
NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.
NOTE: *Failure to disconnect the battery when instructed results in false resistance readings. Refer to Section 414-01.*
K1 CHECK THE ACM VOLTAGE SUPPLY CIRCUITS FOR AN OPEN
• Ignition OFF. • Disconnect: ACM C290A or C240A (4.2-inch screen or 8-inch touchscreen) .

- Ignition ON.
- If equipped with AM/FM radio with single CD, measure the voltage between the ACM C290A Pin 1, circuit SBP29 (GY/RD), harness side and ground.



- N0058708

- If equipped with 4.2-inch screen or 8-inch touchscreen, measure the voltage between the ACM C240A Pin 1, circuit SBP29 (GY/RD), harness side and ground.



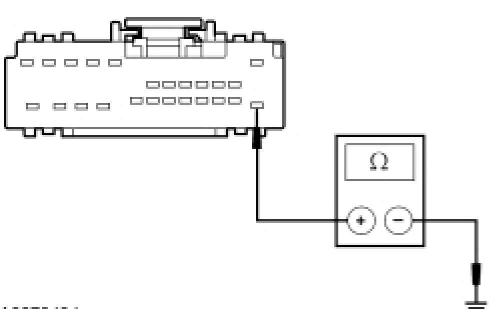
- N0148538

Is the voltage greater than 10 volts?

Yes	GO to K2.
No	VERIFY the Body Control Module (BCM) fuse 29 (20A) is OK. If OK, REPAIR the circuit. If not OK, REFER to the Wiring Diagrams manual to identify the possible causes of the circuit short. CLEAR the DTCs. REPEAT the network test with the scan tool.

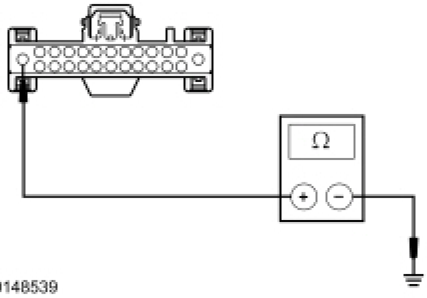
K2 CHECK THE ACM GROUND CIRCUIT FOR AN OPEN

- Ignition OFF.
- Disconnect: Negative Battery Cable .
- If equipped with AM/FM radio with single CD, measure the resistance between the ACM C290A Pin 13, circuit GD114 (BK/BU), harness side and ground.



- A0073494

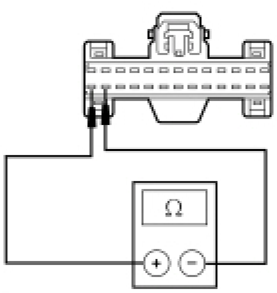
- If equipped with 4-inch screen or 8-inch touchscreen, measure the resistance between the ACM C240A Pin 13, circuit GD114 (BK/BU), harness side and ground.

- 
N0148539

Is the resistance less than 5 ohms?

Yes	If equipped with AM/FM radio with single CD, GO to K5. If equipped with 4-inch screen or 8-inch touchscreen, GO to K3.
No	REPAIR the circuit. CONNECT the negative battery cable. CLEAR the DTCs. REPEAT the network test with the scan tool.

K3 CHECK THE I-CAN TERMINATION RESISTANCE

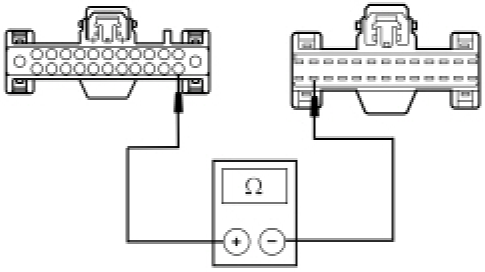
- Ignition OFF.
- Disconnect: Negative Battery Cable .
- Disconnect: IPC C220 .
- Measure the resistance between the IPC C220 Pin 25 and IPC C220 Pin 26, harness side.
- 
N0150481

Is the resistance 120 ohms?

Yes	CONNECT the IPC and the battery cable. GO to K6.
No	GO to K4.

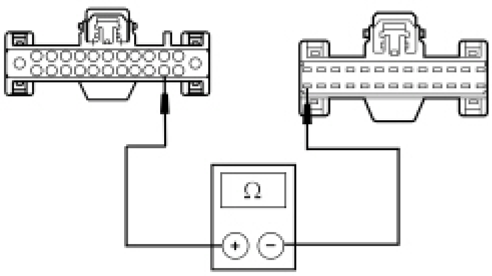
K4 CHECK THE I-CAN CIRCUITS BETWEEN THE ACM AND THE IPC FOR AN OPEN

- Measure the resistance between the ACM C240A Pin 14, circuit VDB13 (BU/GY), harness side and the Instrument Panel Cluster (IPC) C220 Pin 25, circuit VDB13 (BU/GY), harness side.



- N0150484

- Measure the resistance between the ACM C240A Pin 15, circuit VDB14 (VT/GY), harness side and the IPC C220 Pin 26, circuit VDB14 (VT/GY), harness side.



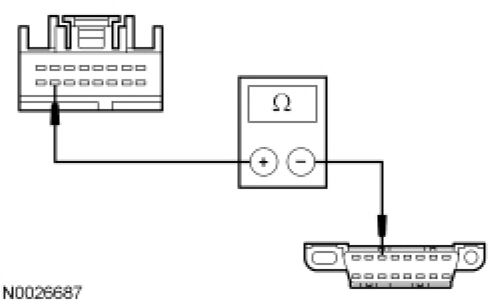
- N0150485

Are the resistances less than 5 ohms?

Yes	GO to K6.
No	REPAIR the circuit in question. CONNECT the IPC and negative battery cable. CLEAR the DTCs. REPEAT the network test with the scan tool.

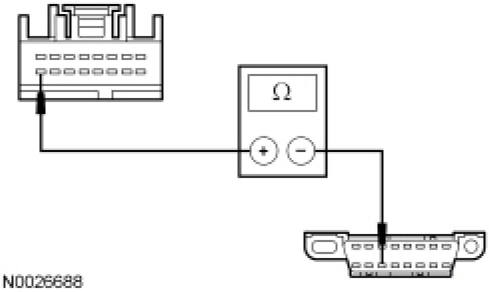
K5 CHECK THE MS-CAN CIRCUITS BETWEEN THE ACM AND THE DLC FOR AN OPEN

- Disconnect: ACM C290B .
- Measure the resistance between the ACM C290B Pin 15, circuit VDB06 (GY/OG), harness side and the DLC C251 Pin 3, circuit VDB06 (GY/OG), harness side.



- N0026687

- Measure the resistance between the ACM C290B Pin 16, circuit VDB07 (VT/OG), harness side and the DLC C251 Pin 11, circuit VDB07 (VT/OG), harness side.



- N0026688

Are the resistances less than 5 ohms?

Yes	CONNECT the negative battery cable. GO to K6.
No	REPAIR the circuit in question. CONNECT the negative battery cable. CLEAR the DTCs. REPEAT the network test with the scan tool.

K6 CHECK FOR CORRECT ACM OPERATION

- Disconnect all the ACM connectors.
- Check for:
  - corrosion
  - damaged pins
  - pushed-out pins
- Connect all the ACM connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new ACM . Refer to the appropriate section in Group 415 for the procedure. CLEAR the DTCs. REPEAT the network test with the scan tool. TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test L: The Audio Digital Signal Processing (DSP) Module Does Not Respond To The Scan Tool

Refer to Wiring Diagrams Cell 14 , Module Communications Network for schematic and connector information.

Refer to Wiring Diagrams Cell 130 , Audio System/Navigation for schematic and connector information.

Normal Operation

The audio Digital Signal Processing (DSP) module communicates with other modules on the I-CAN , but communicates with the scan tool through the IPC (gateway) on the HS-CAN .

This pinpoint test is intended to diagnose the following:

- Fuse
- Wiring, terminals or connectors
- Audio DSP module

PINPOINT TEST L : THE AUDIO DSP MODULE DOES NOT RESPOND TO THE SCAN TOOL

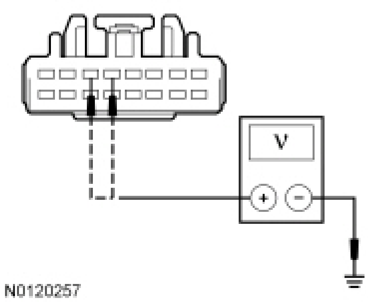
NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

NOTE: *Failure to disconnect the battery when instructed results in false resistance readings. Refer to Section 414-01.*

L1 CHECK THE AUDIO DSP MODULE VOLTAGE SUPPLY CIRCUITS FOR AN OPEN

- Ignition OFF.
- Disconnect: Audio DSP Module C3154C.
- Ignition ON.
- Measure the voltage between the audio DSP module, harness side and ground as follows:

Connector-Pin	Circuit
C3154C Pin 5	SBP19 (BU/RD)
C3154C Pin 6	SBP19 (BU/RD)

- 

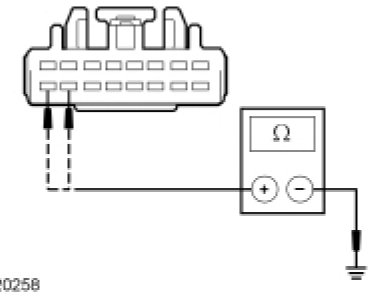
Are the voltages greater than 10 volts?

Yes	GO to L2.
No	VERIFY the Body Control Module (BCM) fuse 19 (20A) is OK. If OK, REPAIR the circuit in question. If not OK, REFER to the Wiring Diagrams manual to identify the possible causes of the circuit short. CLEAR the DTCs. REPEAT the network test with the scan tool.

L2 CHECK THE AUDIO DSP MODULE GROUND CIRCUITS FOR AN OPEN

- Ignition OFF.
- Disconnect: Negative Battery Cable .
- Measure the resistance between the audio DSP module, harness side and ground as follows:

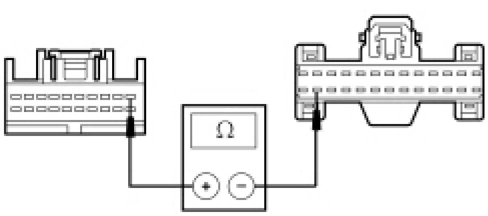
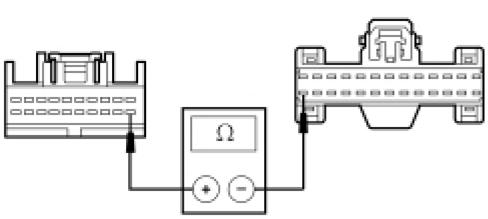
Connector-Pin	Circuit
C3154C Pin 15	GD148 (BK/YE)
C3154C Pin 16	GD148 (BK/YE)

- 
  N0120258

Are the resistances less than 5 ohms?

Yes	GO to L3.
No	REPAIR the circuit in question. CONNECT the negative battery cable. CLEAR the DTCs. REPEAT the network test with the scan tool.

L3 CHECK THE I-CAN CIRCUITS BETWEEN THE AUDIO DSP MODULE AND THE IPC FOR AN OPEN

- Disconnect: Audio DSP Module C3154A.
- Disconnect: IPC C220 .
- Measure the resistance between the audio DSP module C3154A Pin 1, circuit VDB13 (BU/GY), harness side and the IPC C220 Pin 25, circuit VDB13 (BU/GY), harness side.
- 
  N0150486
- Measure the resistance between the audio DSP module C3154A Pin 11, circuit VDB14 (VT/GY), harness side and the IPC C220 Pin 26, circuit VDB14 (VT/GY), harness side.
- 
  N0150487

Are the resistances less than 5 ohms?

Yes	CONNECT the negative battery cable. GO to L4.
No	REPAIR the circuit in question. CONNECT the negative battery cable. CLEAR the DTCs. REPEAT the network test with the scan tool.

L4 CHECK FOR CORRECT AUDIO DSP MODULE OPERATION

- Disconnect all the audio DSP module connectors.
- Check for:
  - corrosion
  - damaged pins
  - pushed-out pins
- Connect all the audio DSP module connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new audio DSP module. REFER to Section 415-00E.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test M: The Front Controls Interface Module (FCIM) Does Not Respond To The Scan Tool

Refer to Wiring Diagrams Cell 14 , Module Communications Network for schematic and connector information.

Refer to Wiring Diagrams Cell 130 , Audio System/Navigation for schematic and connector information.

Normal Operation

For vehicles equipped with AM/FM radio with single CD, the Front Controls Interface Module (FCIM) communicates with the scan tool through the Medium Speed Controller Area Network (MS-CAN) . For vehicles equipped with 4.2-inch screen or 8-inch touchscreen, the Front Controls Interface Module (FCIM) communicates with other modules on the I-CAN , but communicates with the scan tool through the IPC (gateway) on the HS-CAN .

This pinpoint test is intended to diagnose the following:

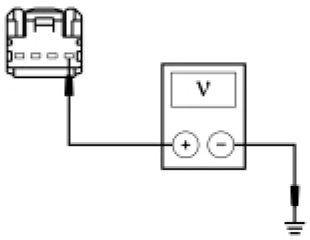
- Fuse
- Wiring, terminals or connectors
- FCIM

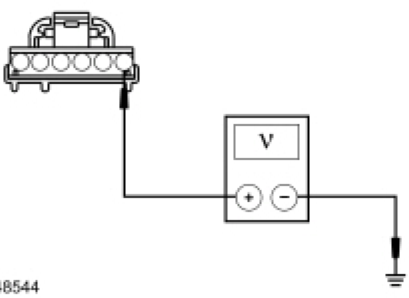
PINPOINT TEST M : THE FCIM DOES NOT RESPOND TO THE SCAN TOOL

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

NOTE: *Failure to disconnect the battery when instructed results in false resistance readings. Refer to Section 414-01.*

M1 CHECK THE FCIM VOLTAGE SUPPLY CIRCUIT FOR AN OPEN

- Ignition OFF.
- Disconnect: FCIM C2402 .
- Ignition ON.
- If equipped with AM/FM radio with single CD, measure the voltage between the FCIM C2402 Pin 1, circuit SBP09 (RD), harness side and ground.
- 

N0122228
- If equipped with 4.2-inch screen or 8-inch touchscreen, measure the voltage between the FCIM C2402 Pin 1, circuit SBP09 (RD), harness side and ground.
- 

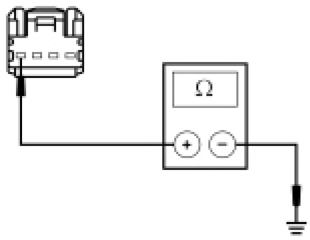
N0148544

Is the voltage greater than 10 volts?

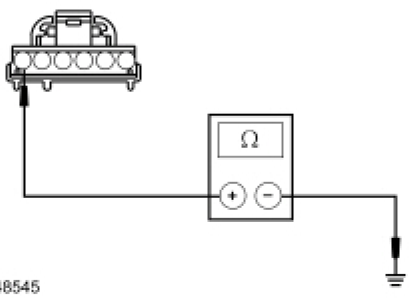
Yes	GO to M2.
No	VERIFY the Body Control Module (BCM) fuse 9 (10A) is OK. If OK, REPAIR the circuit. If not OK, REFER to the Wiring Diagrams manual to identify the possible causes of the circuit short. CLEAR the DTCs. REPEAT the network test with the scan tool.

M2 CHECK THE FCIM GROUND CIRCUIT FOR AN OPEN

- Ignition OFF.
- Disconnect: Negative Battery Cable .
- If equipped with AM/FM radio with single CD, measure the resistance between the FCIM C2402 Pin 4, circuit GD133 (BK), harness side and ground.



- N0122229
- If equipped with 4.2-inch screen or 8-inch touchscreen, measure the resistance between the FCIM C2402 Pin 6, circuit GD133 (BK), harness side and ground.



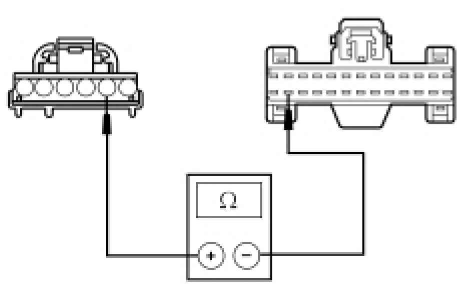
- N0148545

Is the resistance less than 5 ohms?

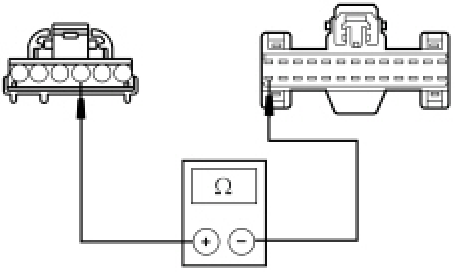
Yes	If equipped with 4.2-inch screen or 8-inch touchscreen, GO to M3. If equipped with AM/FM radio with single CD, GO to M4.
No	REPAIR the circuit. CONNECT the negative battery cable. CLEAR the DTCs. REPEAT the network test with the scan tool.

M3 CHECK THE I-CAN CIRCUITS BETWEEN THE FCIM AND THE IPC FOR AN OPEN

- Disconnect: IPC C220 .
- Measure the resistance between the FCIM C2402 Pin 2, circuit VDB13 (BU/GY), harness side and the IPC C220 Pin 25, circuit VDB13 (BU/GY), harness side.



- N0150488
- Measure the resistance between the FCIM C2402 Pin 3, circuit VDB14 (VT/GY), harness side and the IPC C220 Pin 26, circuit VDB14 (VT/GY), harness side.



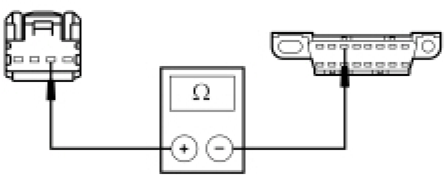
- N0150489

Are the resistances less than 5 ohms?

Yes	CONNECT the negative battery cable. GO to M5.
No	REPAIR the circuit in question. CONNECT the negative battery cable. CLEAR the DTCs. REPEAT the network test with the scan tool.

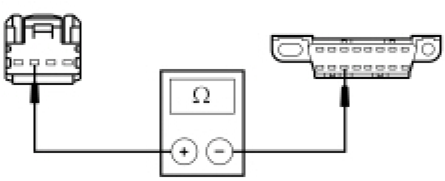
M4 CHECK THE MS-CAN CIRCUITS BETWEEN THE FCIM AND THE DLC FOR AN OPEN

- Measure the resistance between the FCIM C2402 Pin 2, circuit VDB06 (GY/OG), harness side and the Data Link Connector (DLC) C251 Pin 3, circuit VDB06 (GY/OG), harness side.



- N0122230

- Measure the resistance between the FCIM C2402 Pin 3, circuit VDB07 (VT/OG), harness side and the DLC C251 Pin 11, circuit VDB07 (VT/OG), harness side.



- N0122231

Are the resistances less than 5 ohms?

Yes	CONNECT the negative battery cable. GO to M5.
No	REPAIR the circuit in question. CONNECT the negative battery cable. CLEAR the DTCs. REPEAT the network test with the scan tool.

M5 CHECK FOR CORRECT FCIM OPERATION

- Disconnect the FCIM connector.
- Check for:
  - corrosion
  - damaged pins
  - pushed-out pins
- Connect the FCIM connector and make sure it seats correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new FCIM . Refer to the appropriate section in Group 415 for the procedure. CLEAR the DTCs. REPEAT the network test with the scan tool.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test N: The Front Display Interface Module (FDIM) Does Not Respond To The Scan Tool

Refer to Wiring Diagrams Cell 14 , Module Communications Network for schematic and connector information.

Refer to Wiring Diagrams Cell 130 , Audio System/Navigation for schematic and connector information.

Normal Operation

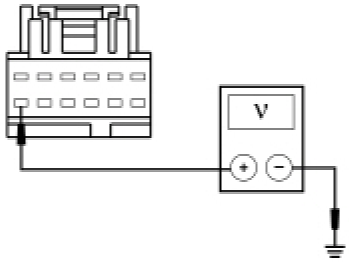
The Front Display Interface Module (FDIM) (vehicles not equipped with 8-inch touchscreen) communicates with the scan tool through the Medium Speed Controller Area Network (MS-CAN) .

This pinpoint test is intended to diagnose the following:

- Fuse
- Wiring, terminals or connectors
- FDIM

PINPOINT TEST N : THE FDIM DOES NOT RESPOND TO THE SCAN TOOL

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.
NOTE: *Failure to disconnect the battery when instructed results in false resistance readings. Refer to Section 414-01.*
N1 CHECK THE FDIM VOLTAGE SUPPLY CIRCUIT FOR AN OPEN
• Ignition OFF. • Disconnect: FDIM C2407 . • Ignition ON. • Measure the voltage between the FDIM C2407 Pin 12, circuit SBP09 (RD), harness side and ground.



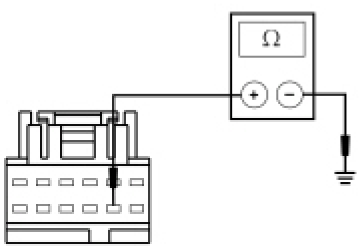
- N0122249

Is the voltage greater than 10 volts?

Yes	GO to N2.
No	VERIFY the Body Control Module (BCM) fuse 9 (10A) is OK. If OK, REPAIR the circuit. If not OK, REFER to the Wiring Diagrams manual to identify the possible causes of the circuit short. CLEAR the DTCs. REPEAT the network test with the scan tool.

N2 CHECK THE FDIM GROUND CIRCUIT FOR AN OPEN

- Ignition OFF.
- Disconnect: Negative Battery Cable .
- Measure the resistance between the FDIM C2407 Pin 8, circuit GD133 (BK), harness side and ground.



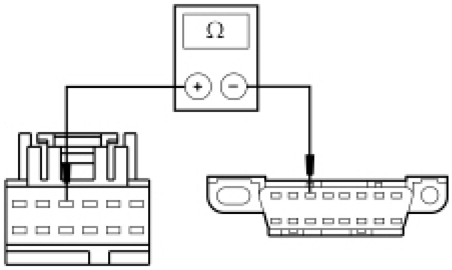
- N0122251

Is the resistance less than 5 ohms?

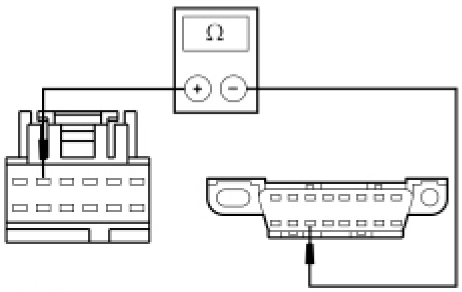
Yes	GO to N3.
No	REPAIR the circuit. CONNECT the negative battery cable. CLEAR the DTCs. REPEAT the network test with the scan tool.

N3 CHECK THE MS-CAN CIRCUITS BETWEEN THE FDIM AND THE DLC FOR AN OPEN

- Measure the resistance between the FDIM C2407 Pin 4, circuit VDB06 (GY/OG), harness side and the Data Link Connector (DLC) C251 Pin 3, circuit VDB06 (GY/OG), harness side.



- N0122252
- Measure the resistance between the FDIM C2407 Pin 5, circuit VDB07 (VT/OG), harness side and the DLC C251 Pin 11, circuit VDB07 (VT/OG), harness side.



- N0122253

Are the resistances less than 5 ohms?

Yes	CONNECT the negative battery cable. GO to N4.
No	REPAIR the circuit in question. CONNECT the negative battery cable. CLEAR the DTCs. REPEAT the network test with the scan tool.

N4 CHECK FOR CORRECT FDIM OPERATION

- Disconnect the FDIM connector.
- Check for:
  - corrosion
  - damaged pins
  - pushed-out pins
- Connect the FDIM connector and make sure it seats correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new FDIM . Refer to the appropriate section in Group 415 for the procedure.. CLEAR the DTCs. REPEAT the network test with the scan tool.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test O: The Front Control/Display Interface Module (FCDIM) Does Not Respond To The Scan Tool

Refer to Wiring Diagrams Cell 14 , Module Communications Network for schematic and connector information.

Refer to Wiring Diagrams Cell 130 , Audio System/Navigation for schematic and connector information.

Normal Operation

The FCDIM communicates with other modules on the I-CAN , but communicates with the scan tool through the IPC (gateway) on the HS-CAN .

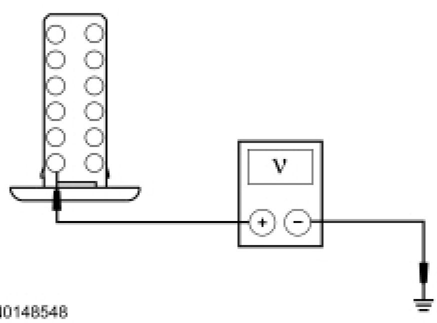
This pinpoint test is intended to diagnose the following:

- Fuse
- Wiring, terminals or connectors
- FCDIM

PINPOINT TEST O : THE FCDIM DOES NOT RESPOND TO THE SCAN TOOL

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.
NOTE: *Failure to disconnect the battery when instructed results in false resistance readings. Refer to Section 414-01.*
O1 CHECK THE FCDIM VOLTAGE SUPPLY CIRCUIT FOR AN OPEN

- Ignition OFF.
- Disconnect: FCDIM C2123 .
- Ignition ON.
- Measure the voltage between the FCDIM C2123 Pin 1, circuit SBP09 (RD), harness side and ground.



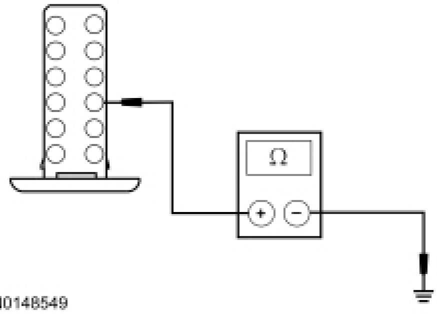
- N0148548

Is the voltage greater than 10 volts?

Yes	GO to O2.
No	VERIFY the Body Control Module (BCM) fuse 9 (10A) is OK. If OK, REPAIR the circuit. If not OK, REFER to the Wiring Diagrams manual to identify the possible causes of the circuit short. CLEAR the DTCs. REPEAT the network test with the scan tool.

O2 CHECK THE FCDIM GROUND CIRCUIT FOR AN OPEN

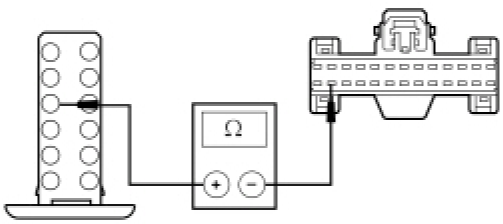
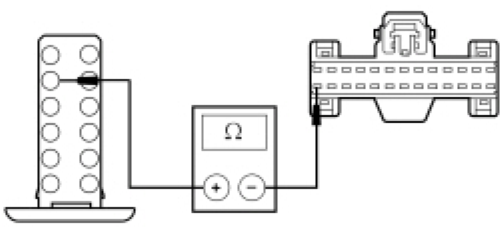
- Ignition OFF.
- Disconnect: Negative Battery Cable .
- Measure the resistance between the FCDIM C2123 Pin 9, circuit GD114 (BK/BU), harness side and ground.

- 
  N0148549

Is the resistance less than 5 ohms?

Yes	GO to O3.
No	REPAIR the circuit. CONNECT the negative battery cable. CLEAR the DTCs. REPEAT the network test with the scan tool.

O3 CHECK THE I-CAN CIRCUITS BETWEEN THE FCDIM AND THE IPC FOR AN OPEN

- Disconnect: IPC C220 .
- Measure the resistance between the FCDIM C2123 Pin 4, circuit VDB13 (BU/GY), harness side and the IPC C220 Pin 25, circuit VDB13 (BU/GY), harness side.
- 
  N0150490
- Measure the resistance between the FCDIM C2123 Pin 5, circuit VDB14 (VT/GY), harness side and the IPC C220 Pin 26, circuit VDB14 (VT/GY), harness side.
- 
  N0150491

Are the resistances less than 5 ohms?

Yes	CONNECT the IPC and the negative battery cable. GO to O4.
No	REPAIR the circuit in question. CONNECT the negative battery cable. CLEAR the DTCs. REPEAT the network test with the scan tool.

O4 CHECK FOR CORRECT FCDIM OPERATION

• Disconnect the FCDIM connector.
• Check for:
  ▪ corrosion
  ▪ damaged pins
  ▪ pushed-out pins
• Connect the FCDIM connector and make sure it seats correctly.
• Operate the system and verify the concern is still present.
 	
Is the concern still present?
 	
Yes — INSTALL a new FCDIM . REFER to Section 415-00C. CLEAR the DTCs. REPEAT the network test with the scan tool.
No — The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test P: The Driver Seat Module (DSM) Does Not Respond To The Scan Tool

Refer to Wiring Diagrams Cell 14 , Module Communications Network for schematic and connector information.

Refer to Wiring Diagrams Cell 123 , Memory Seats for schematic and connector information.

Normal Operation

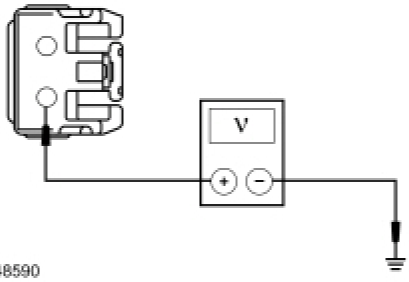
The Driver Seat Module (DSM) communicates with the scan tool through the Medium Speed Controller Area Network (MS-CAN) .

This pinpoint test is intended to diagnose the following:

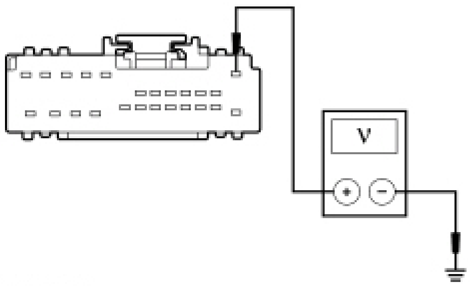
- Fuse
- Wiring, terminals or connectors
- DSM

PINPOINT TEST P : THE DSM DOES NOT RESPOND TO THE SCAN TOOL

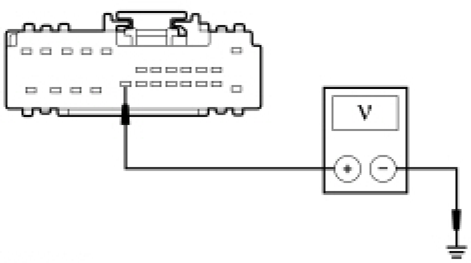
NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.
NOTE: *Failure to disconnect the battery when instructed results in false resistance readings. Refer to Section 414-01.*
P1 CHECK THE DSM VOLTAGE SUPPLY CIRCUITS FOR AN OPEN
• Ignition OFF. • Disconnect: DSM C341A, C341B and C341C. • Ignition ON. • Measure the voltage between the DSM C341A Pin 1, circuit CBX05 (VT/RD), harness side and ground.



- N0148590
- Measure the voltage between the DSM C341B Pin 1, circuit CBX03 (BU/RD), harness side and ground.



- N0068871
- Measure the voltage between the DSM C341C Pin 20, circuit CPP14 (BN/WH), harness side and ground.



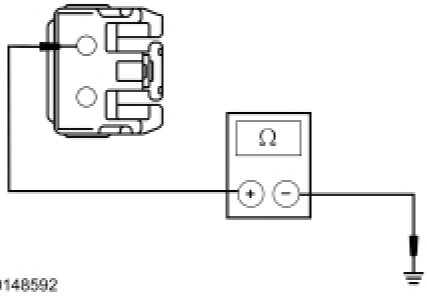
- N0148591

Are the voltages greater than 10 volts?

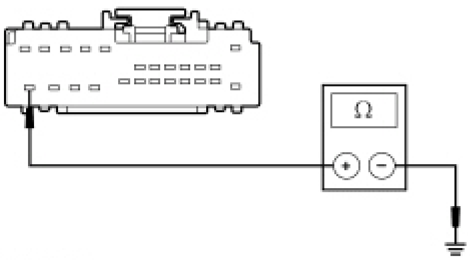
Yes	GO to P2.
No	VERIFY the BJB fuse 74 (30A) or Body Control Module (BCM) fuse 7 (7.5A) is OK. If OK, REPAIR the circuit in question. If not OK, REFER to the Wiring Diagrams manual to identify the possible causes of the circuit short. CLEAR the DTCs. REPEAT the network test with the scan tool.

P2 CHECK THE DSM GROUND CIRCUITS FOR AN OPEN

- Ignition OFF.
- Disconnect: Negative Battery Cable .
- Measure the resistance between the DSM C341A Pin 2, circuit GD906 (BK/GY), harness side and ground.



- N0148592
- Measure the resistance between the DSM C341B Pin 24, circuit GD905 (BK/GN), harness side and ground.



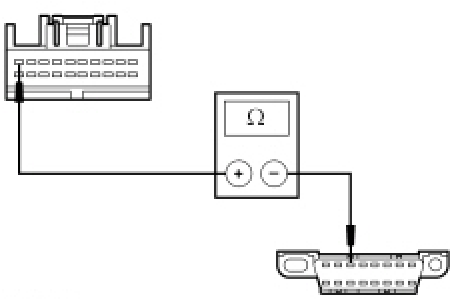
- N0068872

Are the resistances less than 5 ohms?

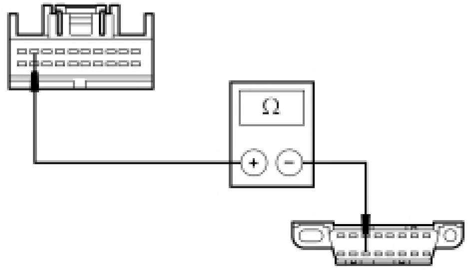
Yes	GO to P3.
No	REPAIR the circuit in question. CONNECT the negative battery cable. CLEAR the DTCs. REPEAT the network test with the scan tool.

P3 CHECK THE MS-CAN CIRCUITS BETWEEN THE DSM AND THE DLC FOR AN OPEN

- Disconnect: DSM C341D .
- Measure the resistance between the DSM C341D Pin 10, circuit VDB06 (GY/OG), harness side and the Data Link Connector (DLC) C251 Pin 3, circuit VDB06 (GY/OG), harness side.



- N0090369
- Measure the resistance between the DSM C341D Pin 9, circuit VDB07 (VT/OG), harness side and the DLC C251 Pin 11, circuit VDB07 (VT/OG), harness side.



- N0090370

Are the resistances less than 5 ohms?

Yes	CONNECT the negative battery cable. GO to P4.
No	REPAIR the circuit in question. CONNECT the negative battery cable. CLEAR the DTCs. REPEAT the network test with the scan tool.

P4 CHECK FOR CORRECT DSM OPERATION

- Disconnect all the DSM connectors.
- Check for:
  - corrosion
  - damaged pins
  - pushed-out pins
- Connect all the DSM connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new DSM . REFER to Section 501-10. REPEAT the network test with the scan tool.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test Q: The Dual Climate Controlled Seat Module (DCSM) Does Not Respond To The Scan Tool

Refer to Wiring Diagrams Cell 14 , Module Communications Network for schematic and connector information.

Refer to Wiring Diagrams Cell 119 , Climate Controlled Seats for schematic and connector information.

Normal Operation

The Dual Climate Controlled Seat Module (DCSM) communicates with the scan tool through the Medium Speed Controller Area Network (MS-CAN) .

This pinpoint test is intended to diagnose the following:

- Fuse

- Wiring, terminals or connectors
- DCSM

PINPOINT TEST Q : THE DCSM DOES NOT RESPOND TO THE SCAN TOOL

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

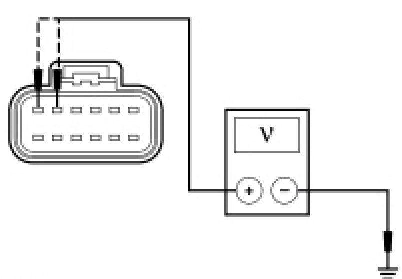
NOTE: *The air bag warning lamp illuminates when the Restraints Control Module (RCM) fuse is removed and the ignition switch is on. This is normal operation and does not indicate an SRS fault.*

NOTE: *The SRS must be fully operational and free of faults before releasing the vehicle to the customer.*

NOTE: *Failure to disconnect the battery when instructed results in false resistance readings. Refer to Section 414-01.*

Q1 CHECK THE DCSM VOLTAGE SUPPLY CIRCUITS FOR AN OPEN

- Ignition OFF.
- For vehicles with seat side air bags, carry out the following:
  - Depower the SRS . Refer to Section 501-20B.
  - Disconnect the driver seat side air bag module.
  - Disconnect the passenger seat side air bag module.
- **WARNING: Make sure no one is in the vehicle and there is nothing blocking or placed in front of any airbag when the battery is connected. Failure to follow these instructions may result in serious personal injury in the event of an accidental deployment.**

  Connect the battery ground cable. Refer to Section 414-01.
- Disconnect: DCSM C3265A .
- Ignition ON.
- Measure the voltage between the DCSM C3265A Pin E, circuit CBX01 (GN/RD), harness side and ground; and between the DCSM C3265A Pin F, circuit CBX01 (GN/RD), harness side and ground.
- 

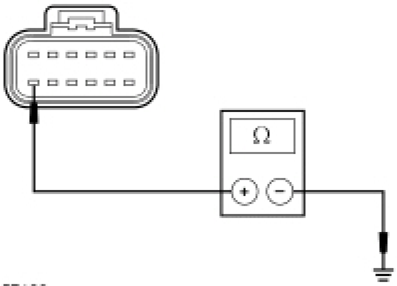
N0084105

Are the voltages greater than 10 volts?

Yes	GO to Q2.
No	VERIFY the Battery Junction Box (BJB) fuse 69 (30A) is OK. If OK, REPAIR the circuit in question. If not OK, REFER to the Wiring Diagrams manual to identify the possible causes of the circuit short. CLEAR the DTCs. REPEAT the network test with the scan tool. REPOWER the SRS . REFER to Section 501-20B. CLEAR the DTCs. REPEAT the network test with the scan tool.

Q2 CHECK THE DCSM GROUND CIRCUIT FOR AN OPEN

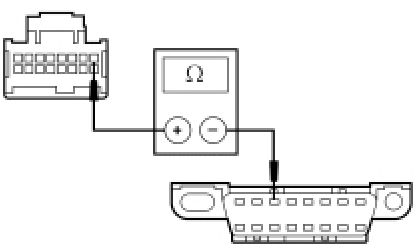
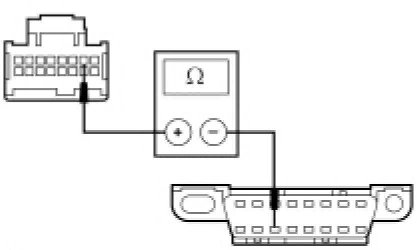
- Ignition OFF.

- Disconnect: Negative Battery Cable .
- Measure the resistance between the DCSM C3265A Pin M, circuit GD903 (BK/WH), harness side and ground.
- 
N0057120

Is the resistance less than 5 ohms?

Yes	GO to Q3.
No	REPAIR the circuit. CONNECT the negative battery cable. REPOWER the SRS . REFER to Section 501-20B. CLEAR the DTCs. REPEAT the network test with the scan tool.

Q3 CHECK THE MS-CAN CIRCUITS BETWEEN THE DCSM AND THE DLC FOR AN OPEN

- Disconnect: DCSM C3265C .
- Measure the resistance between the DCSM C3265C Pin 1, circuit VDB06 (GY/OG), harness side and the Data Link Connector (DLC) C251 Pin 3, circuit VDB06 (GY/OG), harness side.
- 
N0058709
- Measure the resistance between the DCSM C3265C Pin 2, circuit VDB07 (VT/OG), harness side and the DLC C251 Pin 11, circuit VDB07 (VT/OG), harness side.
- 
N0058710

Are the resistances less than 5 ohms?

Yes	CONNECT the negative battery cable. GO to Q4.

No	REPAIR the circuit in question. CONNECT the negative battery cable. REPOWER the SRS . REFER to Section 501-20B. CLEAR the DTCs. REPEAT the network test with the scan tool.

Q4 CHECK FOR CORRECT DCSM OPERATION

• Disconnect all the DCSM connectors.
• Check for:
  ▪ corrosion
  ▪ damaged pins
  ▪ pushed-out pins
• Connect all the DCSM connectors and make sure they seat correctly.
• Operate the system and verify the concern is still present.
 	
Is the concern still present?

Yes	INSTALL a new DCSM . REFER to Section 501-10. CLEAR the DTCs. REPEAT the network test with the scan tool.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test R: The Power Running Board (PRB) Module Does Not Respond To The Scan Tool

Refer to Wiring Diagrams Cell 14 , Module Communications Network for schematic and connector information.

Refer to Wiring Diagrams Cell 109 , Retractable Running Boards for schematic and connector information.

Normal Operation

The Power Running Board (PRB) module communicates with the scan tool through the Medium Speed Controller Area Network (MS-CAN) .

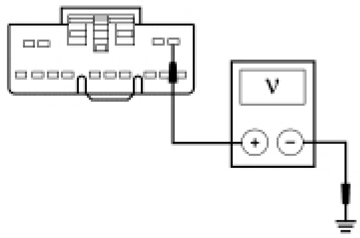
This pinpoint test is intended to diagnose the following:

- Fuse
- Wiring, terminals or connectors
- PRB module

PINPOINT TEST R : THE PRB MODULE DOES NOT RESPOND TO THE SCAN TOOL

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.
NOTE: *Failure to disconnect the battery when instructed results in false resistance readings. Refer to Section 414-01.*
R1 CHECK THE PRB MODULE VOLTAGE SUPPLY CIRCUIT FOR AN OPEN
• Ignition OFF. • Disconnect: PRB Module C3313B.

- Ignition ON.
- Measure the voltage between the PRB module C3313B Pin 1, circuit SBB11 (BU/RD), harness side and ground.



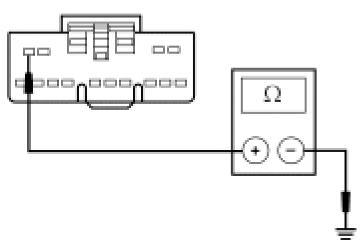
- N0060072

Is the voltage greater than 10 volts?

Yes	GO to R2.
No	VERIFY the Battery Junction Box (BJB) fuse 11 (30A) is OK. If OK, REPAIR the circuit. If not OK, REFER to the Wiring Diagrams manual to identify the possible causes of the circuit short. CLEAR the DTCs. REPEAT the network test with the scan tool.

R2 CHECK THE PRB MODULE GROUND CIRCUIT FOR AN OPEN

- Ignition OFF.
- Disconnect: Negative Battery Cable .
- Measure the resistance between the PRB module C3313B Pin 4, circuit GD139 (BK/YE), harness side and ground.



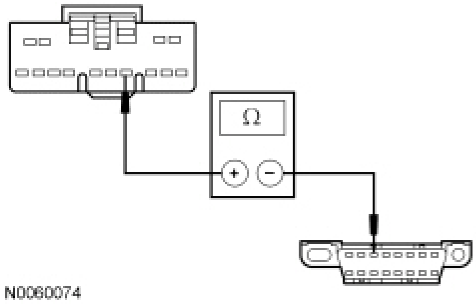
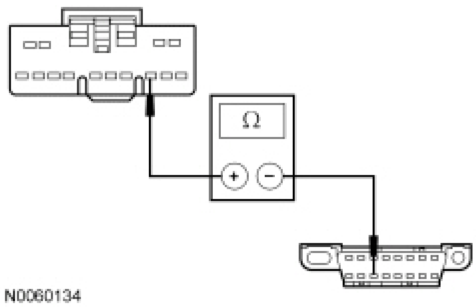
- N0060073

Is the resistance less than 5 ohms?

Yes	GO to R3.
No	REPAIR the circuit. CONNECT the negative battery cable. CLEAR the DTCs. REPEAT the network test with the scan tool.

R3 CHECK THE MS-CAN CIRCUITS BETWEEN THE PRB MODULE AND THE DLC FOR AN OPEN

- Measure the resistance between the PRB module C3313B Pin 8, circuit VDB06 (GY/OG), harness side and the Data Link Connector (DLC) C251 Pin 3, circuit VDB06 (GY/OG), harness side.

- 
  N0060074
- Measure the resistance between the PRB module C3313B Pin 7, circuit VDB07 (VT/OG), harness side and the DLC C251 Pin 11, circuit VDB07 (VT/OG), harness side.
- 
  N0060134

Are the resistances less than 5 ohms?

Yes	CONNECT the negative battery cable. GO to R4.
No	REPAIR the circuit in question. CONNECT the negative battery cable. CLEAR the DTCs. REPEAT the network test with the scan tool.

R4 CHECK FOR CORRECT PRB MODULE OPERATION

- Disconnect all the PRB module connectors.
- Check for:
  - corrosion
  - damaged pins
  - pushed-out pins
- Connect all the PRB module connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new PRB module. REFER to Section 501-08. CLEAR the DTCs. REPEAT the network test with the scan tool.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test S: The Parking Aid Module (PAM) Does Not Respond To The Scan Tool

Refer to Wiring Diagrams Cell 14 , Module Communications Network for schematic and connector information.

Refer to Wiring Diagrams Cell 131 , Parking Aid for schematic and connector information.

Normal Operation

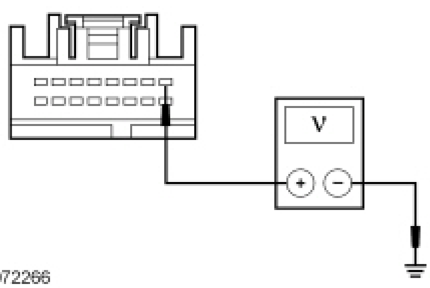
The Parking Aid Module (PAM) communicates with the scan tool through the High Speed Controller Area Network (HS-CAN) .

This pinpoint test is intended to diagnose the following:

- Fuse
- Wiring, terminals or connectors
- PAM

PINPOINT TEST S : THE PAM DOES NOT RESPOND TO THE SCAN TOOL

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.
NOTE: *Failure to disconnect the battery when instructed results in false resistance readings. Refer to Section 414-01.*
S1 CHECK THE PAM VOLTAGE SUPPLY CIRCUIT FOR AN OPEN

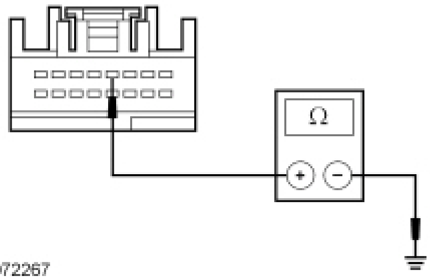
- Ignition OFF.
- Disconnect: PAM C2023 .
- Ignition ON.
- Measure the voltage between the PAM C2023 Pin 1, circuit CBP34 (VT/BN), harness side and ground.
- 

Is the voltage greater than 10 volts?

Yes	GO to S2.
No	VERIFY the Body Control Module (BCM) fuse 34 (10A) is OK. If OK, REPAIR the circuit. If not OK, REFER to the Wiring Diagrams manual to identify the possible causes of the circuit short. TEST the system for normal operation.

S2 CHECK THE PAM GROUND CIRCUIT FOR AN OPEN

- Ignition OFF.
- Disconnect: Negative Battery Cable .
- Measure the resistance between the PAM C2023 Pin 4, circuit GD134 (BK/VT), harness side and ground.



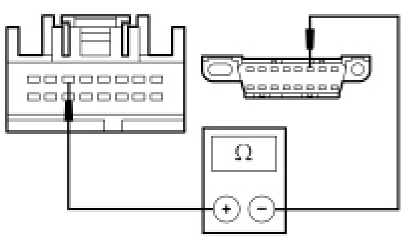
- A0072267

Is the resistance less than 5 ohms?

Yes	GO to S3.
No	REPAIR the circuit. CONNECT the negative battery cable. CLEAR the DTCs. REPEAT the network test with the scan tool.

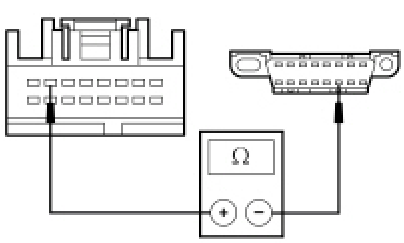
S3 CHECK THE HS-CAN CIRCUITS BETWEEN THE PAM AND THE DLC FOR AN OPEN

- Measure the resistance between the PAM C2023 Pin 6, circuit VDB04 (WH/BU), harness side and the Data Link Connector (DLC) C251 Pin 6, circuit VDB04 (WH/BU), harness side.



- N0120214

- Measure the resistance between the PAM C2023 Pin 7, circuit VDB05 (WH), harness side and the DLC C251 Pin 14, circuit VDB05 (WH), harness side.



- N0120213

Are the resistances less than 5 ohms?

Yes	CONNECT the negative battery cable. GO to S4.
No	REPAIR the circuit in question. CONNECT the negative battery cable. CLEAR the DTCs. REPEAT the network test with the scan tool.

S4 CHECK FOR CORRECT PAM OPERATION

- Disconnect the PAM connector.
- Check for:
  - corrosion
  - damaged pins
  - pushed-out pins
- Connect the PAM connector and make sure it seats correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new PAM . REFER to Section 413-13. CLEAR the DTCs. REPEAT the network test with the scan tool.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test T: The Global Positioning System Module (GPSM) Does Not Respond To The Scan Tool

Refer to Wiring Diagrams Cell 14 , Module Communications Network for schematic and connector information.

Refer to Wiring Diagrams Cell 130 , Audio System/Navigation for schematic and connector information.

Normal Operation

The Global Positioning System Module (GPSM) communicates with the scan tool through the Medium Speed Controller Area Network (MS-CAN) .

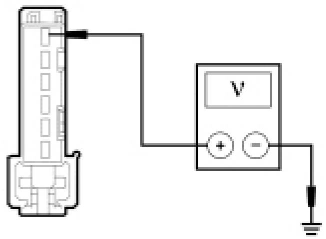
This pinpoint test is intended to diagnose the following:

- Fuse
- Wiring, terminals or connectors
- GPSM

PINPOINT TEST T : THE GPSM DOES NOT RESPOND TO THE SCAN TOOL

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.
NOTE: *Failure to disconnect the battery when instructed results in false resistance readings. Refer to Section 414-01.*
T1 CHECK THE GPSM VOLTAGE SUPPLY CIRCUIT FOR AN OPEN

- Ignition OFF.
- Disconnect: GPSM C2398 .
- Ignition ON.
- Measure the voltage between the GPSM C2398 Pin 1, circuit SBP09 (RD), harness side and ground.



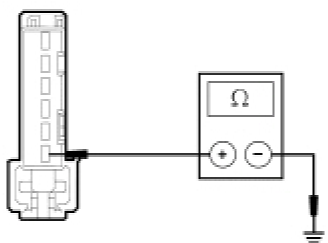
- N0094349

Is the voltage greater than 10 volts?

Yes	GO to T2.
No	VERIFY the Body Control Module (BCM) fuse 9 (10A) is OK. If OK, REPAIR the circuit. If not OK, REFER to the Wiring Diagrams manual to identify the possible causes of the circuit short. CLEAR the DTCs. REPEAT the network test with the scan tool.

T2 CHECK THE GPSM GROUND CIRCUIT FOR AN OPEN

- Ignition OFF.
- Disconnect: Negative Battery Cable .
- Measure the resistance between the GPSM C2398 Pin 6, circuit GD114 (BK/BU), harness side and ground.



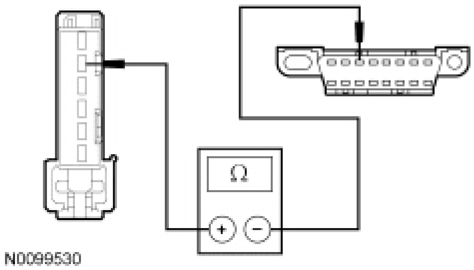
- N0094350

Is the resistance less than 5 ohms?

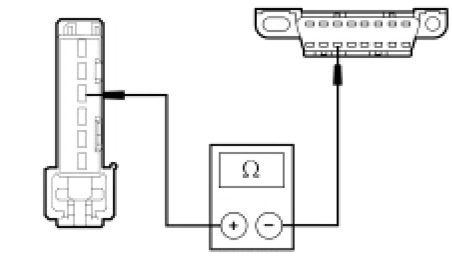
Yes	GO to T3.
No	REPAIR the circuit. CONNECT the negative battery cable. CLEAR the DTCs. REPEAT the network test with the scan tool.

T3 CHECK THE MS-CAN CIRCUITS BETWEEN THE GPSM AND THE DLC FOR AN OPEN

- Measure the resistance between the GPSM C2398 Pin 2, circuit VDB06 (GY/OG), harness side and the Data Link Connector (DLC) C251 Pin 3, circuit VDB06 (GY/OG), harness side.

- 
  N0099530

- Measure the resistance between the GPSM C2398 Pin 3, circuit VDB07 (VT/OG), harness side and the DLC C251 Pin 11, circuit VDB07 (VT/OG), harness side.

- 
  N0099531

Are the resistances less than 5 ohms?

Yes	CONNECT the negative battery cable. GO to T4.
No	REPAIR the circuit in question. CONNECT the negative battery cable. CLEAR the DTCs. REPEAT the network test with the scan tool.

T4 CHECK FOR CORRECT GPSM OPERATION

- Disconnect the GPSM connector.
- Check for:
  - corrosion
  - damaged pins
  - pushed-out pins
- Connect the GPSM connector and make sure it seats correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new GPSM . Refer to the appropriate section in Group 415 for the procedure.. CLEAR the DTCs. REPEAT the network test with the scan tool.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test U: The Accessory Protocol Interface Module (APIM) Does Not Respond To The Scan Tool

Refer to Wiring Diagrams Cell 14 , Module Communications Network for schematic and connector information.

Refer to Wiring Diagrams Cell 130 , Audio System/Navigation for schematic and connector information.

Normal Operation

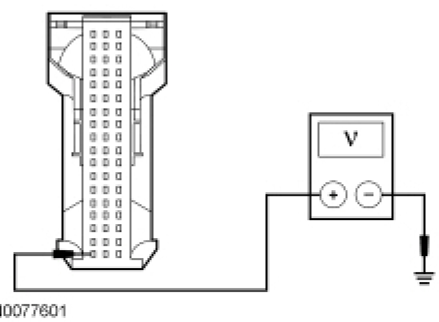
The Accessory Protocol Interface Module (APIM) communicates with the scan tool through the High Speed Controller Area Network (HS-CAN) . For vehicles equipped with 4.2-inch screen and 8-inch touchscreen, the APIM communicates on the Infotainment Controller Area Network (I-CAN) .

This pinpoint test is intended to diagnose the following:

- Fuse
- Wiring, terminals or connectors
- APIM

PINPOINT TEST U : THE APIM DOES NOT RESPOND TO THE SCAN TOOL

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.
NOTE: *Failure to disconnect the battery when instructed results in false resistance readings. Refer to Section 414-01.*
U1 CHECK THE APIM VOLTAGE SUPPLY CIRCUIT FOR AN OPEN

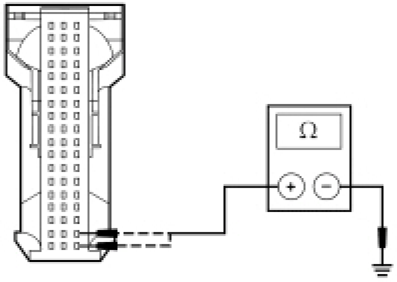
- Disconnect: APIM C2383 .
- Ignition ON.
- Measure the voltage between the APIM C2383 Pin 1, circuit SBP02 (YE/RD), harness side and ground.
- 

Is the voltage greater than 10 volts?

Yes	GO to U2.
No	VERIFY the Body Control Module (BCM) fuse 2 (15A) is OK. If OK, REPAIR the circuit. If not OK, REFER to the Wiring Diagrams manual to identify the possible causes of the circuit short. CLEAR the DTCs. REPEAT the network test with the scan tool.

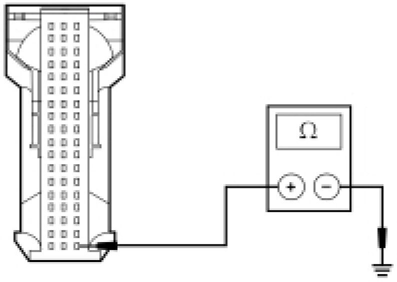
U2 CHECK THE APIM GROUND CIRCUITS FOR AN OPEN

- Ignition OFF.
- Disconnect: Negative Battery Cable .
- For 4.2-inch display, measure the resistance between the APIM C2383 Pin 37, circuit GD114 (BK/BU), harness side and ground; and between the APIM C2383 Pin 38, circuit GD114 (BK/BU), harness side and ground.



- N0077602

- For 8-inch display, measure the resistance between the APIM C2383 Pin 37, circuit GD114 (BK/BU), harness side and ground.



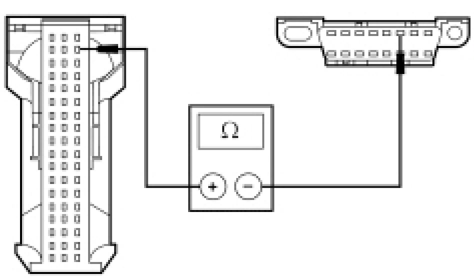
- N0152486

Are the resistances less than 5 ohms?

Yes	CONNECT the negative battery cable. GO to U3.
No	REPAIR the circuit in question. CLEAR the DTCs. REPEAT the network test with the scan tool.

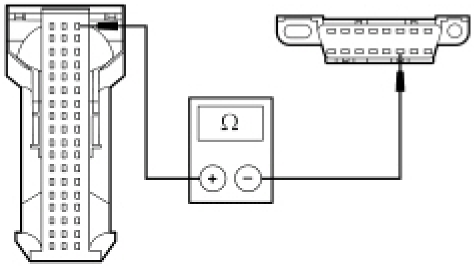
U3 CHECK THE HS-CAN CIRCUITS BETWEEN THE APIM AND THE DLC FOR AN OPEN

- Measure the resistance between the APIM C2383 Pin 53, circuit VDB04 (WH/BU), harness side and the Data Link Connector (DLC) C251 Pin 6, circuit VDB04 (WH/BU), harness side.



- N0077603

- Measure the resistance between the APIM C2383 Pin 54, circuit VDB05 (WH), harness side and the DLC C251 Pin 14, circuit VDB05 (WH), harness side.

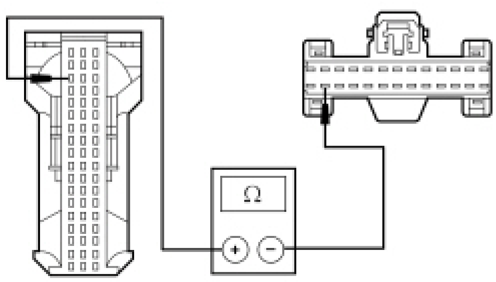
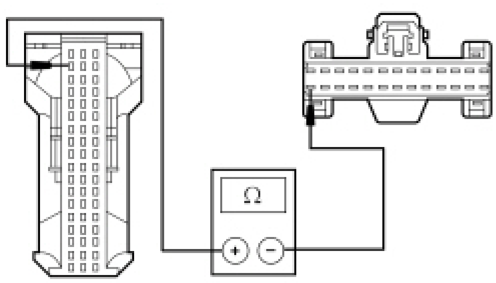
- 
  N0077604

Are the resistances less than 5 ohms?

Yes	If equipped with 4.2-inch screen, GO to U4. If equipped with 8-inch touchscreen, GO to U5.
No	REPAIR the circuit in question. CONNECT the negative battery cable. CLEAR the DTCs. REPEAT the network test with the scan tool.

U4 CHECK THE I-CAN CIRCUITS BETWEEN THE APIM AND THE IPC FOR AN OPEN

NOTE: *For vehicles equipped with 4.2-inch display.*

- Disconnect: IPC C220 .
- Measure the resistance between the APIM C2383 Pin 16, circuit VDB13 (BU/GY), harness side and the Instrument Panel Cluster (IPC) C220 Pin 25, circuit VDB13 (BU/GY), harness side.
- 
  N0150492
- Measure the resistance between the APIM C2383 Pin 17, circuit VDB14 (VT/GY), harness side and the IPC C220 Pin 26, circuit VDB14 (VT/GY), harness side.
- 
  N0150493

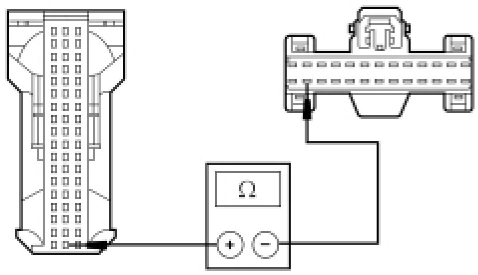
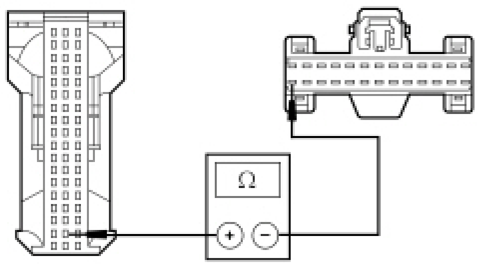
Are the resistances less than 5 ohms?

Yes	GO to U7.
No	REPAIR the circuit in question. CONNECT the IPC and negative battery cable. CLEAR the DTCs.

	REPEAT the network test with the scan tool.

U5 CHECK THE I-CAN CIRCUITS BETWEEN THE APIM AND THE IPC FOR AN OPEN

NOTE: *For vehicles equipped with 8-inch display.*

- Disconnect: IPC C220 .
- Measure the resistance between the APIM C2383 Pin 19, circuit VDB13 (BU/GY), harness side and the Instrument Panel Cluster (IPC) C220 Pin 25, circuit VDB13 (BU/GY), harness side.
- 
  N0150494
- Measure the resistance between the APIM C2383 Pin 20, circuit VDB14 (VT/GY), harness side and the IPC C220 Pin 26, circuit VDB14 (VT/GY), harness side.
- 
  N0150495

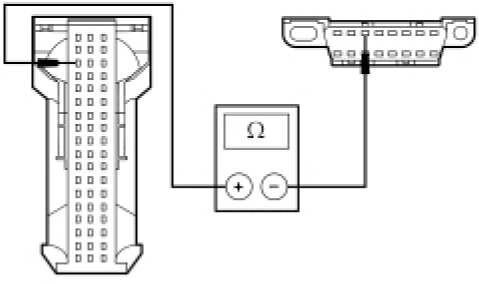
Are the resistances less than 5 ohms?

Yes	GO to U6.
No	REPAIR the circuit in question. CONNECT the IPC and negative battery cable. CLEAR the DTCs. REPEAT the network test with the scan tool.

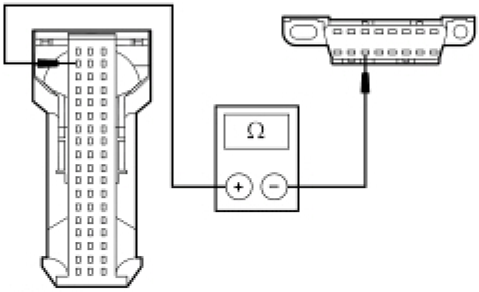
U6 CHECK THE MS-CAN CIRCUITS BETWEEN THE APIM AND THE DLC FOR AN OPEN

NOTE: *For vehicles equipped with 8-inch touchscreen.*

- Measure the resistance between the APIM C2383 Pin 16, circuit VDB06 (GY/OG), harness side and the DLC C251 Pin 3, circuit VDB06 (GY/OG), harness side.

- 
  N0077605

- Measure the resistance between the APIM C2383 Pin 17, circuit VDB07 (VT/OG), harness side and the DLC C251 Pin 11, circuit VDB07 (VT/OG), harness side.

- 
  N0077606

Are the resistances less than 5 ohms?

Yes	GO to U7.
No	REPAIR the circuit in question. CONNECT the negative battery cable. CLEAR the DTCs. REPEAT the network test with the scan tool.

U7 CHECK FOR CORRECT APIM OPERATION

- Disconnect the APIM connector.
- Check for:
  - corrosion
  - damaged pins
  - pushed-out pins
- Connect the APIM connector and make sure it seats correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	PERFORM the APIM Hardware Test. INSTALL a new APIM. Refer to the appropriate Removal and Installation procedure in Section 415-00.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test V: The Steering Column Control Module (SCCM) Does Not Respond To The Scan Tool

Refer to Wiring Diagrams Cell 14 , Module Communications Network for schematic and connector information.

Refer to Wiring Diagrams Cell 44 , Horn/Cigar Lighter for schematic and connector information.

Normal Operation

The Steering Column Control Module (SCCM) communicates with the scan tool through the High Speed Controller Area Network (HS-CAN) .

This pinpoint test is intended to diagnose the following:

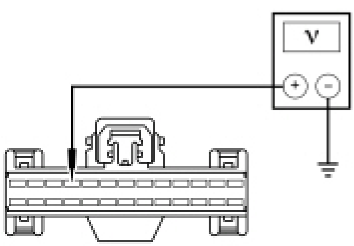
- Fuse
- Wiring, terminals or connectors
- SCCM

PINPOINT TEST V : THE SCCM DOES NOT RESPOND TO THE SCAN TOOL

NOTE: *Failure to disconnect the battery when instructed results in false resistance readings. Refer to Section 414-01.*

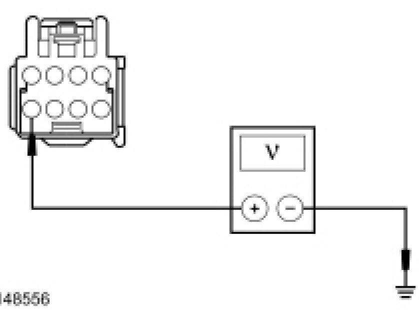
V1 CHECK THE SCCM VOLTAGE SUPPLY CIRCUIT FOR AN OPEN

- Disconnect: SCCM C2414A and C2414B.
- Ignition ON.
- Measure the voltage between the SCCM C2414A Pin 10, circuit SBP23 (WH/RD), harness side and ground.



- N0114048

- Measure the voltage between the SCCM C2414B Pin 8, circuit SBP24 (VT/RD), harness side and ground.



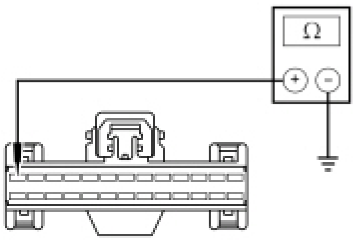
- N0148556

Are the voltages greater than 10 volts?

Yes	GO to V2.
No	VERIFY the BCM fuse 23 (15A) or BCM fuse 24 (15A) is OK. If OK, REPAIR the circuit in question. If not OK, REFER to the Wiring Diagrams manual to identify the possible causes of the circuit short. CLEAR the DTCs. REPEAT the network test with the scan tool.

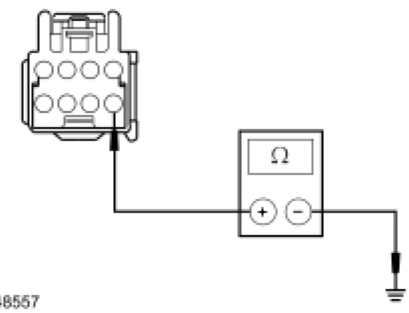
V2 CHECK THE SCCM GROUND CIRCUIT FOR AN OPEN

- Ignition OFF.
- Disconnect: Negative Battery Cable .
- Measure the resistance between the SCCM C2414A Pin 13, circuit GD133 (BK), harness side and ground.



- N0114050

- Measure the resistance between the SCCM C2414B Pin 5, circuit GD138 (BK/WH), harness side and ground.



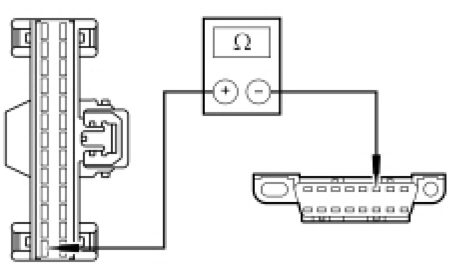
- N0148557

Are the resistances less than 5 ohms?

Yes	GO to V3.
No	REPAIR the circuit in question. CONNECT the negative battery cable. CLEAR the DTCs. REPEAT the network test with the scan tool.

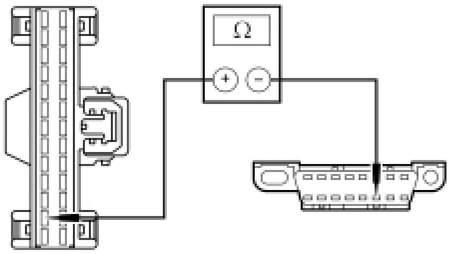
V3 CHECK THE HS-CAN CIRCUITS BETWEEN THE SCCM AND THE DLC FOR AN OPEN

- Measure the resistance between the SCCM C2414A Pin 14, circuit VDB04 (WH/BU), harness side and the Data Link Connector (DLC) C251 Pin 6, circuit VDB04 (WH/BU), harness side.



- N0114053

- Measure the resistance between the SCCM C2414A Pin 15, circuit VDB05 (WH), harness side and the DLC C251 Pin 14, circuit VDB05 (WH), harness side.



Are the resistances less than 5 ohms?

Yes	GO to V4.
No	REPAIR the circuit in question. CONNECT the negative battery cable. CLEAR the DTCs. REPEAT the network test with the scan tool.

V4 CHECK FOR CORRECT SCCM OPERATION

- Disconnect all the SCCM connectors.
- Check for:
  - corrosion
  - damaged pins
  - pushed-out pins
- Connect all the SCCM connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new SCCM . REFER to Section 211-05. CLEAR the DTCs. REPEAT the network test with the scan tool.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test W: The Tire Pressure Monitor (TPM) Module Does Not Respond To The Scan Tool

Refer to Wiring Diagrams Cell 14 , Module Communications Network for schematic and connector information.

Refer to Wiring Diagrams Cell 118 , Tire Pressure Monitor System for schematic and connector information.

NOTE: *The TPM module is also known as the Radio Transceiver Module (RTM).*

Normal Operation

The Tire Pressure Monitor (TPM) module communicates with the scan tool through the Medium Speed Controller Area Network (MS-CAN) .

This pinpoint test is intended to diagnose the following:

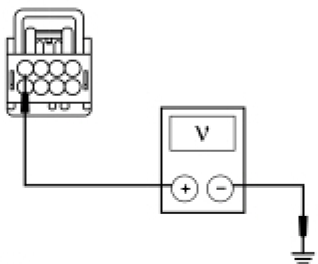
- Fuse
- Wiring, terminals or connectors
- TPM module

PINPOINT TEST W : THE TPM MODULE DOES NOT RESPOND TO THE SCAN TOOL

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

NOTE: *Failure to disconnect the battery when instructed results in false resistance readings. Refer to Section 414-01.*

W1 CHECK THE TPM MODULE VOLTAGE SUPPLY CIRCUIT FOR AN OPEN

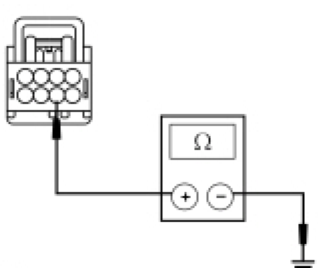
- Ignition OFF.
- Disconnect: TPM C2415 .
- Ignition ON.
- Measure the voltage between the TPM module C2415 Pin 4, circuit SBP26 (YE/RD), harness side and ground.
- 

N0119434

Is the voltage greater than 10 volts?

Yes	GO to W2.
No	VERIFY the Body Control Module (BCM) fuse 26 (5A) is OK. If OK, REPAIR the circuit. If not OK, REFER to the Wiring Diagrams manual to identify the possible causes of the circuit short. CLEAR the DTCs. REPEAT the network test with the scan tool.

W2 CHECK THE TPM MODULE GROUND CIRCUITS FOR AN OPEN

- Ignition OFF.
- Disconnect: Negative Battery Cable .
- Measure the resistance between the TPM module C2415 Pin 6, circuit GD133 (BK), harness side and ground.
- 

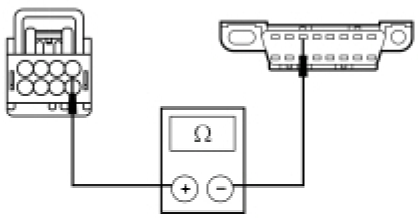
N0119450

Is the resistance less than 5 ohms?

Yes	CONNECT the negative battery cable. GO to W3.
No	REPAIR the circuit. CLEAR the DTCs. REPEAT the network test with the scan tool.

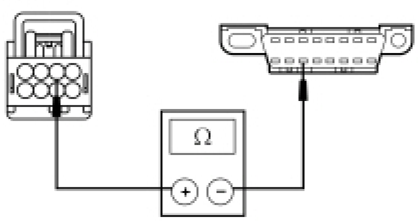
W3 CHECK THE MS-CAN CIRCUITS BETWEEN THE TPM MODULE AND THE DLC FOR AN OPEN

- Measure the resistance between the TPM module C2415 Pin 1, circuit VDB06 (GY/OG), harness side and the DLC C251 Pin 3, circuit VDB06 (GY/OG), harness side.



- N0119451

- Measure the resistance between the TPM module C2415 Pin 2, circuit VDB07 (VT/OG), harness side and the DLC C251 Pin 11, circuit VDB07 (VT/OG), harness side.



- N0119452

Are the resistances less than 5 ohms?

Yes	CONNECT the negative battery cable. GO to W4.
No	REPAIR the circuit in question. CONNECT the negative battery cable. CLEAR the DTCs. REPEAT the network test with the scan tool.

W4 CHECK FOR CORRECT TPM MODULE OPERATION

- Disconnect the TPM module connector.
- Check for:
  - corrosion
  - damaged pins
  - pushed-out pins
- Connect the TPM module connector and make sure it seats correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new TPM module. REFER to Section 204-04. CLEAR the DTCs. REPEAT the network test with the scan tool.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. CLEAR the DTCs. REPEAT the network test with the scan tool.

Pinpoint Test X: No Medium Speed Controller Area Network (MS-CAN) Communication, All Modules Are Not Responding

Refer to Wiring Diagrams Cell 14 , Module Communications Network for schematic and connector information.

NOTE: *The TPM module is also known as the Radio Transceiver Module (RTM).*

Normal Operation

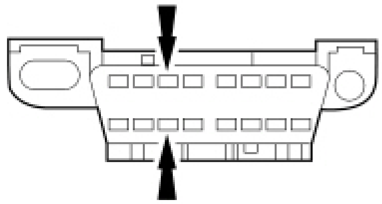
The Medium Speed Controller Area Network (MS-CAN) uses an unshielded twisted pair cable.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Accessory Protocol Interface Module (APIM) (if equipped)
- Audio Front Control Module (ACM)
- Body Control Module (BCM)
- Driver Seat Module (DSM) (if equipped)
- Dual Climate Controlled Seat Module (DCSM) (if equipped)
- Front Controls Interface Module (FCIM) (if equipped)
- Front Display Interface Module (FDIM) (if equipped)
- Global Positioning System Module (GPSM) (if equipped)
- HVAC module
- Power Running Board (PRB) module (if equipped)
- Tire Pressure Monitor (TPM) module (Radio Transceiver Module [RTM])

PINPOINT TEST X : NO MS-CAN COMMUNICATION, ALL MODULES ARE NOT RESPONDING

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.
NOTE: *Most faults are due to connector and/or wiring concerns. Carry out a thorough inspection and verification before proceeding with the pinpoint test.*
NOTE: *Failure to disconnect the battery when instructed results in false resistance readings. Refer to Section 414-01.*
X1 CHECK THE DLC PINS FOR DAMAGE
• Ignition OFF. • Disconnect the scan tool cable from the Data Link Connector (DLC) . • Inspect DLC pins 3 and 11 for damage.



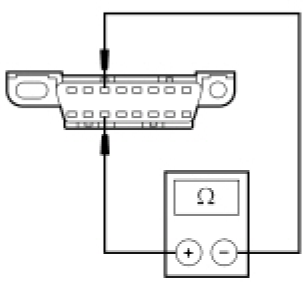
- N0053178

Are DLC pins 3 and 11 OK?

Yes	GO to X2.
No	REPAIR the DLC as necessary. CLEAR the DTCs. REPEAT the network test with the scan tool.

X2 CHECK THE MS-CAN TERMINATION RESISTANCE

- Ignition OFF.
- Disconnect: Negative Battery Cable .
- Measure the resistance between the DLC C251 Pin 3, circuit VDB06 (GY/OG), harness side and the DLC C251 Pin 11, circuit VDB07 (VT/OG), harness side.



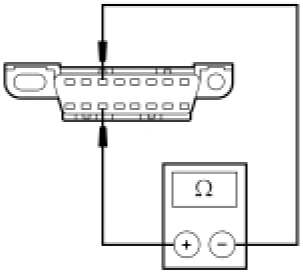
- N0050701

Is the resistance between 54 and 66 ohms?

Yes	GO to X11.
No	GO to X3.

X3 CHECK THE MS-CAN TERMINATION RESISTOR

- Measure the resistance between the DLC C251 Pin 3, circuit VDB06 (GY/OG), harness side and the DLC C251 Pin 11, circuit VDB07 (VT/OG), harness side.



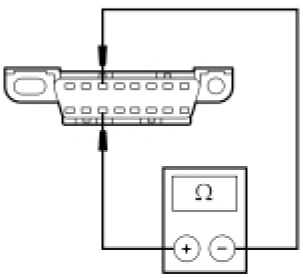
- N0050701

Is the resistance between 108 and 132 ohms?

Yes	GO to X4.
No	GO to X8.

X4 CHECK THE MS-CAN TERMINATION RESISTOR WITH THE BCM DISCONNECTED

- Disconnect: Body Control Module (BCM) C2280B .
- Measure the resistance between the DLC C251 Pin 3, circuit VDB06 (GY/OG), harness side and the DLC C251 Pin 11, circuit VDB07 (VT/OG), harness side.



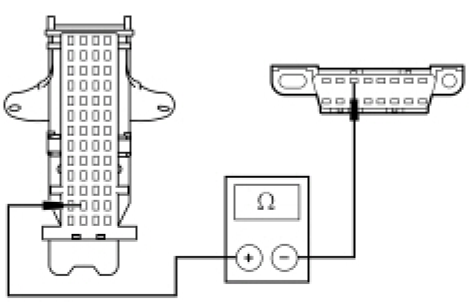
- N0050701

Is the resistance between 108 and 132 ohms?

Yes	GO to X5.
No	GO to X6.

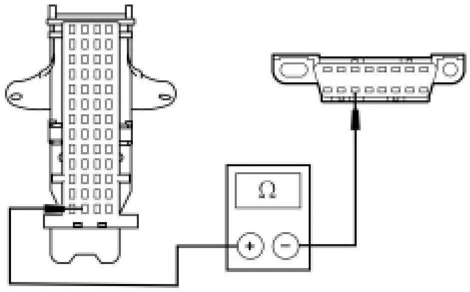
X5 CHECK THE MS-CAN CIRCUITS BETWEEN THE BCM AND THE DLC FOR AN OPEN

- Measure the resistance between the BCM C2280B Pin 38, circuit VDB06 (GY/OG), harness side and the DLC C251 Pin 3, circuit VDB06 (GY/OG), harness side.



- N0114057

- Measure the resistance between the BCM C2280B Pin 39, circuit VDB07 (VT/OG), harness side and the DLC C251 Pin 11, circuit VDB07 (VT/OG), harness side.



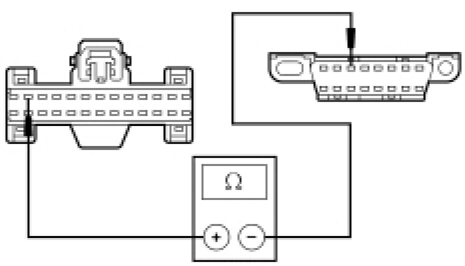
- N0114058

Are the resistances less than 5 ohms?

Yes	CONNECT the negative battery cable. GO to X29.
No	REPAIR the circuit in question. CONNECT the negative battery cable. CLEAR the DTCs. REPEAT the network test with the scan tool.

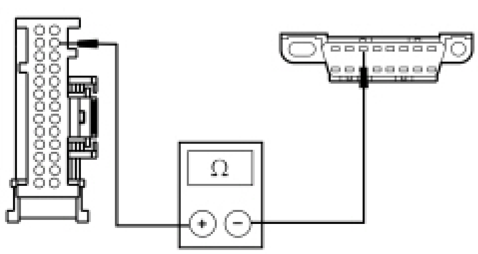
X6 CHECK THE MS-CAN (+) CIRCUIT BETWEEN THE HVAC MODULE AND THE DLC FOR AN OPEN

- For EMTC , measure the resistance between the HVAC module C294A Pin 12, circuit VDB06 (GY/OG), harness side and the DLC C251 Pin 3, circuit VDB06 (GY/OG), harness side.



- N0104911

- For EATC , measure the resistance between the HVAC module C228A Pin 25, circuit VDB06 (GY/OG), harness side and the DLC C251 Pin 3, circuit VDB06 (GY/OG), harness side.



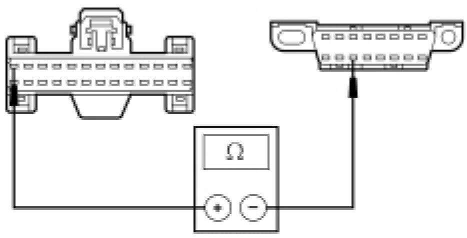
- N0148816

Is the resistance less than 5 ohms?

Yes	CONNECT the negative battery cable. GO to X7.
No	REPAIR the circuit. CONNECT the negative battery cable. CLEAR the DTCs. REPEAT the network test with the scan tool.

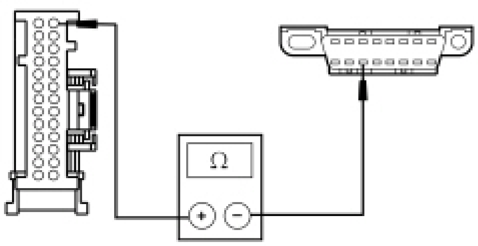
X7 CHECK THE MS-CAN (-) CIRCUIT BETWEEN THE HVAC MODULE AND THE DLC FOR AN OPEN

- For EMTC , measure the resistance between the HVAC module C294A Pin 13, circuit VDB07 (VT/OG), harness side and the DLC C251 Pin 11, circuit VDB07 (VT/OG), harness side.



- N0104912

- For EATC , measure the resistance between the HVAC module C228A Pin 26, circuit VDB07 (VT/OG), harness side and the DLC C251 Pin 11, circuit VDB07 (VT/OG), harness side.



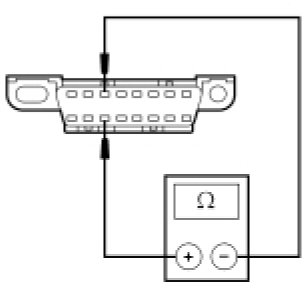
- N0148817

Is the resistance less than 5 ohms?

Yes	CONNECT the negative battery cable. GO to X35.
No	REPAIR the circuit. CONNECT the negative battery cable. CLEAR the DTCs. REPEAT the network test with the scan tool.

X8 CHECK THE MS-CAN (+) AND MS-CAN (-) CIRCUITS FOR A SHORT TOGETHER

- Measure the resistance between the DLC C251 Pin 3, circuit VDB06 (GY/OG), harness side and the DLC C251 Pin 11, circuit VDB07 (VT/OG), harness side.



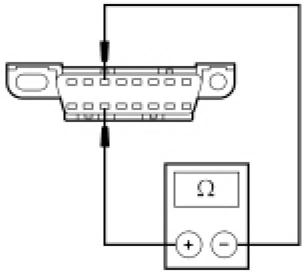
- N0050701

Is the resistance less than 5 ohms?

Yes	GO to X10.

No	GO to X9.

X9 CHECK THE MS-CAN (+) AND MS-CAN (-) CIRCUITS FOR AN OPEN

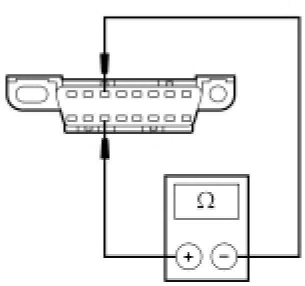
- Measure the resistance between the DLC C251 Pin 3, circuit VDB06 (GY/OG), harness side and the DLC C251 Pin 11, circuit VDB07 (VT/OG), harness side.
- 

N0050701

Is the resistance greater than 10,000 ohms?

Yes	REPAIR the DLC or the circuit in question. CONNECT all modules. CLEAR the DTCs. REPEAT the network test with the scan tool.
No	A capacitor internal to a module may still be draining causing irregular resistance readings. WAIT 5 minutes. REPEAT the pinpoint test.

X10 CHECK THE MS-CAN (+) AND MS-CAN (-) CIRCUITS FOR A SHORT TOGETHER WITH MODULES DISCONNECTED

- While measuring the resistance between the DLC C251 Pin 3, circuit VDB06 (GY/OG), harness side and the DLC C251 Pin 11, circuit VDB07 (VT/OG), harness side, disconnect the following modules one at a time until the resistance is greater than 5 ohms.
  - APIM C2383 (if equipped)
  - ACM C290B (if equipped)
  - BCM C2280B
  - DSM C341D (if equipped)
  - DCSM C3265C (if equipped)
  - FCIM C2402 (if equipped)
  - FDIM C2407 (if equipped)
  - GPSM C2398 (if equipped)
  - HVAC module C294A (EMTC) or C228A (EATC)
  - PRB module C3313B (if equipped)
  - TPM module (Radio Transceiver Module [RTM]) C2415
- 

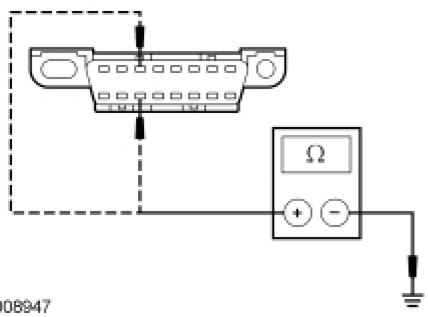
N0050701

Is the resistance less than 5 ohms?

Yes	CONNECT the negative battery cable. For the APIM , GO to X27. For the ACM , GO to X28. For the BCM , GO to X29. For the DSM , GO to X30. For the DCSM , GO to X31. For the FCIM , GO to X32. For the FDIM , GO to X33. For the GPSM , GO to X34. For the HVAC module, GO to X35. For the PRB module, GO to X36. For the TPM module (Radio Transceiver Module [RTM]), GO to X37.
No	REPAIR circuits VDB06 (GY/OG) and VDB07 (VT/OG) for being shorted together. CONNECT all modules. CONNECT the negative battery cable. CLEAR the DTCs. REPEAT the Network Test with the scan tool.

X11 CHECK THE MS-CAN (+) AND MS-CAN (-) CIRCUITS FOR A SHORT TO GROUND

- Measure the resistance between the DLC C251 Pin 3, circuit VDB06 (GY/OG), harness side and ground; and between the DLC C251 Pin 11, circuit VDB07 (VT/OG), harness side and ground.



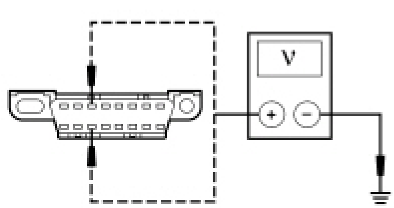
- N0008947

Are the resistances greater than 1,000 ohms?

Yes	CONNECT the negative battery cable. GO to X12.
No	GO to X13.

X12 CHECK THE MS-CAN (+) AND MS-CAN (-) CIRCUITS FOR A SHORT TO VOLTAGE

- Ignition ON.
- Measure the voltage between the DLC C251 Pin 3, circuit VDB06 (GY/OG), harness side and ground; and between the DLC C251 Pin 11, circuit VDB07 (VT/OG), harness side and ground.



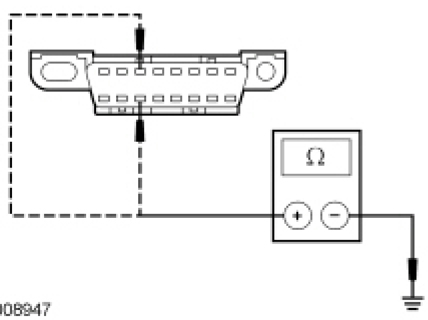
- N0050702

Is the voltage greater than 6 volts?

Yes	REPAIR the circuit. CLEAR the DTCs. REPEAT the network test with the scan tool.
No	GO to X14.

X13 CHECK THE MS-CAN (+) AND MS-CAN (-) CIRCUITS FOR A SHORT TO GROUND WITH MODULES DISCONNECTED

- Measure the resistance between the DLC C251 Pin 3, circuit VDB06 (GY/OG), harness side and ground; and between the DLC C251 Pin 11, circuit VDB07 (VT/OG), harness side and ground. Check for the short to ground on both circuits after disconnecting each module. Disconnect the following modules one at a time until the resistance to ground is greater than 1,000 ohms.
  - APIM C2383 (if equipped)
  - ACM C290B (if equipped)
  - BCM C2280B
  - DSM C341D (if equipped)
  - DCSM C3265C (if equipped)
  - FCIM C2402 (if equipped)
  - FDIM C2407 (if equipped)
  - GPSM C2398 (if equipped)
  - HVAC module C294A (EMTC) or C228A (EATC)
  - PRB module C3313B (if equipped)
  - TPM module (Radio Transceiver Module [RTM]) C2415



- N0008947

Is the resistance greater than 1,000 ohms?

Yes	CONNECT the negative battery cable. For the APIM , GO to X27. For the ACM , GO to X28. For the BCM , GO to X29. For the DSM , GO to X30. For the DCSM , GO to X31. For the FCIM , GO to X32. For the FDIM , GO to X33. For the GPSM , GO to X34. For the HVAC module, GO to X35. For the PRB module, GO to X36. For the TPM module (Radio Transceiver Module [RTM]), GO to X37.
No	REPAIR the circuit in question. CLEAR the DTCs. REPEAT the Network Test with the scan tool.

X14 CHECK FOR RESTORED NETWORK COMMUNICATION WITH THE BCM DISABLED

- Disconnect: BCM Fuse 45 (10A) .

- Enter the following diagnostic mode on the scan tool: Network Test .
- Repeat the network test.

Do all other MS-CAN modules pass the network test?

Yes	INSTALL the removed fuse. GO to Pinpoint Test I.
No	INSTALL the removed fuse. GO to X15.

X15 CHECK FOR RESTORED NETWORK COMMUNICATION WITH THE ACM DISABLED

- Disconnect: BCM Fuse 29 (20A) .
- Enter the following diagnostic mode on the scan tool: Network Test .
- Repeat the network test.

Do all other MS-CAN modules pass the network test?

Yes	INSTALL the removed fuse. GO to Pinpoint Test K.
No	INSTALL the removed fuse. GO to X16.

X16 CHECK FOR RESTORED NETWORK COMMUNICATION WITH THE FCIM , FDIM AND GPSM DISABLED

- Disconnect: BCM Fuse 9 (10A) .
- Enter the following diagnostic mode on the scan tool: Network Test .
- Repeat the network test.

Do all other MS-CAN modules pass the network test?

Yes	INSTALL the removed fuse. GO to X17.
No	INSTALL the removed fuse. GO to X18.

X17 CHECK FOR RESTORED NETWORK COMMUNICATION WITH THE FCIM , FDIM AND GPSM DISCONNECTED

- Disconnect modules one at a time until other MS-CAN modules pass the network test. Skip to the end once network operation is restored.
- Disconnect: GPSM C2398 (if equipped) .
- Enter the following diagnostic mode on the scan tool: Network Test .
- Disconnect: FDIM C2407 (if equipped) .
- Enter the following diagnostic mode on the scan tool: Network Test .

- Repeat the network test.
- Disconnect: FCIM C2402 (if equipped) .
- Enter the following diagnostic mode on the scan tool: Network Test .

Do all other MS-CAN modules pass the network test?

Yes	CONNECT all modules. For restored operation with the FCIM disconnected, GO to Pinpoint Test M. For restored operation with the FDIM disconnected, GO to Pinpoint Test N. For restored operation with the GPSM disconnected, GO to Pinpoint Test T.
No	CONNECT all modules. The MS-CAN tests within specification. VERIFY the correct operation of the scan tool on a known good vehicle.

X18 CHECK FOR RESTORED NETWORK COMMUNICATION WITH THE HVAC MODULE DISABLED

- Disconnect: BCM Fuse 46 (10A) .
- Enter the following diagnostic mode on the scan tool: Network Test .
- Repeat the network test.

Do all other MS-CAN modules pass the network test?

Yes	INSTALL the removed fuse. GO to Pinpoint Test J.
No	INSTALL the removed fuse. If the vehicle is equipped with an Accessory Protocol Interface Module (APIM) , GO to X19. If the vehicle is not equipped with an APIM , GO to X20.

X19 CHECK FOR RESTORED NETWORK COMMUNICATION WITH THE APIM DISABLED

- Disconnect: BCM Fuse 2 (15A) .
- Enter the following diagnostic mode on the scan tool: Network Test .
- Repeat the network test.

Do all other MS-CAN modules pass the network test?

Yes	INSTALL the removed fuse. GO to Pinpoint Test U.
No	INSTALL the removed fuse. GO to X20.

X20 VERIFY VEHICLE EQUIPMENT — PRB MODULE

- Verify whether the vehicle is equipped with a Power Running Board (PRB) module.

Is the vehicle equipped with a PRB module?

Yes	GO to X21.
No	GO to X22.

X21 CHECK FOR RESTORED NETWORK COMMUNICATION WITH THE PRB MODULE DISABLED

- Disconnect: Battery Junction Box (BJB) Fuse 11 (30A) .
- Enter the following diagnostic mode on the scan tool: Network Test .
- Repeat the network test.

Do all other MS-CAN modules pass the network test?

Yes	INSTALL the removed fuse. GO to Pinpoint Test R.
No	INSTALL the removed fuse. GO to X22.

X22 VERIFY VEHICLE EQUIPMENT — DSM

- Verify whether the vehicle is equipped with a Driver Seat Module (DSM) .

Is the vehicle equipped with a DSM ?

Yes	GO to X23.
No	GO to X24.

X23 CHECK FOR RESTORED NETWORK COMMUNICATION WITH THE DSM DISABLED

- Disconnect: BCM Fuse 74 (30A) .
- Enter the following diagnostic mode on the scan tool: Network Test .
- Repeat the network test.

Do all other MS-CAN modules pass the network test?

Yes	INSTALL the removed fuse. GO to Pinpoint Test P.
No	INSTALL the removed fuse. GO to X24.

X24 VERIFY VEHICLE EQUIPMENT — DCSM

- Verify whether the vehicle is equipped with a Dual Climate Controlled Seat Module (DCSM) .

Is the vehicle equipped with a DCSM ?

Yes	GO to X25.
No	GO to X26.

X25 CHECK FOR RESTORED NETWORK COMMUNICATION WITH THE DCSM DISABLED

- Disconnect: BJB Fuse 69 (30A) .
- Enter the following diagnostic mode on the scan tool: Network Test .
- Repeat the network test.

Do all other MS-CAN modules pass the network test?

Yes	INSTALL the removed fuse. GO to Pinpoint Test Q.
No	GO to X26.

X26 CHECK FOR RESTORED NETWORK COMMUNICATION WITH THE TPM MODULE (RADIO TRANSCEIVER MODULE [RTM] DISABLED

- Disconnect: BCM Fuse 26 (5A) .
- Enter the following diagnostic mode on the scan tool: Network Test .
- Repeat the network test.

Do all other MS-CAN modules pass the network test?

Yes	INSTALL the removed fuse. GO to Pinpoint Test W.
No	INSTALL the removed fuse. The MS-CAN tests within specification. VERIFY the correct operation of the scan tool on a known good vehicle.

X27 CHECK FOR CORRECT APIM OPERATION

- Disconnect the APIM connector.
- Check for:
  - corrosion
  - damaged pins
  - pushed-out pins
- Connect the APIM connector and make sure it seats correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new APIM. Refer to the appropriate Removal and Installation procedure in Section 415-00.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. CONNECT all modules.

X28 CHECK FOR CORRECT ACM OPERATION

- Disconnect all the ACM connectors.
- Check for:
  - corrosion
  - damaged pins
  - pushed-out pins
- Connect all the ACM connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new ACM . Refer to the appropriate section in Group 415 for the procedure.. CONNECT all modules. CLEAR the DTCs. REPEAT the network test with the scan tool.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. CONNECT all modules.

X29 CHECK FOR CORRECT BCM OPERATION

- Disconnect all the BCM connectors.
- Check for:
  - corrosion
  - damaged pins
  - pushed-out pins
- Connect all the BCM connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . CLEAR the DTCs. REPEAT the network test with the scan tool.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

X30 CHECK FOR CORRECT DSM OPERATION

- Disconnect all the DSM connectors.
- Check for:
  - corrosion
  - damaged pins
  - pushed-out pins
- Connect all the DSM connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new DSM . REFER to Section 501-10. CONNECT all modules. CLEAR the DTCs. REPEAT the network test with the scan tool.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. CONNECT all modules.

X31 CHECK FOR CORRECT DCSM OPERATION

- Disconnect all the DCSM connectors.
- Check for:
  - corrosion
  - damaged pins
  - pushed-out pins
- Connect all the DCSM connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new DCSM . REFER to Section 501-10. CONNECT all modules. CLEAR the DTCs. REPEAT the network test with the scan tool.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. CONNECT all modules.

X32 CHECK FOR CORRECT FCIM OPERATION

- Disconnect the FCIM connector.
- Check for:
  - corrosion
  - damaged pins
  - pushed-out pins
- Connect the FCIM connector and make sure it seats correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new FCIM . Refer to the appropriate section in Group 415 for the procedure.. CLEAR the DTCs. REPEAT the network test with the scan tool.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

X33 CHECK FOR CORRECT FDIM OPERATION

- Disconnect the FDIM connector.

- Check for:
  - corrosion
  - damaged pins
  - pushed-out pins
- Connect the FDIM connector and make sure it seats correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new FDIM . Refer to the appropriate section in Group 415 for the procedure.. CONNECT all modules. CLEAR the DTCs. REPEAT the network test with the scan tool.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. CONNECT all modules.

X34 CHECK FOR CORRECT GPSM OPERATION

- Disconnect the GPSM connector.
- Check for:
  - corrosion
  - damaged pins
  - pushed-out pins
- Connect the GPSM connector and make sure it seats correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new GPSM . Refer to the appropriate section in Group 415 for the procedure.. CONNECT all modules. CLEAR the DTCs. REPEAT the network test with the scan tool.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. CONNECT all modules.

X35 CHECK FOR CORRECT HVAC MODULE OPERATION

- Disconnect all the HVAC module connectors.
- Check for:
  - corrosion
  - damaged pins
  - pushed-out pins
- Connect all the HVAC module connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new HVAC module. REFER to Section 412-01. CONNECT all modules. CLEAR the DTCs. REPEAT the network test with the scan tool.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. CONNECT all modules.

X36 CHECK FOR CORRECT PRB MODULE OPERATION

- Disconnect all the PRB module connectors.
- Check for:
  - corrosion
  - damaged pins
  - pushed-out pins
- Connect all the PRB module connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new PRB module. REFER to Section 501-08. CLEAR the DTCs. CONNECT all modules. REPEAT the network test with the scan tool.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. CLEAR the DTCs. CONNECT all modules. REPEAT the network test with the scan tool.

X37 CHECK FOR CORRECTTPM MODULE (RADIO TRANSCEIVER MODULE [RTM] OPERATION

- Disconnect the TPM module (Radio Transceiver Module [RTM]) connector.
- Check for:
  - corrosion
  - damaged pins
  - pushed-out pins
- Connect the TPM module (Radio Transceiver Module [RTM]) connector and make sure it seats correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new TPM module (Radio Transceiver Module [RTM]). REFER to Section 204-04. CONNECT all modules. CLEAR the DTCs. REPEAT the network test with the scan tool.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. CONNECT all modules.

Pinpoint Test Y: No High Speed Controller Area Network (HS-CAN) Communication, All Modules Are Not Responding

Refer to Wiring Diagrams Cell 14 , Module Communications Network for schematic and connector information.

Normal Operation

The High Speed Controller Area Network (HS-CAN) uses an unshielded twisted pair cable.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- TCCM (if equipped)
- ABS module
- Accessory Protocol Interface Module (APIM) (if equipped)
- Body Control Module (BCM)
- Instrument Panel Cluster (IPC)
- Occupant Classification System Module (OCSM)
- Parking Aid Module (PAM) (if equipped)
- PCM
- Power Steering Control Module (PSCM) (if equipped)
- Restraints Control Module (RCM)
- Steering Column Control Module (SCCM)
- Trailer Brake Control (TBC) module (if equipped)

PINPOINT TEST Y : NO HS-CAN COMMUNICATION, ALL MODULES ARE NOT RESPONDING

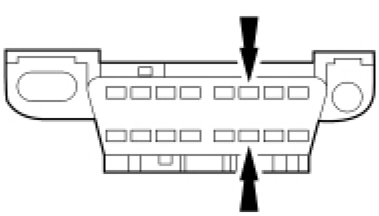
NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

NOTE: *Most faults are due to connector and/or wiring concerns. Carry out a thorough inspection and verification before proceeding with the pinpoint test.*

NOTE: *Failure to disconnect the battery when instructed results in false resistance readings. Refer to Section 414-01.*

Y1 CHECK THE DLC PINS FOR DAMAGE

- Ignition OFF.
- Disconnect the scan tool cable from the Data Link Connector (DLC) .
- Inspect the DLC pins 6 and 14 for damage.

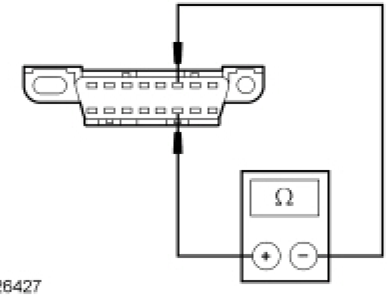


- A0093867

Are DLC pins 6 and 14 OK?

Yes	GO to Y2.
No	REPAIR the DLC as necessary. CLEAR the DTCs. REPEAT the network test with the scan tool.

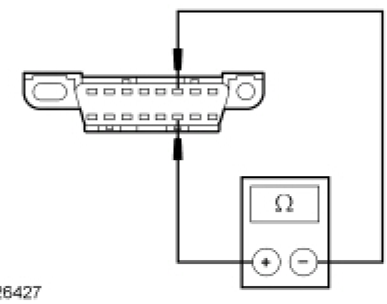
Y2 CHECK THE HS-CAN TERMINATION RESISTANCE

- Ignition OFF.
- Disconnect: Negative Battery Cable .
- Measure the resistance between the DLC C251 Pin 6, circuit VDB04 (WH/BU), harness side and the DLC C251 Pin 14, circuit VDB05 (WH), harness side.
- 
N0026427

Is the resistance between 54 and 66 ohms?

Yes	GO to Y11.
No	GO to Y3.

Y3 CHECK THE HS-CAN TERMINATION RESISTOR

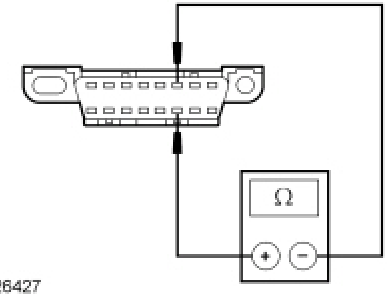
- Measure the resistance between the DLC C251 Pin 6, circuit VDB04 (WH/BU), harness side and the DLC C251 Pin 14, circuit VDB05 (WH), harness side.
- 
N0026427

Is the resistance between 108 and 132 ohms?

Yes	GO to Y4.
No	GO to Y8.

Y4 CHECK THE HS-CAN TERMINATION RESISTOR WITH THE PCM DISCONNECTED

- Disconnect: PCM C175B (3.7L), C1551B (3.5L), or C1381B (5.0L and 6.2L) .
- Measure the resistance between the DLC C251 Pin 6, circuit VDB04 (WH/BU), harness side and the DLC C251 Pin 14, circuit VDB05 (WH), harness side.



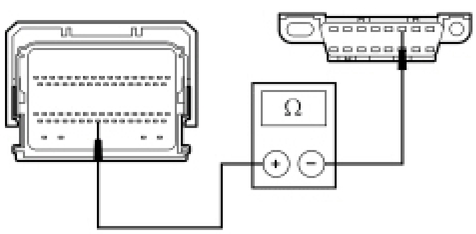
- N0026427

Is the resistance between 108 and 132 ohms?

Yes	For 3.7L, 5.0L or 6.2L, GO to Y5. For 3.5L, GO to Y6.
No	GO to Y7.

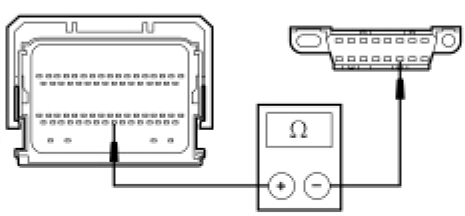
Y5 CHECK THE HS-CAN CIRCUITS BETWEEN THE PCM AND THE DLC FOR AN OPEN

- Measure the resistance between the PCM C175B Pin 59 (3.7L) or C1381B Pin 59 (5.0L and 6.2L), circuit VDB04 (WH/BU), harness side and the DLC C251 Pin 6, circuit VDB04 (WH/BU), harness side.



- N0077607

- Measure the resistance between the PCM C175B Pin 58 (3.7L) or C1381B Pin 58 (5.0L and 6.2L), circuit VDB05 (WH), harness side and the DLC C251 Pin 14, circuit VDB05 (WH), harness side.



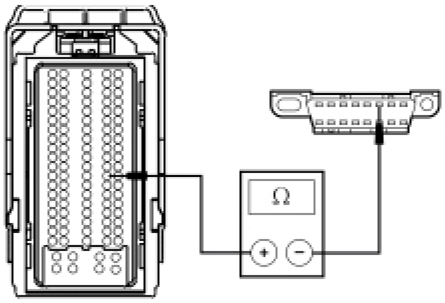
- N0113941

Are the resistances less than 5 ohms?

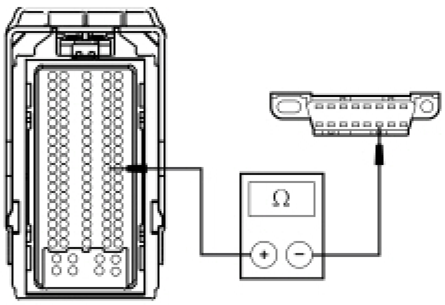
Yes	CONNECT the negative battery cable. GO to Y34.
No	REPAIR the circuit in question. CONNECT the negative battery cable. CLEAR the DTCs. REPEAT the network test with the scan tool.

Y6 CHECK THE HS-CAN CIRCUITS BETWEEN THE PCM AND THE DLC FOR AN OPEN

- Measure the resistance between the PCM C1551B Pin 69, circuit VDB04 (WH/BU), harness side and the DLC C251 Pin 6, circuit VDB04 (WH/BU), harness side.

- 
  N0123791

- Measure the resistance between the PCM C1551B Pin 68, circuit VDB05 (WH), harness side and the DLC C251 Pin 14, circuit VDB05 (WH), harness side.

- 
  N0123790

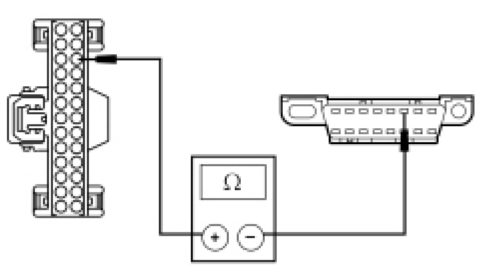
Are the resistances less than 5 ohms?

Yes	CONNECT the negative battery cable. GO to Y34.
No	REPAIR the circuit in question. CONNECT the negative battery cable. CLEAR the DTCs. REPEAT the network test with the scan tool.

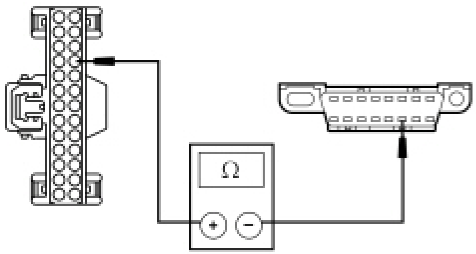
Y7 CHECK THE HS-CAN CIRCUITS BETWEEN THE DLC AND THE BCM FOR AN OPEN

- Disconnect: BCM C2280F .

- Measure the resistance between the BCM C2280F Pin 16, circuit VDB04 (WH/BU), harness side and the Data Link Connector (DLC) C251 Pin 6, circuit VDB06 (WH), harness side.

- 
  N0115338

- Measure the resistance between the BCM C2280F Pin 17, circuit VDB05 (WH), harness side and the DLC C251 Pin 14, circuit VDB05 (WH), harness side.



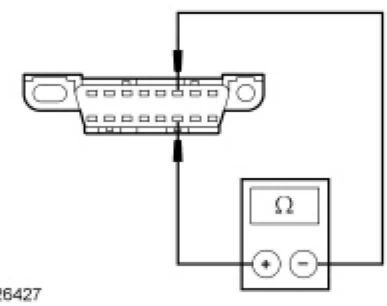
- N0115339

Are the resistances less than 5 ohms?

Yes	CONNECT the negative battery cable. GO to Y29.
No	REPAIR the circuit in question. CONNECT the negative battery cable. CLEAR the DTCs. REPEAT the network test with the scan tool.

Y8 CHECK THE HS-CAN (+) AND HS-CAN (-) CIRCUITS FOR A SHORT TOGETHER

- Measure the resistance between the DLC C251 Pin 6, circuit VDB04 (WH/BU), harness side and the DLC C251 Pin 14, circuit VDB05 (WH), harness side.



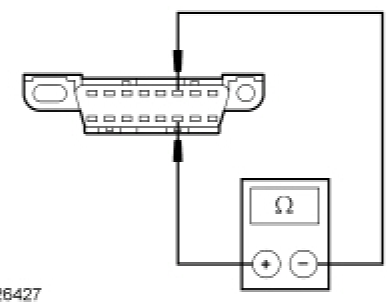
- N0026427

Is the resistance less than 5 ohms?

Yes	GO to Y10.
No	GO to Y9.

Y9 CHECK THE HS-CAN (+) ANDHS-CAN (-) CIRCUITS FOR AN OPEN

- Measure the resistance between the DLC C251 Pin 6, circuit VDB04 (WH/BU), harness side and the DLC C251 Pin 14, circuit VDB05 (WH), harness side.



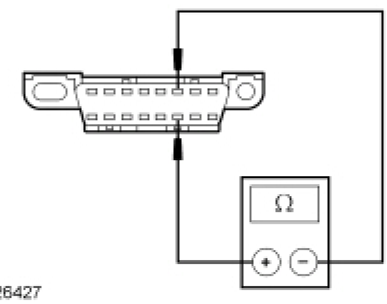
- N0026427

Is the resistance greater than 10,000 ohms?

Yes	REPAIR the DLC or the circuit in question. CONNECT all modules. CLEAR the DTCs. REPEAT the network test with the scan tool.
No	A capacitor internal to a module may still be draining, causing irregular resistance readings. WAIT 5 minutes. REPEAT the pinpoint test.

Y10 CHECK THE HS-CAN (+) AND HS-CAN (-) CIRCUITS FOR A SHORT TOGETHER

- While measuring the resistance between the DLC C251 Pin 6, circuit VDB04 (WH/BU), harness side and the DLC C251 Pin 14, circuit VDB05 (WH), harness side, disconnect the following modules one at a time until the resistance is greater than 5 ohms.
  - TCCM C2371A (if equipped)
  - APIM C2383 (if equipped)
  - ABS module C135
  - BCM C2280F
  - IPC C220
  - OCSM C3159
  - PAM C2023 (if equipped)
  - PSCM C1463B (if equipped)
  - PCM C175B, C1381B or C1551B
  - RCM C310B
  - SCCM C2414A
  - TBC module C2142 (if equipped)



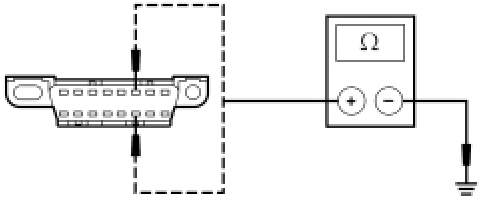
- N0026427

Is the resistance greater than 5 ohms with one of the modules disconnected?

Yes	CONNECT the negative battery cable. For the TCCM , GO to Y26. For the APIM , GO to Y27. For the ABS module, GO to Y28. For the BCM , GO to Y29. For the IPC , GO to Y30. For the OCSM , GO to Y31. For the PAM , GO to Y32. For the PSCM , GO to Y33. For the PCM, GO to Y34. For the RCM , GO to Y35. For the SCCM , GO to Y36. For the TBC module, GO to Y37.
No	REPAIR circuits VDB04 (WH/BU) and VDB05 (WH) for being shorted together. CLEAR the DTCs. REPEAT the Network Test with the scan tool.

Y11 CHECK THE HS-CAN (+) AND HS-CAN (-) CIRCUITS FOR A SHORT TO GROUND

- Measure the resistance between the DLC C251 Pin 6, circuit VDB04 (WH/BU), harness side and ground; and between the DLC C251 Pin 14, circuit VDB05 (WH), harness side and ground.



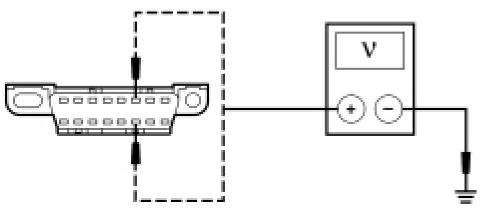
- N0002963

Are the resistances greater than 1,000 ohms?

Yes	CONNECT the negative battery cable. GO to Y12.
No	GO to Y13.

Y12 CHECK THE HS-CAN (+) AND HS-CAN (-) CIRCUITS FOR A SHORT TO VOLTAGE

- Ignition ON.
- Measure the voltage between the DLC C251 Pin 6, circuit VDB04 (WH/BU), harness side and ground; and between the DLC C251 Pin 14, circuit VDB05 (WH), harness side and ground.



- N0002964

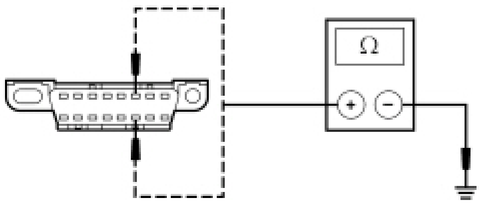
Are the voltages greater than 6 volts?

Yes	REPAIR the circuit in question. CLEAR the DTCs. REPEAT the network test with the scan tool.
No	GO to Y14.

Y13 CHECK THE HS-CAN (+) AND HS-CAN (-) CIRCUITS FOR A SHORT TO GROUND WITH THE MODULES DISCONNECTED

- While measuring the resistance between the DLC C251 Pin 6, circuit VDB04 (WH/BU), harness side and ground; and between the DLC C251 Pin 14, circuit VDB05 (WH), harness side and ground, check for the short to ground on both circuits after disconnecting each module. Disconnect the following modules one at a time until the resistance to ground is greater than 1,000 ohms.
  - TCCM C2371A (if equipped)
  - APIM C2383 (if equipped)
  - ABS module C135
  - BCM C2280F
  - IPC C220
  - OCSM C3159

- PAM C2023 (if equipped)
- PSCM C1463B (if equipped)
- PCM C175B, C1381B or C1551B
- RCM C310B
- SCCM C2414A
- TBC module C2142 (if equipped)



- N0002963

Is the resistance greater than 1,000 ohms with one of the modules disconnected?

Yes	CONNECT the negative battery cable. For the TCCM , GO to Y26. For the APIM , GO to Y27. For the ABS module, GO to Y28. For the BCM , GO to Y29. For the IPC , GO to Y30. For the OCSM , GO to Y31. For the PAM , GO to Y32. For the PSCM , GO to Y33. For the PCM, GO to Y34. For the RCM , GO to Y35. For the SCCM , GO to Y36. For the TBC module, GO to Y37.
No	REPAIR the circuit in question. CONNECT all modules. CONNECT the negative battery cable. CLEAR the DTCs. REPEAT the Network Test with the scan tool.

Y14 CHECK FOR RESTORED COMMUNICATION WITH THE PCM DISABLED

- Disconnect: Battery Junction Box (BJB) Fuses 26 (10A) and 75 (15A) (25A for 3.5L engine) .
- Enter the following diagnostic mode on the scan tool: Network Test .
- Repeat the network test.

Do all other HS-CAN modules pass the network test?

Yes	INSTALL the removed fuses. GO to Pinpoint Test A.
No	GO to Y15.

Y15 CHECK FOR RESTORED NETWORK COMMUNICATION WITH THE PSCM DISABLED

- Disconnect: High Current Battery Junction Box (BJB) fuse 1 (125A) and Battery Junction Box (BJB) Fuse 52 (5A) .

- Enter the following diagnostic mode on the scan tool: Network Test .
- Repeat the network test.

Do all other HS-CAN modules pass the network test?

Yes	INSTALL the removed fuse. GO to Pinpoint Test H.
No	GO to Y16.

Y16 CHECK FOR RESTORED NETWORK COMMUNICATION WITH THE ABS MODULE AND TCCM DISABLED

- Disconnect: BJB Fuses 20 (20A)(ESOF only), 36 (30A), 47 (60A), 54 (5A) and 68 (25A) .
- Enter the following diagnostic mode on the scan tool: Network Test .
- Repeat the network test.

Do all other HS-CAN modules pass the network test?

Yes	INSTALL the removed fuses. GO to Y17.
No	INSTALL the removed fuses. If the vehicle is equipped with a TCCM , GO to Y17. If the vehicle is not equipped with a TCCM , GO to Y18.

Y17 CHECK FOR RESTORED NETWORK COMMUNICATION WITH THE TCCM DISCONNECTED

- Disconnect: TCCM .
- Enter the following diagnostic mode on the scan tool: Network Test .
- Repeat the network test.

Do all other HS-CAN modules pass the network test?

Yes	INSTALL the removed fuses. GO to Pinpoint Test F .
No	INSTALL the removed fuses. GO to Pinpoint Test B.

Y18 CHECK FOR RESTORED NETWORK COMMUNICATION WITH THE RCM AND OCSM DISABLED

- Disconnect: Body Control Module (BCM) Fuse 36 (10A) .
- Enter the following diagnostic mode on the scan tool: Network Test .
- Repeat the network test.

Do all other HS-CAN modules pass the network test?

Yes	INSTALL the removed fuse. GO to Y19.

No	INSTALL the removed fuse. If the vehicle is equipped with a PAM , GO to Y20. If the vehicle is not equipped with a PAM , GO to Y21.

Y19 CHECK FOR RESTORED NETWORK COMMUNICATION WITH THE RCM AND OCSM DISCONNECTED

- Disconnect: RCM C310B .
- Enter the following diagnostic mode on the scan tool: Network Test .
- Disconnect: OCSM C3159 .
- Enter the following diagnostic mode on the scan tool: Network Test .
- Repeat the network test.

Do all other HS-CAN modules pass the network test?

Yes	CONNECT all modules. For restored operation with the RCM disconnected, GO to Pinpoint Test D. For restored operation with the OCSM disconnected, GO to Pinpoint Test E.
No	CONNECT all modules. The HS-CAN tests within specification. VERIFY the correct operation of the scan tool on a known good vehicle.

Y20 CHECK FOR RESTORED NETWORK COMMUNICATION WITH THE PAM DISABLED

- Disconnect: Body Control Module (BCM) Fuse 34 (10A) .
- Enter the following diagnostic mode on the scan tool: Network Test .
- Repeat the network test.

Do all other HS-CAN modules pass the network test?

Yes	INSTALL the removed fuse. GO to Pinpoint Test S.
No	GO to Y21.

Y21 CHECK FOR RESTORED NETWORK COMMUNICATION WITH THE SCCM DISABLED

- Disconnect: BCM Fuse 23 (15A) .
- Enter the following diagnostic mode on the scan tool: Network Test .
- Repeat the network test.

Do all other HS-CAN modules pass the network test?

Yes	INSTALL the removed fuse. GO to Pinpoint Test V.
No	INSTALL the removed fuse. If the vehicle is equipped with an Accessory Protocol Interface Module (APIM) , GO to Y22.

	If the vehicle is not equipped with an APIM , GO to Y23.

Y22 CHECK FOR RESTORED NETWORK COMMUNICATION WITH THE APIM DISABLED

- Disconnect: BCM Fuse 2 (15A) .
- Enter the following diagnostic mode on the scan tool: Network Test .
- Repeat the network test.

Do all other HS-CAN modules pass the network test?

Yes	INSTALL the removed fuse. GO to Pinpoint Test U.
No	INSTALL the removed fuse. GO to Y23.

Y23 CHECK FOR RESTORED NETWORK COMMUNICATION WITH THE BCM DISABLED

- Disconnect: BJB Fuse 45 (10A) .
- Enter the following diagnostic mode on the scan tool: Network Test .
- Repeat the network test.

Do all other HS-CAN modules pass the network test?

Yes	INSTALL the removed fuse. GO to Pinpoint Test I.
No	INSTALL the removed fuse. If the vehicle is equipped with an Trailer Brake Control (TBC) , GO to Y24. If the vehicle is not equipped with a TBC , GO to Y25.

Y24 CHECK FOR RESTORED NETWORK COMMUNICATION WITH THE TBC MODULE DISABLED

- Disconnect: BCM Fuse 37 (10A), BJB Fuse 17 (30A) .
- Enter the following diagnostic mode on the scan tool: Network Test .
- Repeat the network test.

Do all other HS-CAN modules pass the network test?

Yes	INSTALL the removed fuses. GO to Pinpoint Test G.
No	INSTALL the removed fuses. GO to Y25.

Y25 CHECK FOR RESTORED NETWORK COMMUNICATION WITH THE IPC DISABLED

- Disconnect: BCM Fuse 11 (10A) .
- Enter the following diagnostic mode on the scan tool: Network Test .
- Repeat the network test.

Do all other HS-CAN modules pass the network test?

Yes	INSTALL the removed fuse. GO to Pinpoint Test C.
No	INSTALL the removed fuse. The HS-CAN tests within specification. VERIFY the correct operation of the scan tool on a known good vehicle.

Y26 CHECK FOR CORRECT TCCM OPERATION

- Disconnect all the TCCM connectors.
- Check for:
  - corrosion
  - damaged pins
  - pushed-out pins
- Connect all the TCCM connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new TCCM . REFER to Section 308-07A. CONNECT all modules. CLEAR the DTCs. REPEAT the network test with the scan tool.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. CONNECT all modules.

Y27 CHECK FOR CORRECT APIM OPERATION

- Disconnect the APIM connector.
- Check for:
  - corrosion
  - damaged pins
  - pushed-out pins
- Connect the APIM connector and make sure it seats correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new APIM. Refer to the appropriate Removal and Installation procedure in Section 415-00.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. CONNECT all modules.

Y28 CHECK FOR CORRECT ABS MODULE OPERATION

- Disconnect the ABS module connector.

- Check for:
  - corrosion
  - damaged pins
  - pushed-out pins
- Connect the ABS module connector and make sure it seats correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new ABS module. REFER to Section 206-09. CONNECT all modules. CLEAR the DTCs. REPEAT the network test with the scan tool.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. CONNECT all modules.

Y29 CHECK FOR CORRECT BCM OPERATION

- Disconnect all the BCM connectors.
- Check for:
  - corrosion
  - damaged pins
  - pushed-out pins
- Connect all the BCM connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . CLEAR the DTCs. REPEAT the network test with the scan tool.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Y30 CHECK FOR CORRECT IPC OPERATION

- Disconnect the IPC connector.
- Check for:
  - corrosion
  - damaged pins
  - pushed-out pins
- Connect the IPC connector and make sure it seats correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. CONNECT all modules. CLEAR the DTCs. REPEAT the network test with the scan tool.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. CONNECT all modules.

Y31 CHECK FOR CORRECT OCSM OPERATION

- Disconnect the OCSM connector.
- Check for:
  - corrosion
  - damaged pins
  - pushed-out pins
- Connect the OCSM connector and make sure it seats correctly.
- Verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new OCSM . REFER to Section 501-20B. CONNECT all modules. CLEAR the DTCs. REPEAT the network test with the scan tool.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. CONNECT all modules.

Y32 CHECK FOR CORRECT PAM OPERATION

- Disconnect the PAM connector.
- Check for:
  - corrosion
  - damaged pins
  - pushed-out pins
- Connect the PAM connector and make sure it seats correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new PAM . REFER to Section 413-13. CONNECT all modules. CLEAR the DTCs. REPEAT the network test with the scan tool.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. CONNECT all modules.

Y33 CHECK FOR CORRECT PSCM OPERATION

- Disconnect all the PSCM connectors.
- Check for:
  - corrosion
  - damaged pins

- pushed-out pins

- Connect all the PSCM connectors and make sure they seat correctly.
- Verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new EPAS steering gear. REFER to Section 211-02. CONNECT all modules. CLEAR the DTCs. REPEAT the network test with the scan tool.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. CONNECT all modules.

Y34 CHECK FOR CORRECT PCM OPERATION

- Disconnect all the PCM connectors.
- Check for:
  - corrosion
  - damaged pins
  - pushed-out pins
- Connect all the PCM connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new PCM. REFER to Section 303-14. CLEAR the DTCs. REPEAT the network test with the scan tool.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Y35 CHECK FOR CORRECT RCM OPERATION

- Disconnect all the RCM connectors.
- Check for:
  - corrosion
  - damaged pins
  - pushed-out pins
- Connect all the RCM connectors and make sure they seat correctly.
- Verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new RCM . REFER to Section 501-20B. CONNECT all modules. CLEAR the DTCs. REPEAT the network test with the scan tool.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. CONNECT all modules.

Y36 CHECK FOR CORRECT SCCM OPERATION

• Disconnect all the SCCM connectors.
• Check for:
  ▪ corrosion
  ▪ damaged pins
  ▪ pushed-out pins
• Connect all the SCCM connectors and make sure they seat correctly.
• Verify the concern is still present.
 	
Is the concern still present?

Yes	INSTALL a new SCCM . REFER to Section 211-05. CONNECT all modules. CLEAR the DTCs. REPEAT the network test with the scan tool.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. CONNECT all modules.

Y37 CHECK FOR CORRECT TBC MODULE OPERATION

• Disconnect the TBC module connector.
• Check for:
  ▪ corrosion
  ▪ damaged pins
  ▪ pushed-out pins
• Connect the TBC module connector and make sure it seats correctly.
• Verify the concern is still present.
 	
Is the concern still present?

Yes	INSTALL a new TBC module. REFER to Section 206-10. CONNECT all modules. CLEAR the DTCs. REPEAT the network test with the scan tool.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. CONNECT all modules.

Pinpoint Test Z: No Infotainment Controller Area Network (I-CAN) Communication, All Modules Are Not Responding

Refer to Wiring Diagrams Cell 14 , Module Communications Network for schematic and connector information.

Normal Operation

The Infotainment Controller Area Network (I-CAN) uses an unshielded twisted pair cable.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Accessory Protocol Interface Module (APIM) (if equipped)
- Audio Front Control Module (ACM)
- Audio DSP module (if equipped)
- Front Controls Interface Module (FCIM) (if equipped)
- Front Control/Display Interface Module (FCDIM) (if equipped)
- Instrument Panel Cluster (IPC)

PINPOINT TEST Z : NO I-CAN COMMUNICATION, ALL MODULES ARE NOT RESPONDING

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

NOTE: *Most faults are due to connector and/or wiring concerns. Carry out a thorough inspection and verification before proceeding with the pinpoint test.*

NOTE: *Failure to disconnect the battery when instructed results in false resistance readings. Refer to Section 414-01.*

Z1 CHECK FOR RESTORED NETWORK COMMUNICATION WITH THE ACM DISCONNECTED

NOTE: *When re-running the network test, close the network test application first or the screen display reverts back to the previously run network test results.*

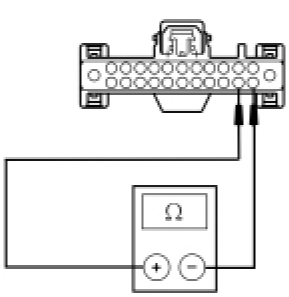
- Disconnect: ACM C240A .
- Using a scan tool, perform a network test.

Do all other I-CAN modules pass the network test?

Yes	GO to Z10.
No	GO to Z2.

Z2 CHECK THE I-CAN TERMINATION RESISTANCE

- Ignition OFF.
- Disconnect: Negative Battery Cable .
- Measure the resistance between the ACM C240A Pin 14, circuit VDB13 (BU/GY), harness side and the ACM C240A Pin 15, circuit VDB14 (VT/GY), harness side.



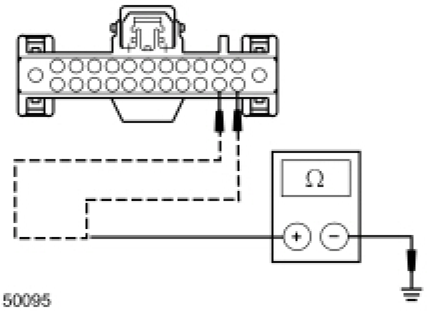
- N0150065

Is the resistance between 108 and 132 ohms?

Yes	GO to Z3.

No	GO to Z6.

Z3 CHECK THE I-CAN (+) AND I-CAN (-) CIRCUITS FOR A SHORT TO GROUND	

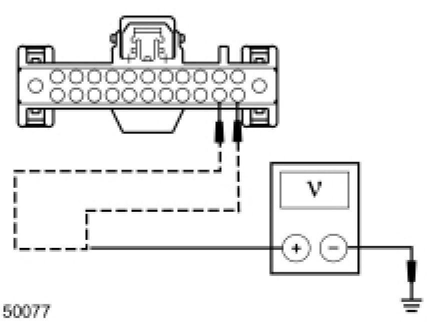
- Ignition OFF.
- Measure the resistance between the ACM C240A Pin 14, circuit VDB13 (BU/GY), harness side and ground; and between the ACM C240A Pin 15, circuit VDB14 (VT/GY), harness side and ground.
- 

N0150095

Are the resistances greater than 1,000 ohms?

Yes	GO to Z4.
No	REPAIR the circuit in question.

Z4 CHECK THE I-CAN (+) AND I-CAN (-) CIRCUITS FOR A SHORT TO VOLTAGE

- Connect: negative battery cable .
- Ignition ON.
- Measure the voltage between the ACM C240A Pin 14, circuit VDB13 (BU/GY), harness side and ground; and between the ACM C240A Pin 15, circuit VDB14 (VT/GY), harness side and ground.
- 

N0150077

Is the voltage greater than 6 volts?

Yes	REPAIR the circuit in question.
No	GO to Z5.

Z5 CHECK FOR RESTORED NETWORK COMMUNICATION

NOTE: *When re-running the network test, close the network test application first or the screen display reverts back to the previously run network test results.*

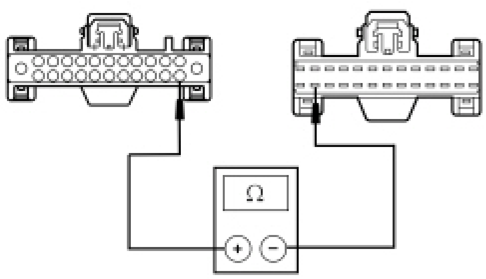
- Ignition OFF.
- Connect all module(s). Make sure all connectors seat and latch correctly.
- Ignition ON.
- Using a scan tool, perform the network test.

Do all other I-CAN modules pass the network test?

Yes	The I-CAN tests within specification. VERIFY the correct operation of the scan tool on a known good vehicle.
No	GO to Z14.

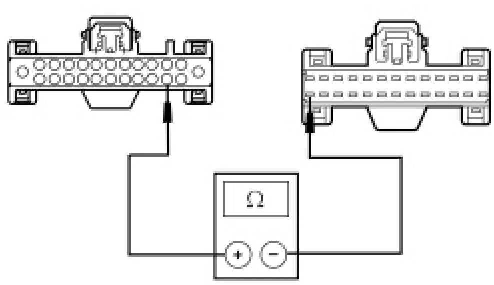
Z6 CHECK THE I-CAN CIRCUITS BETWEEN THE ACM AND THE IPC FOR AN OPEN

- Disconnect: IPC C220 .
- Measure the resistance between the ACM C240A Pin 14, circuit VDB13 (BU/GY), harness side and the IPC C220 Pin 25, circuit VDB13 (BU/GY), harness side.



- N0150484

- Measure the resistance between the ACM C240A Pin 15, circuit VDB14 (VT/GY), harness side and the IPC C220 Pin 26, circuit VDB14 (VT/GY), harness side.

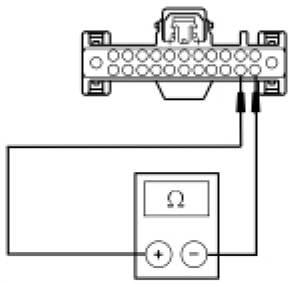


- N0150485

Are the resistances less than 5 ohms?

Yes	CONNECT the negative battery cable. GO to Z7.
No	REPAIR the circuit in question.

Z7 CHECK THE I-CAN (+) AND I-CAN (-) CIRCUITS FOR A SHORT TOGETHER

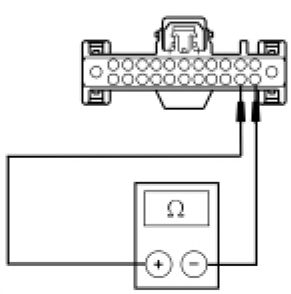
- Measure the resistance between the ACM C240A Pin 14, circuit VDB13 (BU/GY), harness side and the ACM C240A Pin 15, circuit VDB14 (VT/GY), harness side.
- 
N0150065

Is the resistance greater than 1,000 ohms?

Yes	GO to Z8.
No	REPAIR the circuit in question.

Z8 CHECK THE I-CAN (+) AND I-CAN (-) CIRCUITS FOR A SHORT TOGETHER WITH MODULES DISCONNECTED

NOTE: *Skip to the end of this test once the suspect module(s) has been identified.*

- Disconnect: negative battery cable .
- While measuring the resistance between the ACM C290A Pin 14, circuit VDB13 (BU/GY), harness side and the ACM C290A Pin 15, circuit VDB14 (VT/GY), harness side, disconnect the following modules one at a time until the resistance is greater than 5 ohms.
  - APIM C2298 (if equipped)
  - Audio DSP module C3154 (if equipped)
  - FCIM C2402 (if equipped)
  - FCDIM C2123 (if equipped)
  - IPC C220
- 
N0150065

Did the resistance change to between 108 and 132 ohms with one of the modules disconnected?

Yes	CONNECT the negative battery cable. For the APIM , GO to Z9. For the audio DSP module, GO to Z11. For the FCIM , GO to Z12. For the FCDIM , GO to Z13. For the IPC , GO to Z14.
No	REPAIR circuits in question for a short together. CONNECT the modules. CONNECT the negative

	battery cable. CLEAR the DTCs.

Z9 CHECK FOR CORRECT APIM OPERATION

- Disconnect the APIM connector.
- Check for:
  - corrosion
  - damaged pins
  - pushed-out pins
- Connect the APIM connector and make sure it seats correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new APIM. Refer to the appropriate Removal and Installation procedure in Section 415-00.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. CONNECT all modules.

Z10 CHECK FOR CORRECT ACM OPERATION

- Disconnect all the ACM connectors.
- Check for:
  - corrosion
  - damaged pins
  - pushed-out pins
- Connect all the ACM connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new ACM . Refer to the appropriate section in Group 415 for the procedure. CONNECT all modules. CLEAR the DTCs. REPEAT the network test with the scan tool.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. CONNECT all modules.

Z11 CHECK FOR CORRECT AUDIO DSP MODULE OPERATION

- Disconnect the audio DSP module connector.
- Check for:
  - corrosion
  - damaged pins
  - pushed-out pins
- Connect the audio DSP module connector and make sure it seats correctly.

- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new audio DSP module. REFER to Section 415-00E. CONNECT all modules. CLEAR the DTCs. REPEAT the network test with the scan tool.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. CONNECT all modules.

Z12 CHECK FOR CORRECT FCIM OPERATION

- Disconnect the FCIM connector.
- Check for:
  - corrosion
  - damaged pins
  - pushed-out pins
- Connect the FCIM connector and make sure it seats correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new FCIM . REFER to Refer to the appropriate section in Group 415 for the procedure. CLEAR the DTCs. REPEAT the network test with the scan tool.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Z13 CHECK FOR CORRECT FCDIM OPERATION

- Disconnect the FCDIM connector.
- Check for:
  - corrosion
  - damaged pins
  - pushed-out pins
- Connect the FCDIM connector and make sure it seats correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new FCDIM . REFER to Section 415-00C. CLEAR the DTCs. REPEAT the network test with the scan tool.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Z14 CHECK FOR CORRECT IPC OPERATION

- Disconnect the IPC connector.
- Check for:
  - corrosion
  - damaged pins
  - pushed-out pins
- Connect the IPC connector and make sure it seats correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new IPC. Refer to the appropriate Removal and Installation procedure in Section 413-01. CONNECT all modules. CLEAR the DTCs. REPEAT the network test with the scan tool.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. CONNECT all modules.

Pinpoint Test AA: No Power To The Scan Tool

Refer to Wiring Diagrams Cell 14 , Module Communications Network for schematic and connector information.

Normal Operation

The scan tool is connected to the Data Link Connector (DLC) to communicate with the High Speed Controller Area Network (HS-CAN) , Medium Speed Controller Area Network (MS-CAN) and Infotainment Controller Area Network (I-CAN) communications networks.

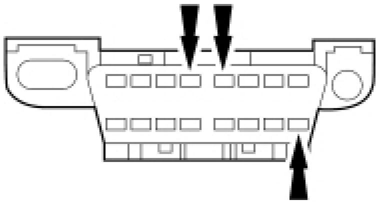
A loss of ground or poor ground at the DLC may result in HS-CAN , MS-CAN or I-CAN faults while the scan tool is connected.

This pinpoint test is intended to diagnose the following:

- Fuse
- Wiring, terminals or connectors
- Scan tool
- DLC

PINPOINT TEST AA : NO POWER TO THE SCAN TOOL

NOTE: *Most faults are due to connector and/or wiring concerns. Carry out a thorough inspection and verification before proceeding with the pinpoint test.*
NOTE: *Failure to disconnect the battery when instructed results in false resistance readings. Refer to Section 414-01.*
AA1 CHECK THE DLC PINS FOR DAMAGE
• Disconnect the scan tool cable from the DLC . • Inspect the DLC pins 4, 5 and 16 for damage.



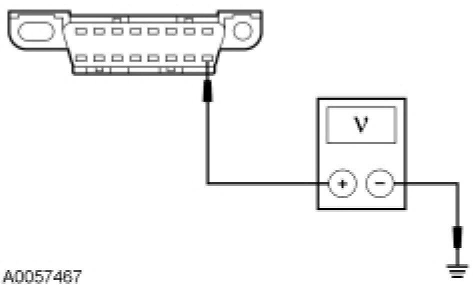
- N0050767

Are DLC pins 4, 5 and 16 OK?

Yes	GO to AA2.
No	REPAIR the DLC as necessary. CLEAR the DTCs. REPEAT the network test with the scan tool.

AA2 CHECK THE DLC VOLTAGE SUPPLY CIRCUIT FOR AN OPEN

- Measure the voltage between the DLC C251 Pin 16, circuit SBP24 (VT/RD), harness side and ground.



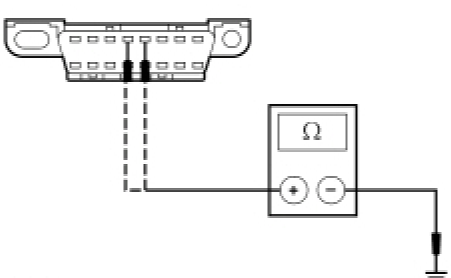
- A0057467

Is the voltage greater than 10 volts?

Yes	GO to AA3.
No	VERIFY the Body Control Module (BCM) fuse 24 (15A) is OK. If OK, REPAIR the circuit. If not OK, REFER to the Wiring Diagrams manual to identify the possible causes of the circuit short. REPEAT the network test with the scan tool.

AA3 CHECK THE DLC GROUND CIRCUITS FOR AN OPEN

- Disconnect: Negative Battery Cable .
- Measure the resistance between the DLC C251 Pin 4, circuit GD133 (BK), harness side and ground; and between the DLC C251 Pin 5, circuit GD133 (BK), harness side and ground.



- A0060571

Are the resistances less than 5 ohms?

Yes	REPAIR the scan tool. CONNECT the negative battery cable. REPEAT the network test with the scan tool.
No	REPAIR the circuit in question. CONNECT the negative battery cable. REPEAT the network test with the scan tool.